

THE 4 COMPONENTS OF A COMPUTER SYSTEM

Every computer system works together using these four essential components.

1 HARDWARE

The physical parts you can see and touch.



EXAMPLES

- Monitor
- Keyboard
- Mouse
- Motherboard
- RAM
- CPU
- Hard Drive / SSD
- Graphics Card
- Printer
- Webcam



Hardware performs the actual tasks and provides input, output, processing and storage.

2 SOFTWARE

The programs and instructions that tell the hardware what to do.



EXAMPLES

- Operating Systems (Windows, macOS, Linux)
- Applications (Word, Excel, Chrome)
- Drivers
- Utilities (Antivirus, Backup)
- Programming Languages
- Games



Software manages hardware resources and provides functions, tools and interfaces for users.

3 STORAGE

Stores data and information for short-term or long-term use.



EXAMPLES

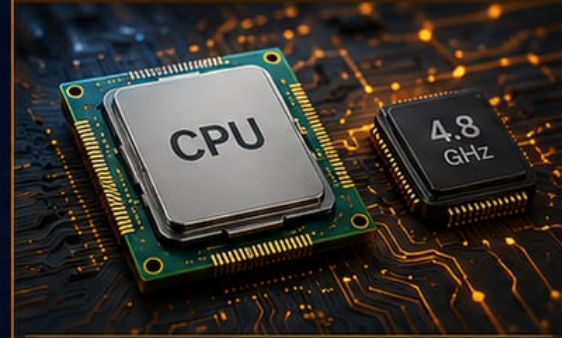
- Hard Disk Drives (HDD)
- Solid State Drives (SSD)
- USB Flash Drives
- SD / MicroSD Cards
- Optical Discs (CD/DVD)
- Cloud Storage



Storage retains data, programs and system files — even when the power is off.

4 PROCESSING

Performs calculations and executes instructions.



EXAMPLES

- CPU (Central Processing Unit)
- GPU (Graphics Processing Unit)
- ALU (Arithmetic Logic Unit)
- Control Unit
- Cache Memory
- Registers



Processing is the brain of the system — it thinks, calculates and controls all operations.

HOW THEY WORK TOGETHER



Hardware provides the physical components.



Software tells the hardware what to do and how.



Storage saves data and makes it available when needed.



Processing carries out instructions and returns results.



Together, they deliver useful information and powerful experiences.

REMEMBER



No single component works alone. When hardware, software, storage and processing work together, computers become powerful tools that help us learn, communicate, create and solve problems.

MOTHERBOARD COMPONENTS

The essential parts that connect, power and control your PC.



Connects everything on your motherboard



Enables communication between components



Delivers power where it's needed



Supports performance and expandability



Ensures stability and reliability

1 CPU SOCKET



Holds the CPU in place and provides electrical connections between the CPU and motherboard.



LGA / PGA Type varies (e.g., LGA1700, AM5)

2 CHIPSET



Manages data flow between the CPU, memory, storage, and peripherals.



Controls I/O, USB, SATA, PCIe lanes and more

3 VRM (VOLTAGE REGULATOR MODULE)



Delivers clean, stable power to the CPU by converting and regulating voltage.



More phases = Better power delivery and stability

4 DIMM SLOTS



Holds RAM modules. Supports dual-channel or quad-channel memory.



DDR4 / DDR5 Slots (usually 2 or 4)

5 PCIe SLOTS



Used to install expansion cards such as graphics cards, sound cards, network cards and more.



x1, x4, x8, x16 (varies by slot)

6 M.2 SLOTS



Connects M.2 SSDs for ultra-fast storage using PCIe or SATA interface.



NVMe (PCIe) = Faster
SATA = Compatible

7 SATA PORTS



Connects SATA devices such as HDDs, SSDs and optical drives.



SATA 3.0 (6Gb/s)
Backwards compatible with older devices

8 CMOS BATTERY



Powers the CMOS to retain BIOS settings, date, and time when the PC is off.



Usually CR2032
Replace when BIOS resets or time is wrong

9 BIOS / UEFI CHIP



Stores the firmware that starts the system and provides low-level hardware control.



UEFI = Modern interface
Supports advanced features & security

10 REAR I/O PANEL



External ports for connecting peripherals such as USB devices, monitors, network cables and audio.



Includes USB, audio, network, video and more



WHY THESE COMPONENTS MATTER

- ✓ Work together to ensure your PC runs smoothly and efficiently.
- ✓ A higher quality motherboard = better performance, stability and upgrade potential.
- ✓ Knowing your motherboard helps with building, troubleshooting and upgrading.



Check compatibility (CPU, RAM, GPU, etc.)



Look for good VRM if using a high-end CPU



Ensure enough slots and ports for your needs



Keep BIOS updated for best compatibility



Use quality parts for long-term reliability

TIPS FOR A SMART BUILD

EVERY PART. EVERY CONNECTION. EVERY TIME.



Your motherboard is the foundation—build on it with confidence.

MOTHERBOARD COMPONENTS

CPU SOCKET

The CPU socket is the interface on the motherboard that holds the processor in place and provides electrical connections between the CPU and the motherboard.

WHAT IS A CPU SOCKET?

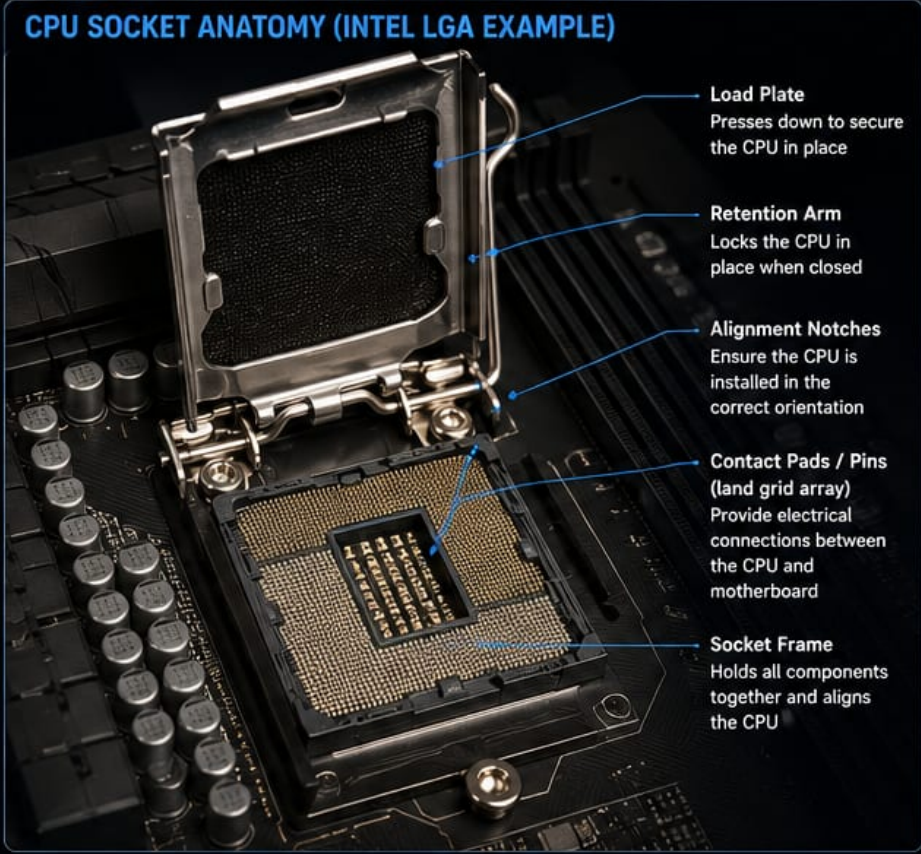


It is a physical and electrical connection point that allows the CPU to communicate with the motherboard. Different CPU generations use different socket types.

KEY FUNCTIONS

-  Holds the CPU securely in place
-  Provides electrical contact for data, power, and control signals
-  Ensures proper alignment and installation
-  Determines CPU compatibility with the motherboard

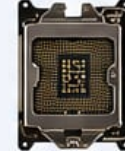
CPU SOCKET ANATOMY (INTEL LGA EXAMPLE)



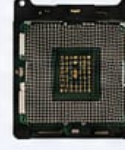
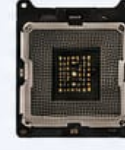
- Load Plate**
Presses down to secure the CPU in place
- Retention Arm**
Locks the CPU in place when closed
- Alignment Notches**
Ensure the CPU is installed in the correct orientation
- Contact Pads / Pins (land grid array)**
Provide electrical connections between the CPU and motherboard
- Socket Frame**
Holds all components together and aligns the CPU

COMMON CPU SOCKET TYPES

INTEL SOCKETS (LGA – Land Grid Array)

 LGA 1700 12th, 13th, 14th Gen Core Processors	 LGA 1200 10th, 11th Gen Core Processors	 LGA 1151 6th, 7th, 8th, 9th Gen Core Processors	 LGA 2066 Intel Core X-Series (High End)
--	--	--	--

AMD SOCKETS (PGA – Pin Grid Array / LGA)

 AM5 AMD Ryzen 7000 Series	 AM4 AMD Ryzen 1000 to 5000 Series	 AM3+ AMD FX / Ryzen (Older)	 TR4 / sTRX4 AMD Threadripper Series
--	--	--	--

CPU SOCKET COMPATIBILITY

The CPU socket determines which processors are compatible with your motherboard. Always check the motherboard manual or vendor website for supported CPUs.

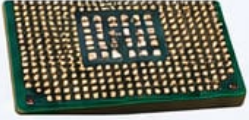
 SOCKET TYPE Must match the CPU	 CHIPSET Determines features and compatibility	 BIOS VERSION May need update to support new CPUs
--	---	--

CPU SOCKET INSTALLATION (LGA EXAMPLE)

- 1** Open the retention arm and lift the load plate.
- 2** Align the CPU notches with the socket.
- 3** Place the CPU gently into the socket.
- 4** Close the load plate and lock the retention arm.



LGA vs PGA SOCKETS

LGA (Land Grid Array)  ✓ Pins are on the motherboard ✓ More durable (CPU has flat pads) ✓ Better for high pin counts ✓ Typical for Intel (modern)	PGA (Pin Grid Array)  ✗ Pins are on the CPU ✗ More fragile (pins can bend) ✗ Older design ✗ Typical for AMD (older)
--	--

TIPS & BEST PRACTICES

-  Always power off and unplug before installing a CPU.
-  Handle the CPU by the edges. Avoid touching the contacts.
-  Ensure proper alignment using the notches or keys.
-  Do not force the CPU into the socket.
-  Apply thermal paste and install a cooler after installation.



DID YOU KNOW?

The "LGA" in Intel sockets stands for Land Grid Array, whereas "PGA" in older AMD sockets stands for Pin Grid Array.

GLOSSARY

LGA: Land Grid Array – Flat contacts on the CPU that land on pads in the socket.
PGA: Pin Grid Array – Pins on the CPU that go into holes in the socket.

VRM: Voltage Regulator Module – Delivers power to the CPU.
Chipset: Manages communication between CPU and other components.



COMPATIBILITY CHECK

Use tools like your motherboard manufacturer's website or PCPartPicker to verify CPU compatibility.

VRM (VOLTAGE REGULATOR MODULE)

The Power Behind Your CPU

The VRM converts 12V power from the PSU into the precise low voltage required by the CPU, ensuring stable power delivery for maximum performance and system stability.

WHAT IS VRM?



VRM

VRM (Voltage Regulator Module) is a circuit on the motherboard responsible for powering the CPU.

- Converts and regulates voltage
- Supplies clean, stable power
- Handles high current loads
- Critical for performance and overclocking

POWER FLOW



WHY VRM MATTERS

- STABILITY**
Delivers consistent power under heavy loads.
- PERFORMANCE**
Better VRM = higher performance potential.
- EFFICIENCY**
High-quality VRMs run cooler and waste less power.
- LONGEVITY**
Protects CPU and extends the lifespan of components.

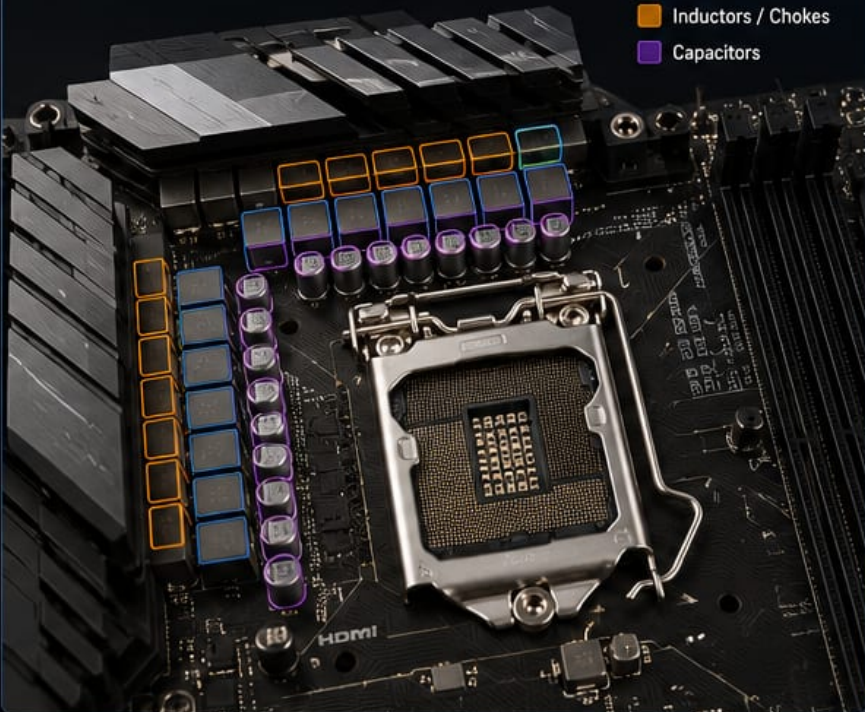


DID YOU KNOW?

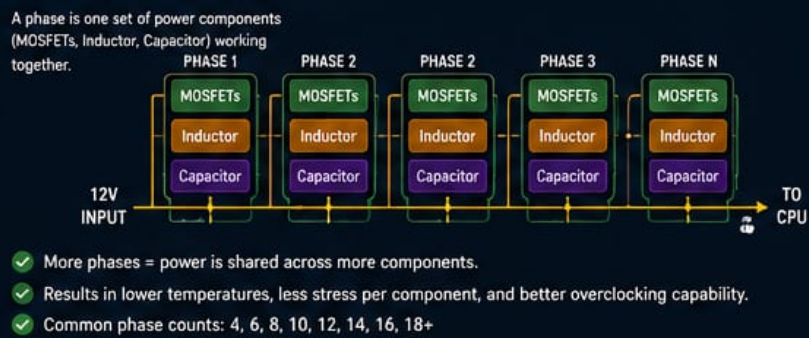
The VRM is one of the most important factors in system stability, especially for high-end CPUs and overclocking.

WHERE IS THE VRM?

VRMs are usually located around the CPU socket.



VRM PHASES EXPLAINED



HOW VRM WORKS

- 12V power from the PSU enters the motherboard.
- The VRM converts 12V to lower voltages.
- The voltage is regulated and delivered to the CPU.



- MOSFET (High/Low Side)** Acts as a switch, controlling power on and off at high speed.
- Inductor (Choke)** Stores energy and smoothens the power delivery.
- Capacitors** Filter and stabilize the voltage, reducing ripple.
- Controller IC** Monitors and controls the VRM operation.

VRM COMPONENTS

MOSFETs (Power Stages)	DrMOS / Power Stage (Integrated)	Inductors / Chokes	Capacitors	VRM Controller IC
				
High-speed switches that control power delivery to the CPU.	Combines MOSFETs and driver IC in one package for higher efficiency.	Store energy and smooth out current fluctuations.	Filter voltage ripple and provide stable power output.	The brain of the VRM. Monitors voltage, current and temperature and controls phases.

GOOD VRM vs POOR VRM

GOOD VRM DESIGN	POOR VRM DESIGN
	
- More phases - High-quality components - Better cooling (heatsinks) - Stable under heavy load - Better for overclocking	- Fewer phases - Lower quality components - Little or no heatsink - Runs hotter - Limited performance potential

EXAMPLE: HIGH-END 12+2 PHASE VRM



KEY SPECIFICATIONS TO CHECK

- Phase count (more is generally better)
- Component quality (DrMOS, capacitors, inductors)
- Heatsink design and VRM cooling
- Power stage current rating (e.g., 50A, 60A, 90A)

WHO NEEDS A STRONG VRM?



Gamers



Overclockers



Content Creators



Workstations

BIOS/UEFI CHIP

THE FIRMWARE THAT STARTS YOUR PC

The BIOS/UEFI chip is a small memory chip on the motherboard that stores the firmware (software) used to start and initialize your computer before the operating system loads.

KEY FUNCTIONS

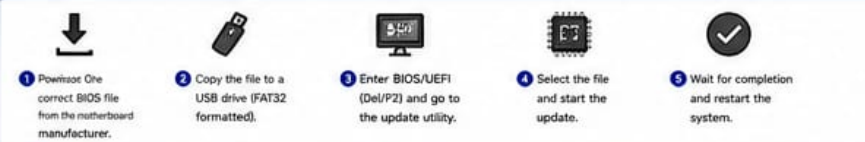
- Starts the System**
Executes Power-On Self Test (POST) and initializes hardware.
- Configures Hardware**
Sets up and configures all connected hardware components.
- Boots the OS**
Loads the bootloader from the selected device to start the operating system.
- Provides Setup Interface**
Stores settings like boot order, date/time, fan control, and security options.
- Firmware Updates**
Can be updated to add features, improve compatibility, and fix issues.

BIOS is traditional firmware using a legacy interface. UEFI (Unified Extensible Firmware Interface) is the modern replacement with more features, faster boot, and support for large drives and secure boot.

HOW IT WORKS (BOOT PROCESS)



HOW TO UPDATE BIOS/UEFI



Do not turn off the power during a BIOS/UEFI update! It can damage the chip.

DID YOU KNOW?

The BIOS/UEFI chip can be reprogrammed or replaced if the firmware becomes corrupted.

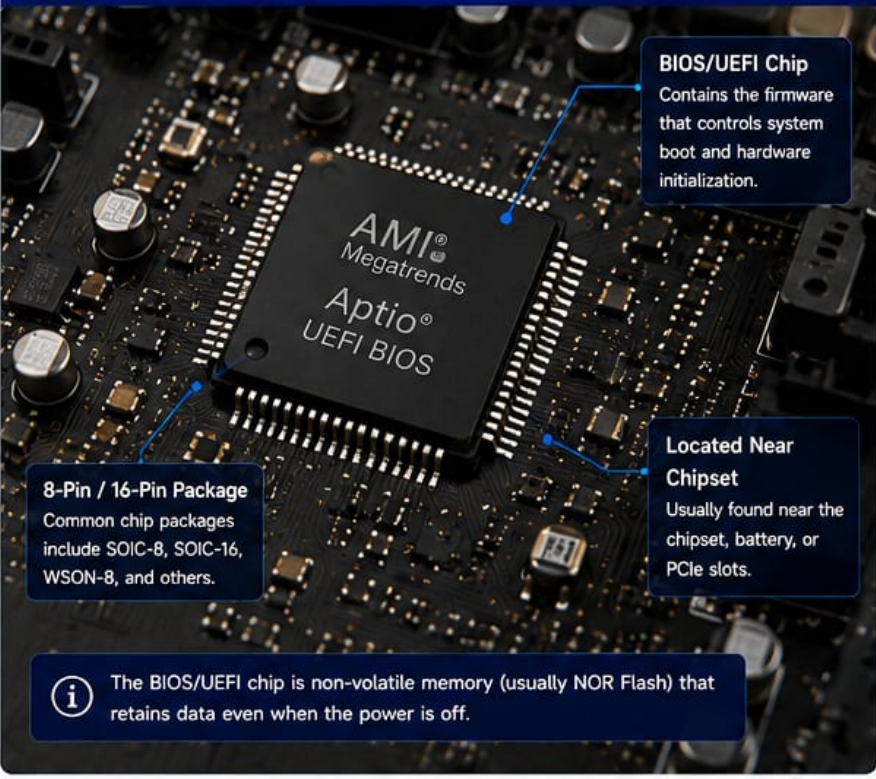
CHIP REMOVAL (IF NEEDED)

You can remove the chip with a chip puller or hot air gun and program it using an external SPI programmer.

BIOS RECOVERY OPTIONS

- Dual BIOS / Backup BIOS
- BIOS Flashback Button
- External BIOS Programmer

BIOS/UEFI CHIP ON A MOTHERBOARD



COMMON BIOS/UEFI CHIP TYPES

Winbond (W25Q128JV) 128 Mbit	Macronix (MX25U128) 128 Mbit	GigaDevice (GD25Q128) 128 Mbit	ISSI (IS25L128) 128 Mbit

BIOS VS UEFI

Feature	BIOS (Legacy)	UEFI (Modern)
Interface	16-bit	32/64-bit
Boot Speed	Slower	Faster
Drive Support	Up to 2.2 TB (MBR)	Over 2.2 TB (GPT)
Security	Basic	Secure Boot, TPM
User Interface	Text-based	Graphical (GUI)
Mouse Support	No	Yes
Network Boot	Limited	Yes

WHERE IS IT LOCATED?

The BIOS/UEFI chip is typically located:

- Near the chipset
- Near the CMOS battery
- Near PCIe slots

It is usually a small 8-pin or 16-pin chip labeled with the manufacturer name.

TIPS & BEST PRACTICES

- Always use the correct BIOS file for your exact motherboard model.
- Ensure stable power during BIOS updates.
- Keep a backup of your current BIOS settings.
- Update BIOS only when necessary (for new CPU support, bug fixes, etc.).
- Some motherboards have Dual BIOS for recovery.

COMMON BIOS/UEFI CHIP SOCKETS

SOIC-8 (8-Pin)

Common for smaller motherboards.

SOIC-16 (16-Pin)

Most common for desktop motherboards.

WSON-8 / DFN

Used on some modern motherboards.

SOME POPULAR MANUFACTURERS

AMI
American Megatrends

Insyde

Phoenix

Coreboot

TianoCore

IMPORTANT!
Flashing the wrong BIOS or interrupting the update can make your motherboard unusable (bricked). Always follow the manufacturer's instructions carefully.

RELATED COMPONENTS

CMOS Battery

Chipset

SPI Flash Chip

Clear CMOS Jumper

CMOS BATTERY

THE POWER SOURCE FOR BIOS SETTINGS

The CMOS battery is a small coin cell battery on the motherboard that powers the CMOS memory. It keeps your BIOS settings, date, and time saved when the computer is, turned off or unplugged.

 **CMOS** stands for **Complementary Metal-Oxide-Semiconductor**

KEY FUNCTIONS

-  **Maintains BIOS Settings**
Keeps BIOS configuration, boot order, and preferences saved.
-  **Keeps Date & Time**
Powers the real-time clock (RTC) when the system is off.
-  **Stores Hardware Information**
Preserves system hardware settings and configurations.
-  **Enables System Startup**
Ensures the system can start correctly with saved settings.

CMOS BATTERY ON A MOTHERBOARD



CMOS Battery
The small coin cell battery that powers the CMOS memory.

Battery Holder
Holds the battery securely in place. Gently push the clip to release the battery.

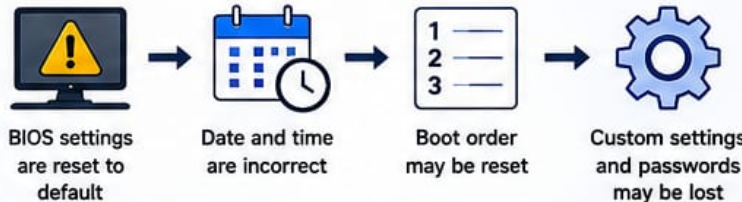
Without power from this battery, BIOS settings will be lost when the computer is turned off or unplugged.

COMMON CMOS BATTERY TYPE



- Type: CR2032 (most common)
- Voltage: 3 Volts
- Chemistry: Lithium
- Diameter: 20 mm
- Thickness: 3.2 mm

WHAT HAPPENS WHEN THE BATTERY IS DEAD?



BIOS settings are reset to default → Date and time are incorrect → Boot order may be reset → Custom settings and passwords may be lost

HOW TO IDENTIFY A LOW OR DEAD CMOS BATTERY

- BIOS settings keep resetting after saving
- Incorrect date and time on every boot
- "CMOS Checksum Error" or "CMOS Battery Low" message
- System asks to press F1/F2 to continue

 **CMOS Checksum Error Defaults Loaded**
Press F1 to Continue

HOW TO REMOVE & REPLACE

- Power off and unplug the computer.
- Open the case and locate the CMOS battery.
- Gently push the retention clip outward.
- Remove the old battery from the holder.
- Insert the new battery (positive side "+" facing up).
- Press it down until it clicks into place.

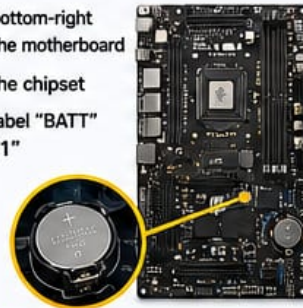


 Always handle the battery by the edges. Avoid touching the metal surfaces.

WHERE IS IT LOCATED?

The CMOS battery is typically found:

- Near the bottom-right corner of the motherboard
- Close to the chipset
- Near the label "BATT" or "JBAT1"



TIPS & BEST PRACTICES

- Replace the battery every 2-5 years.
- Use a high-quality CR2032 battery from a trusted brand.
- Ensure the "+" side is facing up when installing.
- After replacing, you may need to reconfigure BIOS settings.
- Dispose of old batteries properly. Do not recharge or burn.

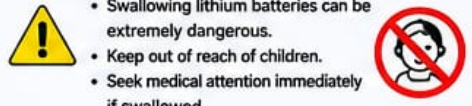


DID YOU KNOW?

A CMOS battery can last 2 to 5 years depending on usage and environment.

LITHIUM BATTERY SAFETY

- Swallowing lithium batteries can be extremely dangerous.
- Keep out of reach of children.
- Seek medical attention immediately if swallowed.

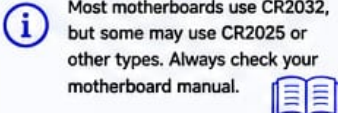


SYMPTOMS vs SOLUTION

Symptoms	Solution
BIOS keeps resetting	Replace the CMOS battery
Wrong date and time	Re-enter BIOS settings
CMOS error message	Save and exit

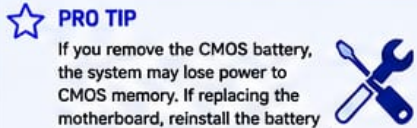
COMPATIBILITY NOTE

Most motherboards use CR2032, but some may use CR2025 or other types. Always check your motherboard manual.



PRO TIP

If you remove the CMOS battery, the system may lose power to CMOS memory. If replacing the motherboard, reinstall the battery before powering on if preconfigured.



02

CPU (Central Processing Unit)

The Brain of the Computer

The CPU is the most important component in your computer. It thinks, calculates, and gives instructions to make everything work.



Simple Idea

The CPU is like the brain of your computer. It processes information, makes decisions, and controls all the other parts.



What is it?

The CPU (Central Processing Unit) is a tiny chip that performs all the calculations and runs the instructions (programs) on your computer.

- ✓ Executes instructions
- ✓ Performs calculations
- ✓ Controls all operations
- ✓ The "brain" of the system



How it works

The CPU fetches instructions, decodes them, processes the data, and returns the results.



1. Fetch

Gets the next instruction from memory.



2. Decode

Understands what the instruction means.



3. Execute

Performs the calculation or task.



4. Write Back

Saves the result back to memory or sends it to another part.



Where it is located



The CPU sits in the CPU socket on the motherboard under a cooler (fan or liquid cooler).



Inside the CPU

Control Unit

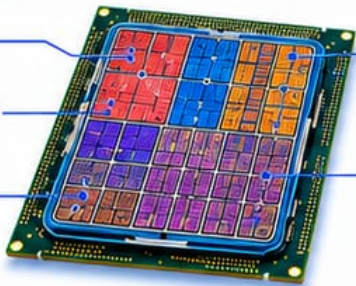
Tells the CPU what to do.

ALU (Arithmetic Logic Unit)

Performs math and logic operations.

Registers

Tiny ultra-fast storage for data the CPU is using right now.



Cache (L1, L2, L3)

Very fast memory that stores data the CPU uses often.

Cores

Each core can work on tasks. More cores = better multitasking.



Key Specifications



Cores

Number of independent processing units. More cores = better multitasking.



Threads

Each core can handle multiple threads. More threads = smoother performance.



Clock Speed (GHz)

How fast a core can work. Higher GHz = faster (but not everything!).



Cache

Built-in memory that makes the CPU faster. More cache = less waiting.



TDP (Thermal Design Power)

Amount of heat (in watts) the CPU produces. Lower TDP = cooler and more efficient.



Popular Modern Examples



Intel Core i9-14900K
High performance for gaming, creation & work.



AMD Ryzen 9 7950X
Excellent for multitasking & productivity.



Apple M3 Pro
Powerful & efficient for Mac computers.



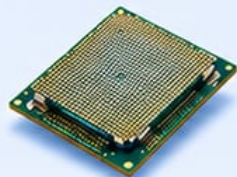
Real-World Uses

- ✓ Running apps (Word, Chrome, Photoshop)
- ✓ Playing games
- ✓ Editing videos and photos
- ✓ Programming and compiling code
- ✓ 3D modeling and animation
- ✓ Artificial Intelligence & Machine Learning



Did You Know?

Modern CPUs have billions of transistors!
A tiny CPU can do billions of calculations every second.



Knowledge Check

What is the main job of the CPU?

- A. Store data
- B. Display images
- C. Process data and run instructions
- D. Provide power to the computer



Analogy

CPU = Brain
Motherboard = Body
RAM = Desk (workspace)
Storage = Bookshelf
GPU = Artist
PSU = Power from food



KEY TAKEAWAY:

The CPU is the brain of your computer. It processes instructions, performs calculations, and controls everything.
A better CPU means a faster and smarter computer!



CPU CACHE (L1, L2, L3)

The Memory Hierarchy Inside the CPU

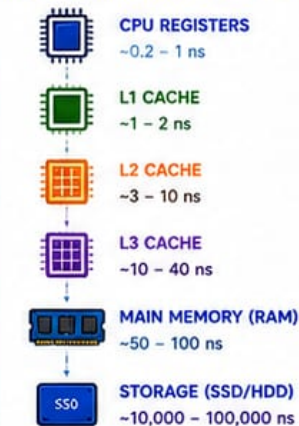
CPU cache is a small, ultra-fast memory located on or very close to the CPU cores. It stores copies of data and instructions that the CPU uses most often, so they can be accessed much faster than from RAM.

WHY DO WE NEED CACHE?

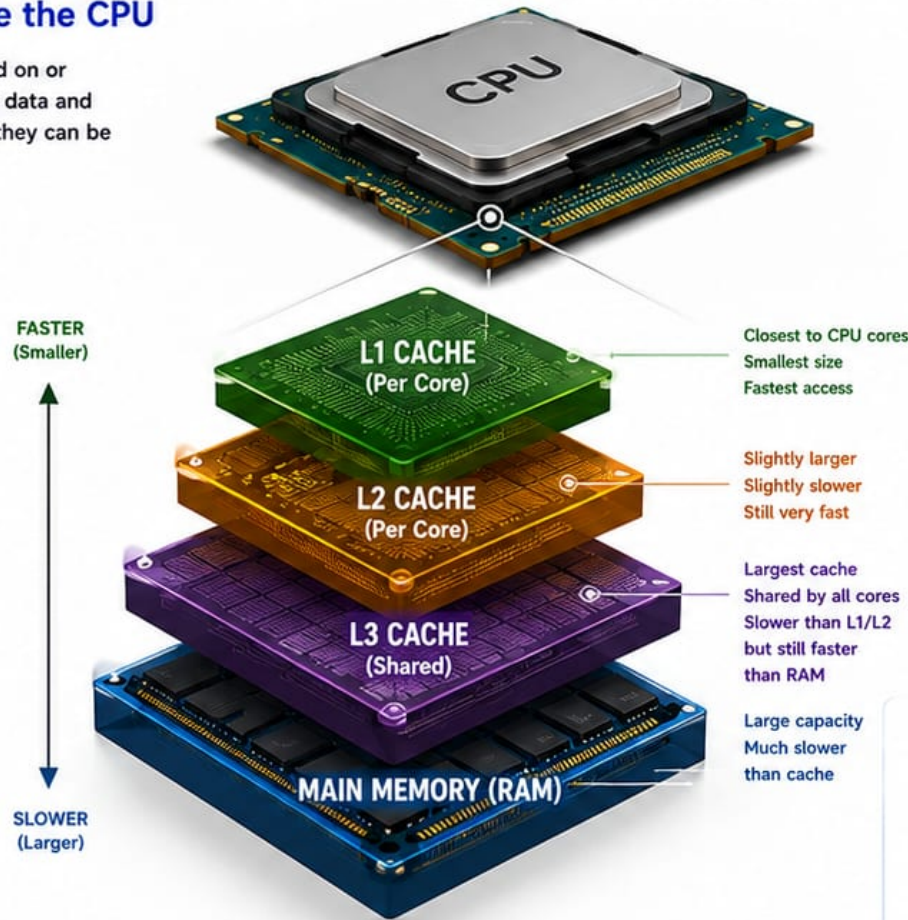
The CPU is extremely fast, but main memory (RAM) is much slower.

Cache acts as a high-speed buffer between the CPU and RAM, reducing the average time to access data.

MEMORY HIERARCHY (FROM FASTEST TO SLOWEST)



WHERE CACHE SITS IN THE CPU



CACHE LEVELS EXPLAINED



L1 CACHE (LEVEL 1)

- Smallest and fastest cache
- Usually split into:
 - L1 Instruction Cache (I-Cache)
 - L1 Data Cache (D-Cache)
- Private to each CPU core
- Size: 16 KB – 128 KB per core (typical)
- Latency: ~1 – 2 nanoseconds

Holds:

- Frequently used instructions
- Frequently used data



L2 CACHE (LEVEL 2)

- Larger than L1, slightly slower
- Private to each CPU core
- Size: 256 KB – 2 MB per core (typical)
- Latency: ~3 – 10 nanoseconds

Holds:

- Less frequently used instructions and data from L1 and RAM



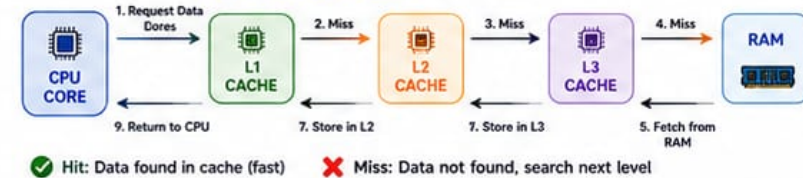
L3 CACHE (LEVEL 3)

- Largest cache, shared by all cores
- Acts as a last-level cache before going to RAM
- Size: 4 MB – 64 MB+ (shared)
- Latency: ~10 – 40 nanoseconds

Holds:

- Data and instructions shared across all cores

HOW CACHE WORKS (Simplified)



KEY BENEFITS OF CACHE



NOTE:
Cache sizes and latencies vary between CPU architectures and generations. Larger cache generally improves performance, but is more expensive and uses more chip area and power.

TYPICAL CACHE SIZES (EXAMPLES)

CPU EXAMPLE	L1 CACHE (Per Core)		L2 CACHE (Per Core)	L3 CACHE (Shared)	CORES
	I-Cache	D-Cache			
Intel Core i5 (12 th Gen)	32 KB	48 KB	1.25 MB	18 MB	6
AMD Ryzen 5 5600X	32 KB	32 KB	512 KB	32 MB	6
Apple M2 (Performance Core)	128 KB	128 KB	4 MB	16 MB	8
Intel Core i9 (14th Gen)	32 KB	48 KB	2 MB	36 MB	24

DIFFERENT CPUs AND GPUs

THE TWO BRAINS OF MODERN COMPUTERS

CPUs handle general tasks and run your operating system and applications.
GPUs handle graphics, gaming, AI, and massively parallel workloads.



SIMPLE IDEA

The CPU is like the brain of your computer. It processes information, makes decisions, and controls all the other parts.



CENTRAL PROCESSING UNITS (CPUs)

DESKTOP CPUs For desktop computers and workstations



Intel Core i9-14900K

High performance for enthusiasts and gamers

- 24 Cores (8 P + 16 E)
- 32 Threads
- Up to 6.0 GHz
- LGA 1700 Socket



Intel Core i5-14400

Great balance of performance and price

- 10 Cores (6 P + 4 E)
- 16 Threads
- Up to 4.7 GHz
- LGA 1700 Socket



AMD Ryzen 9 7950X

Extreme performance for creators and gamers

- 16 Cores
- 32 Threads
- Up to 5.7 GHz
- AM5 Socket



AMD Ryzen 5 7600

Excellent for gaming and everyday use

- 6 Cores
- 12 Threads
- Up to 5.1 GHz
- AM5 Socket



Intel Core Ultra 7 155H

Power efficient hybrid performance

- 16 Cores (6 P + 8 E + 2 LPE)
- 22 Threads
- Up to 4.8 GHz
- Intel 4 Process

MOBILE / LAPTOP CPUs For laptops, thin & light devices



Apple M3

Powerful and efficient for MacBooks

- 8 CPU Cores
- 10 GPU Cores
- 16-Core Neural Engine
- Unified Memory



AMD Ryzen 7 7840U

Great performance with amazing battery life

- 8 Cores
- 16 Threads
- Up to 5.1 GHz
- Radeon 780M Graphics



Intel Core i7-1360P

High performance mobile processor

- 12 Cores (4 P + 8 E)
- 16 Threads
- Up to 5.0 GHz
- Intel Iris Xe Graphics



Qualcomm Snapdragon X Elite

Arm-based chip for Windows laptops

- 12 Cores
- Up to 4.3 GHz
- Adreno GPU
- Very low power



Intel Core Ultra 5 125U

Efficient performance for everyday use

- 12 Cores (2 P + 8 E + 2 LPE)
- 14 Threads
- Up to 4.3 GHz
- Intel 4 Process

GRAPHICS PROCESSING UNITS (GPUs)

CONSUMER / GAMING GPUs For gaming, content creation, and everyday users



NVIDIA GeForce RTX 4090

The ultimate gaming GPU

- 24GB GDDR6X
- 16384 CUDA Cores
- Ray Tracing
- RTRacing
- DLSS 3



NVIDIA GeForce RTX 4070 Ti SUPER

High performance for 1440p / 4K gaming

- 16GB GDDR6X
- 8448 CUDA Cores
- Ray Tracing
- DLSS 3



NVIDIA GeForce RTX 4060

Great for 1080p gaming and content creation

- 8GB GDDR6
- 3072 CUDA Cores
- Ray Tracing
- DLSS 3



AMD Radeon RX 7900 XT

Powerful 4K gaming and streaming

- 20GB GDDR6
- 5376 Stream Processors
- Ray Tracing
- FSR 3



AMD Radeon RX 7600

Great value for 1080p gaming

- 8GB GDDR6
- 2048 Stream Processors
- Ray Tracing
- FSR 3

PROFESSIONAL / WORKSTATION GPUs For professionals, engineers, creators, and AI development



NVIDIA RTX 6000 Ada

High-end professional graphics and compute

- 48GB GDDR6
- 18176 CUDA Cores
- ECC Memory
- Optimized Drivers



NVIDIA RTX A4000

Great for 3D design, rendering, and AI

- 16GB GDDR6
- 6144 CUDA Cores
- Ray Tracing
- ECC Memory



NVIDIA RTX A2000

Compact and efficient workstation GPU

- 12GB GDDR6
- 3328 CUDA Cores
- Ray Tracing
- ECC Memory



NVIDIA L40S

Powerful for AI training and large workloads

- 48GB GDDR6
- 18176 CUDA Cores
- Tensor Cores
- Data Center Ready



AMD Radeon PRO W7900

Professional performance for creators

- 48GB GDDR6
- 6144 Stream Processors
- ECC Memory
- Professional Drivers



WHAT IS A CPU?

The CPU (Central Processing Unit) is the main processor. It executes instructions, runs your operating system, apps, and handles general tasks.

COMMON CPU USES

- Operating System
- Web Browsing
- Productivity Apps
- Multitasking
- Programming
- Calculations

CPU vs GPU		
Feature	CPU	GPU
Designed For	Versatility	Parallel Processing
Cores	Few (4 - 24)	Many (Hundreds - Thousands)
Best At	General tasks	Graphics, AI, Compute
Used For	OS, Apps, Games Logic	Gaming, 3D, AI, Video Editing
Power Use	Lower	Higher
Examples	Intel Core, AMD Ryzen, Apple M	NVIDIA GeForce, RTX, AMD Radeon



WHAT IS A GPU?

The GPU (Graphics Processing Unit) specializes in processing thousands of tasks at once. It renders images, videos, and accelerates AI and scientific workloads.

COMMON GPU USES

- Gaming
- 3D Rendering
- Video Editing
- Live Streaming
- AI / Machine Learning
- Scientific Simulations



DID YOU KNOW?

GPUs have thousands of smaller cores designed to work on many tasks at the same time, which makes them perfect for graphics, AI, and high-performance computing.

GRAPHICS & AI HARDWARE

The Key Processors Powering Modern Graphics, Computing & Artificial Intelligence

1. DEDICATED GPU

Graphics Processing Unit



A powerful, standalone chip designed primarily for high-performance graphics rendering and parallel computation.

BEST FOR

- Gaming
- 3D Rendering
- Video Editing
- AI/ML Training (High Performance)

KEY FEATURES

- Thousands of high-performance cores (CUDA cores)
- High memory bandwidth (GDDR6/6X)
- Designed for massive parallel workloads
- Typically 75W – 600W+ TDP

EXAMPLE PRODUCTS



WHERE USED

Gaming PCs, Workstations, Content Creation, AI Training Servers

2. INTEGRATED GPU

Integrated Graphics



Built into the CPU or SoC and shares system memory. Optimized for efficiency and everyday computing.

BEST FOR

- Everyday Computing
- Web Browsing & Office Apps
- 4K Video Playback
- Light Gaming

KEY FEATURES

- Shares memory with system (RAM)
- Lower power consumption
- Smaller die size & cost-effective
- Good for general-purpose tasks

EXAMPLE PRODUCTS



WHERE USED

Laptops, Ultrabooks, Desktops, Mobile Devices

3. AI ACCELERATOR (NPU)

Neural Processing Unit



Specialized processor designed specifically for AI/ML workloads with high efficiency at low power.

BEST FOR

- On-device AI (Inference)
- Voice & Image Recognition
- Real-time Translation
- Low-power AI Apps

KEY FEATURES

- Optimized for AI inference
- Extremely power efficient
- Built into SoCs (CPU + NPU)
- Handles neural networks efficiently

EXAMPLE PRODUCTS



WHERE USED

Smartphones, Laptops, Tablets, Edge Devices, IoT

4. AI ACCELERATOR CARD (ENTERPRISE)

High-Performance AI Card



Add-in cards designed for data centers and enterprises to accelerate AI training and inference at scale.

BEST FOR

- AI Model Training (Large Scale)
- High-Performance Inference
- Data Analytics
- HPC & Scientific Computing

KEY FEATURES

- Massive compute power & memory (HBM2e/HBM3)
- High bandwidth (NVLink / PCIe)
- Optimized for datacenter workloads
- Scalable across multiple cards

EXAMPLE PRODUCTS



WHERE USED

Data Centers, AI Labs, Cloud Platforms, Enterprise AI Systems

5. TPU (TENSOR PROCESSING UNIT)

Google's AI Accelerator



Custom-built by Google for machine learning workloads, optimized for TensorFlow and large-scale AI training.

BEST FOR

- AI/ML Training (Google Cloud)
- Deep Learning Workloads
- Large-Scale Inference
- Data Center AI at Scale

KEY FEATURES

- Designed specifically for tensor operations
- Very high throughput
- Optimized for TensorFlow
- High energy efficiency

EXAMPLE PRODUCTS



WHERE USED

Google Cloud, AI Research, Machine Learning Infrastructure

QUICK COMPARISON

	Graphics & Parallel Compute	General Purpose Efficiency	AI Inference (Edge, Low Power)	AI Training & Inference (Data Center)	AI Training (Google Cloud)
Power Usage	High	Low	Very Low	High	High (Optimized)
Typical Location	Discrete Card (PCIe)	Built into CPU/SoC	Built into SoC	Add-in Card (PCIe)	Custom Board (Data Center)

Different hardware for different needs — from gaming to AI at scale!

Note: Specifications and examples are representative and may change over time.

MODERN AI PC HARDWARE

Powering the next generation of intelligent, faster, and more efficient experiences.



Smarter Performance
AI tasks run faster



Better Efficiency
More power, less drain



Enhanced Privacy
AI on-device, your data stays with you



Next-Gen Productivity
Create, collaborate, and get more done



Future Ready
Built for tomorrow's AI innovations

1 NEURAL PROCESSING UNIT (NPU)



WHAT IT IS

A dedicated processor built specifically for AI workloads.

WHY IT MATTERS

- Runs AI tasks locally and efficiently
- Improves battery life
- Powers real-time features (e.g., Recall, Live Captions, Studio Effects)



AI Acceleration

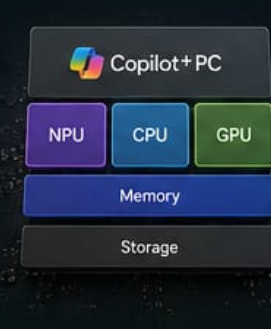


On-Device AI



Better Battery Life

2 COPILOT+ PC ARCHITECTURE



WHAT IT IS

A new class of Windows PCs designed for AI, built around the NPU, powerful CPU, GPU, and fast memory.

WHY IT MATTERS

- Optimized for AI experiences
- Balanced performance and efficiency
- Unlocks exclusive Copilot+ PC features



Optimized Stack



AI-Ready



Smarter by Design

3 DDR5 MEMORY



WHAT IT IS

The latest high-speed memory for desktops and laptops.

WHY IT MATTERS

- Higher bandwidth
- Better multitasking
- Supports demanding AI and creative apps



Faster Performance



Greater Capacity



Improved Efficiency

4 LPDDR5X MEMORY



WHAT IT IS

Low-power memory designed for thin, light devices.

WHY IT MATTERS

- Ultra-fast speeds
- Lower power usage
- Longer battery life in slimmer designs



Low Power



High Speed



Longer Battery Life

5 PCIe 5.0 SSD



WHAT IT IS

Next-gen storage with blazing fast read/write speeds.

WHY IT MATTERS

- Faster boot and load times
- Handles large AI models and datasets
- Future-proof bandwidth



Extreme Speed



High Capacity



Reliable Performance

6 AI GPU



WHAT IT IS

Graphics processors optimized for AI and parallel workloads.

WHY IT MATTERS

- Accelerates AI training and inference
- Great for content creation and 3D rendering
- Enables local AI apps



AI Acceleration



Content Creation



3D & Simulation

7 WI-FI 7 MODULE



WHAT IT IS

The latest Wi-Fi standard offering ultra-fast, low-latency wireless connectivity.

WHY IT MATTERS

- Multi-gigabit speeds
- Lower latency
- Better in crowded environments



Ultra-Fast



Low Latency



More Reliable

8 USB4 / THUNDERBOLT 5 CONTROLLER



WHAT IT IS

Next-gen connectivity controller for super-fast data, video, and power delivery.

WHY IT MATTERS

- Up to 80 Gbps bandwidth
- Connect high-res displays and docks
- Charge devices and transfer data simultaneously



80 Gbps



Display & Docking



Power Delivery

BUILT FOR AI. READY FOR YOU.



Modern AI PCs combine specialized hardware and smarter architecture to deliver powerful on-device AI experiences—securely, efficiently, and intelligently.

WHAT CAN YOU DO?



Real-time translation



AI image generation



Video enhancement



Code completion



Smart collaboration



Personalized experiences

THE AI PC ADVANTAGE

- ✓ Run AI locally, even without the cloud
- ✓ More responsive and seamless experiences
- ✓ Better privacy and data security
- ✓ Optimized power for all-day productivity



Smarter Hardware. Smarter Experiences.



The foundation of the AI-powered future.

Choose an AI PC built for what's next.

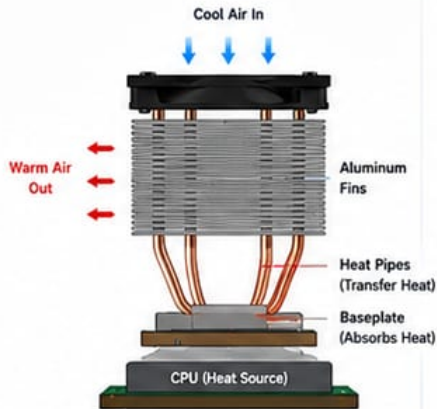
CPU COOLER (AIR COOLER)

Keeping Your CPU Cool for Peak Performance

A CPU air cooler dissipates heat from the processor using a heatsink and fan. It draws heat away from the CPU and transfers it into the air, keeping temperatures low and performance high.

HOW IT WORKS

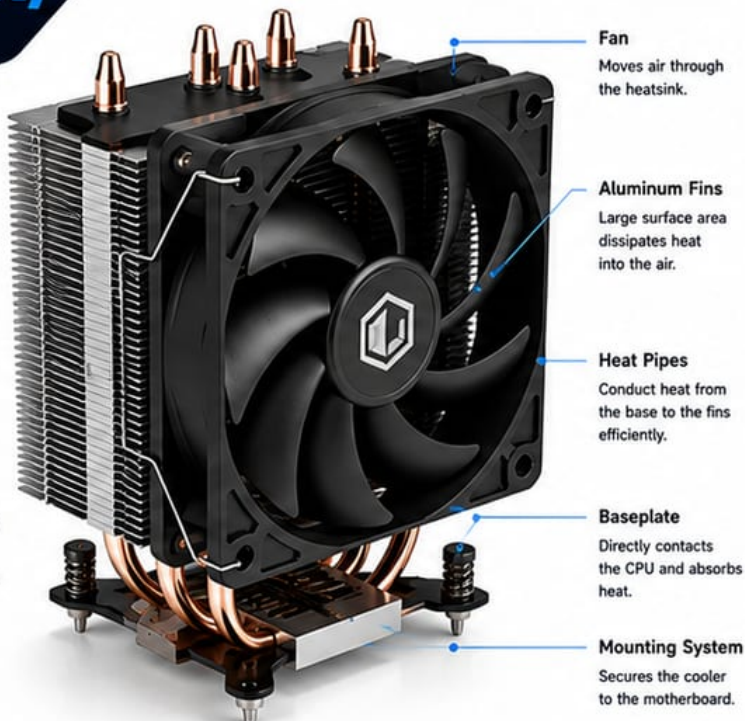
- 1 The CPU generates heat when in use.
- 2 The heat is transferred to the cooler's baseplate.
- 3 Heat travels up through the heat pipes into the aluminum fins.
- 4 The fan pushes cool air through the fins, carrying heat away.
- 5 Warm air is expelled into the case and exhausted by case fans.



AIRFLOW INSIDE THE CASE



AIR COOLER COMPONENTS



Fan
Moves air through the heatsink.

Aluminum Fins
Large surface area dissipates heat into the air.

Heat Pipes
Conduct heat from the base to the fins efficiently.

Baseplate
Directly contacts the CPU and absorbs heat.

Mounting System
Secures the cooler to the motherboard.

TYPES OF AIR COOLERS

TOWER COOLER

Most common type. Offers excellent cooling performance and fits most PC cases.



TOP-FLOW COOLER

Blows air downward onto the motherboard. Ideal for small cases or tight spaces.



KEY FEATURES



EFFICIENT COOLING
Heat pipes and fins work together to dissipate heat effectively.



QUIET OPERATION
Large fans can move more air at lower speeds, reducing noise.



EASY INSTALLATION
Simple mounting systems support all major CPU sockets.



RELIABLE & DURABLE
No moving parts other than the fan, making it long-lasting.



COST EFFECTIVE
Excellent performance at an affordable price compared to liquid coolers.



NO MAINTENANCE
Just install and forget. No refilling or special maintenance required.

COMPATIBILITY

SOCKET SUPPORT (Examples)

INTEL

LGA 1851 / 1700
LGA 1200 / 1151
LGA 1150 / 1155
LGA 1156

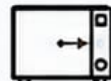
AMD

AM5 / AM4
AM3+ / AM3
AM2+ / AM2
FM2+ / FM2

CLEARANCE CHECK



Check RAM height for clearance with the cooler.



Ensure the cooler height fits your case.

TIPS FOR BEST PERFORMANCE

- ✓ Apply a high-quality thermal paste.
- ✓ Ensure good airflow in your case.
- ✓ Keep the cooler and case fans clean.
- ✓ Choose the right cooler size for your CPU (TDP).
- ✓ Monitor temperatures for optimal performance.



DID YOU KNOW?

A good air cooler can keep high-performance CPUs running cool and stable without the cost and complexity of liquid cooling.



EXAMPLE AIR COOLER SPECIFICATIONS (Typical Tower Cooler)



DIMENSIONS
132 x 76 x 154 mm
(L x W x H)



WEIGHT
650 g



FAN SIZE
120 mm



FAN SPEED
600 - 1500 RPM
(±10%)



AIRFLOW
59 CFM (Max)



NOISE LEVEL
25.6 dBA (Max)



TDP SUPPORT
Up to 180W

LIQUID CPU COOLER (AIO)

Advanced Cooling for Maximum Performance

An All-In-One (AIO) liquid cooler uses liquid to transfer heat away from the CPU, then dissipates it through a radiator. It provides superior cooling compared to air coolers, ideal for high-performance CPUs and overclocking.



AIO COMPONENTS

RADIATOR

Aluminum radiator contains thin fins that maximize surface area for heat dissipation.

FANS

Mounted on the radiator to push air through the fins and remove heat.

TUBES

Durable, low-permeation tubes carry coolant between the pump and radiator.

PUMP

Circulates coolant through the system. Usually located in the pump housing.

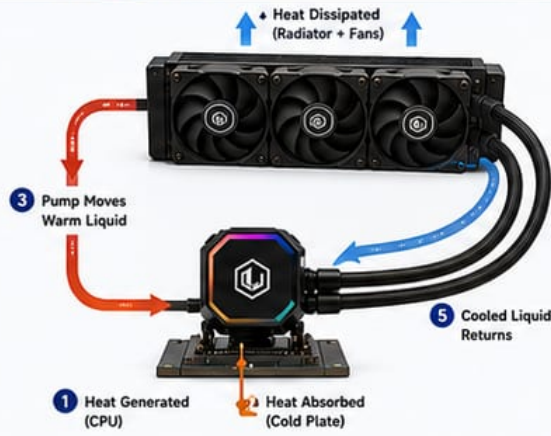
COLD PLATE

Copper plate makes direct contact with the CPU to absorb heat efficiently.



HOW IT WORKS

- 1 The CPU generates heat during operation.
- 2 Coolant in the cold plate absorbs the heat.
- 3 The pump moves the heated coolant through the tubes to the radiator.
- 4 The radiator's fins and fans dissipate the heat into the air.
- 5 The cooled liquid returns to the pump to repeat the cycle.



KEY BENEFITS



SUPERIOR COOLING

More efficient heat transfer than air coolers, lower temperatures.



QUIET OPERATION

Fans run at lower speeds for the same cooling performance.



BETTER FOR OVERCLOCKING

Keeps high-performance CPUs cool under heavy loads.



SLEEK & MODERN LOOK

Clean installation with no bulky heatsink on the CPU.



SPACE SAVING

Frees up space around the CPU socket for better airflow.

AIO vs AIR COOLER

AIO LIQUID COOLER

- ✓ Better cooling performance
- ✓ Handles high TDP CPUs
- ✓ Better for overclocking
- ✓ Quieter at high loads
- ✓ Clean and sleek look
- ✓ Higher cost

VS

AIR COOLER

- ✓ Good cooling performance
- ✗ May struggle at high TDP
- ✗ Limited overclocking headroom
- ✗ Can be noisier under load
- ✗ Bulky and taller
- ✓ More affordable

COMMON RADIATOR SIZES



120mm
(1 Fan)

- Good for low to mid-range CPUs
- Fits in most small cases
- Best for standard builds



240mm
(2 Fans)

- Great balance of performance & size
- Most popular size
- Ideal for gaming PCs



360mm
(3 Fans)

- Best cooling performance
- Ideal for high-end CPUs
- Requires larger case support

COLD PLATE CLOSE-UP



Copper Base

High thermal conductivity for efficient heat transfer.



Micro Fins

Increase surface area to absorb more heat from the CPU.



Pre-Applied Thermal Paste

Ensures optimal contact and easy installation.

INSTALLATION OVERVIEW

1 Prepare

Clean CPU surface and apply thermal paste (if not pre-applied).



2 Mount Brackets

Install the mounting brackets to the motherboard.



3 Mount Pump Block

Secure the pump block onto the CPU using the screws.



4 Mount Radiator

Install the radiator in the case and secure it.



5 Connect Cables

Connect PWM and RGB cables to the motherboard and power.



MAINTENANCE TIPS



Ensure radiator fans and case fans are clean.



Check for air bubbles or unusual pump noise.



Keep your case well-ventilated.



AIO coolers are maintenance-free, but replace if performance drops or leaks occur.



DID YOU KNOW?

A liquid cooler can reduce CPU temperatures by 10-20°C compared to high-end air coolers, depending on the CPU and workload.



SPECIFICATIONS (EXAMPLE 240mm AIO)



Radiator Material
Aluminum



Radiator Size
277 x 120 x 27 mm



Fan Size
120 x 120 x 25 mm



Fan Speed
600 - 1800 RPM
(±10%)



Airflow
21 - 78 CFM



Noise Level
18 - 30 dBA



Pump Speed
2000 - 3000 RPM
(±10%)



Socket Support
Intel: LGA 1851/1700/1200/115X
AMD: AM5 / AM4



Warranty
3 - 6 Years

COOLING & THERMAL COMPONENTS

These components work together to keep your system cool, stable and performing at its best.



Prevent Overheating
Protects components from heat damage.



Maintain Performance
Keeps temperatures low for consistent speed and efficiency.



Reduce Noise
Efficient cooling means quieter operation.

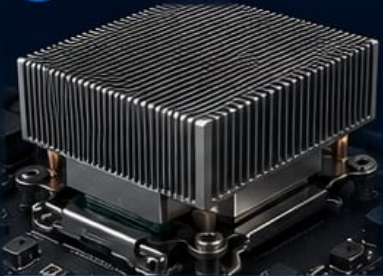


Extend Lifespan
Lower temps = longer life for your hardware.



Improve Efficiency
Better cooling uses less power to do more.

1 HEAT SINK



WHAT IT DOES

Absorbs heat from a component and dissipates it into the air using a large surface area.

WHY IT'S USED

Keeps CPUs, GPUs, and other chips cool by moving heat away from the source.



CPU | GPU | Chipset
VRM | SSD (M.2)



Passive Cooling
(Quiet & Reliable)

2 THERMAL PASTE



WHAT IT DOES

Fills microscopic gaps between the CPU (or chip) and the heatsink to improve heat transfer.

WHY IT'S USED

Improves thermal conductivity so heat moves efficiently from the chip to the cooler.



CPU | GPU



Better Contact =
Lower Temperatures

3 HEAT PIPES



WHAT IT DOES

Transfers heat quickly from the base to the fins using sealed copper tubes with fluid inside.

WHY IT'S USED

Moves heat faster and more evenly across the heatsink for better cooling.



CPU | GPU
Heatsinks



Faster Heat Transfer
High Efficiency

4 VAPOR CHAMBER



WHAT IT DOES

Uses vapor to spread heat evenly across a flat chamber, then condenses to release it.

WHY IT'S USED

Distributes heat more uniformly than heat pipes — ideal for high-performance chips.



High-End CPUs | GPUs
Laptops | Consoles



Even Heat Distribution
Lower Hotspots

5 RADIATOR



WHAT IT DOES

Dissipates heat from liquid coolant into the air through thin fins and a large surface.

WHY IT'S USED

Cools the liquid from a water cooling loop before it returns to the CPU/GPU.



Liquid Cooling Systems
(AIO | Custom Loops)



High Heat Dissipation
Stable Temperatures

6 COOLING FANS



WHAT IT DOES

Moves cool air in and hot air out of the case or across heatsinks and radiators.

WHY IT'S USED

Provides airflow to remove heat and maintain optimal temperatures inside the system.



Case | Heatsinks | Radiators
PSU | VRM



Airflow = Better Cooling
Lower Temps & Noise

WHY COOLING MATTERS



- ✓ Prevents thermal throttling and crashes
- ✓ Protects hardware from heat damage
- ✓ Ensures peak performance under load
- ✓ Extends the life of your components

HOW COOLING WORKS TOGETHER



HEAT IS
GENERATED
CPU / GPU / Chips



HEAT IS
TRANSFERRED
Via Paste, Heat Pipes
or Vapor Chamber



HEAT IS
DISSIPATED
By Radiator
into the Air



AIRFLOW REMOVES
THE HEAT
Fans keep air moving
and temps low

TIPS FOR BETTER COOLING



Keep your system clean and dust-free



Ensure good airflow in your case



Use quality thermal paste



Monitor temperatures regularly



Cool system. Stable performance. Built to last.



The right cooling keeps everything running at its best.

OUTPUT DEVICES

The devices that present information from your computer to your senses

Output devices take digital data from your computer and convert it into something you can see, hear or experience.



See



Hear



Feel/Experience

7 PLOTTER

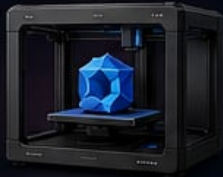


Creates large, precise prints for technical drawings, blueprints and posters.

BEST FOR

- Architecture
- Engineering
- CAD Drawings

8 3D PRINTER



Transforms digital 3D models into physical objects layer by layer.

BEST FOR

- Prototyping
- Product Design
- Education

9 LED DISPLAY / DIGITAL SIGNAGE



Displays bright, dynamic content suitable for public viewing in large spaces.

BEST FOR

- Advertising
- Events & Venues
- Information Displays

10 BRAILLE DISPLAY



Outputs text in Braille for users who are visually impaired.

BEST FOR

- Accessibility
- Education
- Inclusive Workspaces

11 HAPTIC FEEDBACK DEVICE



Provides touch feedback, vibrations or force to simulate the sense of touch.

BEST FOR

- Gaming
- Training & Simulation
- Virtual Reality

12 SMART GLASSES



Displays information in your field of view without blocking the real world.

BEST FOR

- Navigation
- Industrial Work
- Augmented Reality

13 HOLOGRAPHIC DISPLAY (EMERGING)



Projects 3D holograms in mid-air for a futuristic visual experience.

BEST FOR

- Advanced Presentations
- Museums & Exhibits
- Product Showcases

14 LASER SHOW PROJECTOR



Projects powerful light beams and patterns to create stunning visual shows.

BEST FOR

- Concerts
- Events & Festivals
- Theme Parks

15 DIGITAL FRAME



Displays digital photos and slideshows in a continuous loop.

BEST FOR

- Home Décor
- Gifts
- Photo Display

16 SOUND BAR



Delivers enhanced audio from your TV or computer in a compact design.

BEST FOR

- Home Entertainment
- Movies & TV
- Gaming

17 TACTILE DISPLAY



Uses pins to create touch sensations on the skin to display images or text.

BEST FOR

- Assistive Technology
- Education & Research
- Accessibility

18 SCENT OUTPUT DEVICE



Releases scents to enhance digital experiences (e.g., in gaming or VR).

BEST FOR

- Gaming
- Virtual Reality
- Therapy & Wellness

19 AMBIENT LIGHT SYSTEM



Produces light around your screen that reacts to on-screen content.

BEST FOR

- Gaming
- Movies & TV
- Immersive Experiences

20 THERMAL PRINTER



Prints using heat instead of ink — fast, quiet and reliable.

BEST FOR

- Receipts
- Tickets
- Point of Sale



WHY OUTPUT DEVICES MATTER

They turn data into meaningful experiences. The right output device improves understanding, engagement and productivity.

OUTPUT DEVICES APPEAL TO OUR SENSES



VISION

See images, videos and information.



HEARING

Hear sound, music and voice.



TOUCH

Feel feedback, vibrations and interactions.



IMMERSION

Experience virtual and real-world environments.

UNUSUAL BUT POWERFUL



ATMOSPHERIC EFFECTS

Devices like fog machines, wind simulators and motion platforms create realistic environments.



WATER DISPLAY / FOUNTAIN SCREEN

Projects images on water screens for eye-catching visual experiences.

★ From simple screens to immersive technologies, output devices bring digital content to life.



Choose the right output device for the right experience!

DIMM SLOTS

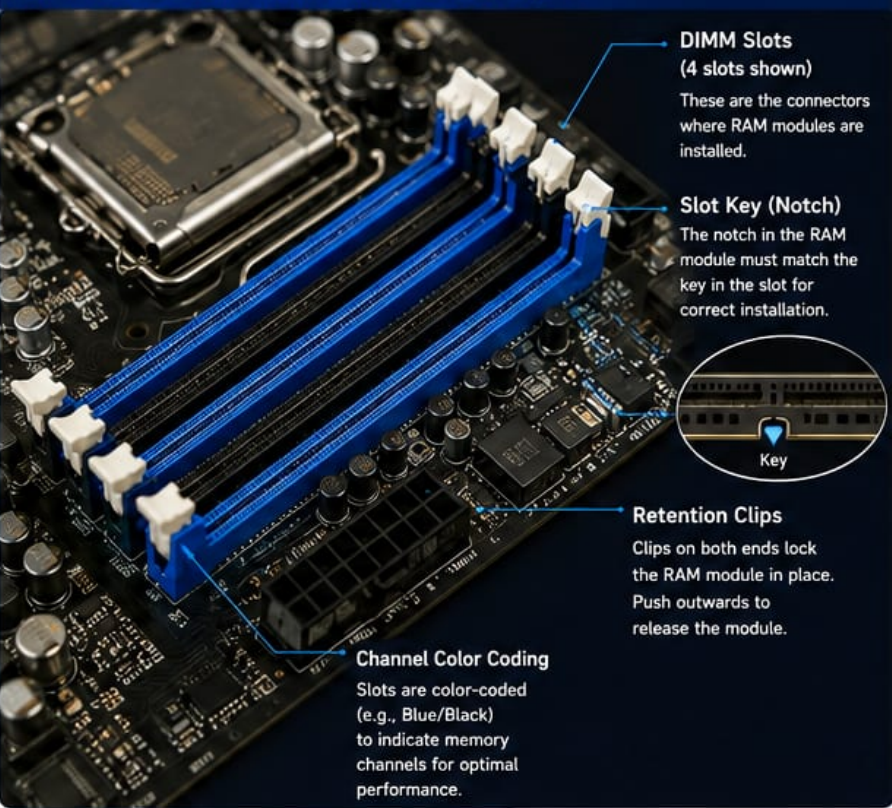
(DUAL IN-LINE MEMORY MODULE) THE MEMORY CONNECTORS ON YOUR MOTHERBOARD

DIMM slots are the long connectors on a motherboard that hold the RAM modules securely in place. They allow your system to access high-speed memory for faster multitasking, applications, and performance.

KEY FUNCTIONS

-  **Secure Connection**
Holds RAM modules firmly in place.
-  **High-Speed Data Transfer**
Enables fast communication between RAM and the CPU.
-  **Supports Multiple Configurations**
Single or dual channel for better performance.
-  **Determines Capacity & Speed**
Supports the maximum memory capacity and speed supported by the board.

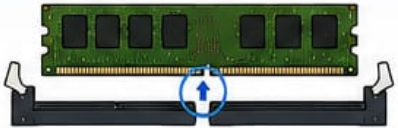
DIMM SLOTS ON A MOTHERBOARD (Typical)



DIMM KEY (NOTCH) – ENSURES CORRECT FIT

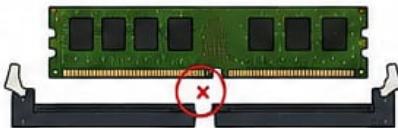
✓ Correct Fit

The notch on the RAM module aligns with the key in the slot.



✗ Incorrect Fit

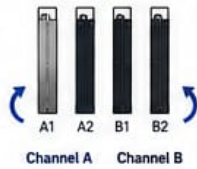
If the notch does not align, the RAM module will not fit.



MEMORY CHANNELS EXPLAINED

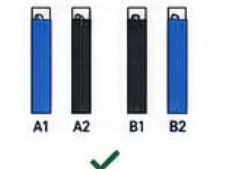
Single Channel

RAM modules installed in one channel only.



Dual Channel (Recommended)

Modules installed in matching slots of both channels.



Quad Channel (High-End)

Requires compatible CPU and motherboard.



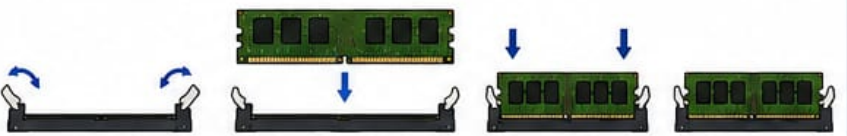
■ Channel A (Blue) ■ Channel B (Black)

TYPES OF DIMM (BY MEMORY GENERATION)

DDR3 DIMM 240-Pin	DDR4 DIMM 288-Pin	DDR5 DIMM 288-Pin
		
Voltage: 1.5V	Voltage: 1.2V	Voltage: 1.1V
Use: Older Systems	Use: Most Common	Use: Latest Systems

 DIMMs are not cross-compatible. Always use the correct type supported by your motherboard.

HOW TO INSTALL A DIMM

- 1 Open the retention clips on both ends of the slot.
 - 2 Align the notch on the RAM module with the key in the slot.
 - 3 Firmly press down on the module evenly until the clips snap into place.
 - 4 Ensure both clips are locked and the module is secure.
- 

 Always handle RAM by the edges and avoid touching the gold contacts.

TIPS & BEST PRACTICES

- ✓ Use the RAM type and speed recommended by your motherboard.
- ✓ Populate slots as per the motherboard manual for best performance.
- ✓ For dual channel, install matching modules in the correct slots (e.g., A2 and B2).
- ✓ Mixing different RAM sizes or speeds may run at the lowest common speed.
- ✓ Ensure DIMMs are fully seated and clips are locked.
- ✓ Keep DIMM slots and modules clean and free from dust.



DID YOU KNOW?

 Using dual channel memory can increase memory bandwidth by up to 100%, significantly improving system performance in many tasks and applications.

QUICK REFERENCE – HOW MANY RAM MODULES?

Slots Available	1 Slot	2 Slots	4 Slots
Best Practice	1 DIMM	2 DIMMs (Dual Channel)	2 or 4 DIMMs (Dual / Quad Channel)

COMMON DIMM SLOT ISSUES

-  System not posting: RAM may not be fully seated.
-  Lower performance: RAM not in the correct slots for the channel.
-  Beep codes or errors: Incompatible or faulty RAM.

★ PRO TIP

Always refer to your motherboard manual for the exact DIMM slot population guide and limitations.



STORAGE HARDWARE

Different types of storage devices for every need

Storage hardware is used to store, access, and protect data. From ultra-fast NVMe SSDs to high-capacity HDDs and NAS drives, each type is designed for specific use cases.

KEY INTERFACES

NVMe / PCIe
Ultra-fast

Ultra-fast
SATA III, SATA II

SATA

Widely compatible
SATA III, SATA II, SATA I

1 NVMe SSD (M.2 PCIe)



Uses PCIe interface via M.2 slot for the fastest performance.

- Blazing fast read/write speeds
- Low latency
- Ideal for demanding applications and gaming

Best for:



Interface

PCIe 3.0 / 4.0 / 5.0 (x4)



Form Factor

M.2 2280 / 2230



2 SATA SSD (2.5-inch)



Uses SATA interface for broad compatibility.

- Much faster than HDDs
- Reliable and durable
- Great for everyday computing upgrades

Best for:



Interface

SATA III (6 Gb/s)



Form Factor

2.5-inch (7mm)



3 EXTERNAL SSD (Portable)



Portable SSD with USB connection for on-the-go storage.

- Fast transfer via USB 3.2 / USB-C
- Rugged and compact
- Ideal for backups and travel

Best for:



Interface

USB 3.2 Gen 2 / USB-C



Form Factor

Portable (Pocket Size)



4 ENTERPRISE SSD (Data Center)



Built for 24/7 workloads in enterprise and data center environments.

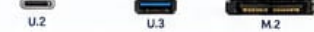
- High endurance (DWPD)
- Consistent performance
- Power loss protection & advanced reliability

Best for:



Interface

SATA / SAS / PCIe (U.2, U.3, M.2)



Form Factor

2.5-inch / U.2 / U.3 / M.2



5 NAS DRIVE (Network Attached Storage)



Designed for NAS systems and always-on network environments.

- Optimized for 24/7 operation
- High reliability and vibration resistant
- Supports RAID and multi-user access

Best for:



Interface

SATA III (6 Gb/s)



Form Factor

3.5-inch



6 HDD (Hard Disk Drive)



Traditional spinning disk drive for high-capacity storage.

- Large storage capacity
- Lowest cost per TB
- Ideal for archives and backups

Best for:



Interface

SATA III (6 Gb/s)



Form Factor

3.5-inch / 2.5-inch



COMPARISON AT A GLANCE

Type	Interface	Speed	Capacity	Durability	Best Use
NVMe SSD	PCIe (NVMe)	★★★★★ (Up to 14 GB/s+)	Up to 8 TB	High	High performance, gaming, content creation
SATA SSD	SATA III (6 Gb/s)	★★★★☆ (Up to 560 MB/s)	Up to 4 TB	High	Daily use, upgrades
External SSD	USB 3.2 / USB-C	★★★★☆ (Up to 2000 MB/s)	Up to 8 TB	High	Portability, backups, media transfer
Enterprise SSD	SATA / SAS / PCIe	★★★★☆ (Consistent)	Up to 15 TB	Very High	Servers, databases, data centers
NAS Drive	SATA III	★★★★☆ (Up to 200 MB/s)	Up to 20 TB+	High	NAS, always-on systems, file sharing
HDD	SATA III	★★★☆☆ (Up to 200 MB/s)	Up to 24 TB+	Medium	Archiving, backups, bulk storage

HOW THEY CONNECT



TIPS

- ✓ Use NVMe SSDs for the fastest system and application performance.
- ✓ Use SATA SSDs for a great balance of speed and price.
- ✓ Back up important data regularly to an external drive or NAS.
- ✓ Choose enterprise or NAS drives for 24/7 reliability.
- ✓ HDDs are best for large, infrequently accessed data.



DID YOU KNOW?

A 1TB HDD can hold about 250 movies, while a 1TB NVMe SSD can transfer those same movies in just a few seconds!



SSD (NVMe M.2)

Blazing Fast Storage for Modern Computers

An NVMe M.2 SSD is a high-performance solid-state drive that uses the NVMe (Non-Volatile Memory Express) protocol and connects directly to the motherboard via the M.2 slot for ultra-fast data transfer and low latency.

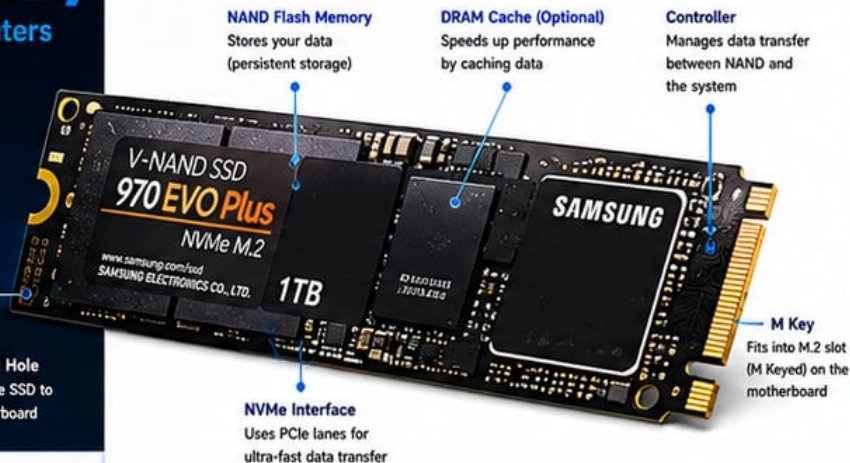
WHAT IS AN SSD?



SSD stands for Solid State Drive. Unlike traditional hard drives (HDDs) that use moving parts, SSDs use flash memory chips to store data. This makes them faster, quieter, more reliable, and more energy efficient.

Mounting Hole
Secures the SSD to the motherboard

NVMe M.2 SSD OVERVIEW



NVMe vs SATA: WHAT'S THE DIFFERENCE?

SATA SSD

Uses SATA Interface



- ❌ Max speed: ~550 MB/s
- ❌ Uses SATA cable
- ❌ AHCI protocol
- ❌ Higher latency
- ❌ Older technology

VS

NVMe M.2 SSD

Uses PCIe Interface



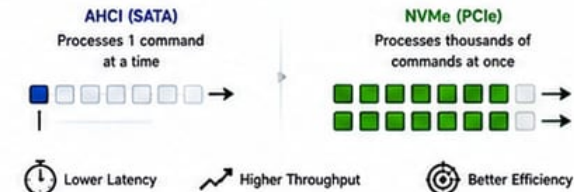
- ✅ Max speed: 3,000 – 7,000+ MB/s
- ✅ Connects directly to M.2 slot
- ✅ NVMe protocol
- ✅ Lower latency
- ✅ Modern & future-ready

KEY SPECIFICATIONS

Interface	PCIe (Gen3 x4 / Gen4 x4 / Gen5 x4)
Protocol	NVMe (Non-Volatile Memory Express)
Form Factor	M.2 (2280, 2260, 2242, etc.)
Capacities	250GB, 500GB, 1TB, 2TB, 4TB, 8TB+
Sequential Read Speed	Up to 7,000+ MB/s (Gen4)
Sequential Write Speed	Up to 6,500+ MB/s (Gen4)
Endurance (TBW)	150 TBW – 4800+ TBW (varies by model)
MTTF (Reliability)	Up to 1.5 Million Hours

NVMe PROTOCOL ADVANTAGE

NVMe is designed specifically for SSDs and allows thousands of commands to be processed in parallel.



M.2 KEY TYPES

M.2 connectors come in different "keys" to prevent incorrect insertion.



HOW NVMe M.2 SSD WORKS



Data flows directly between the CPU and the NVMe SSD over PCIe lanes, eliminating the slower SATA interface and reducing latency.

KEY BENEFITS OF NVMe M.2 SSD



ULTRA FAST SPEEDS

Up to 7,000+ MB/s read speeds – much faster than SATA SSDs and HDDs.



LOW LATENCY

NVMe protocol is optimized for speed and responsiveness.



HIGH RELIABILITY

No moving parts means less wear and tear and better durability.



ENERGY EFFICIENT

Consumes less power and runs cooler than traditional drives.

WHERE IS IT USED?



OPERATING SYSTEM

Faster boot time and quick application launches.



GAMING

Reduce load times and smoother gameplay.



CONTENT CREATION

Handle large files, 4K/8K video editing, and rendering.



DATA STORAGE

Ideal for databases, virtual machines, and heavy workloads.

M.2 SSD FORM FACTORS (BY LENGTH)

All are 22mm wide, but lengths vary.

2230 22mm x 30mm	22mm x 30mm	Common in compact devices and handhelds
2242 22mm x 42mm	22mm x 42mm	Tablets and ultrabooks
2260 22mm x 60mm	22mm x 60mm	Some laptops and small PCs
2280 22mm x 80mm	22mm x 80mm	Most common for desktops and laptops

DID YOU KNOW?

An NVMe M.2 SSD can be up to 12x faster than a SATA SSD and up to 100x faster than a traditional HDD!



M.2 SLOTS

HIGH-PERFORMANCE STORAGE IN A SMALL FORM FACTOR

M.2 slots allow you to install M.2 SSDs that provide ultra-fast storage performance in a compact form factor. They connect directly to the motherboard and use PCIe lanes (or SATA) for data transfer.

KEY FUNCTIONS

-  **High Speed**
Supports PCIe NVMe SSDs with exceptional read/write speeds.
-  **Compact Design**
M.2 SSDs are much smaller than traditional 2.5" drives.
-  **Versatility**
Supports different lengths and both PCIe and SATA interfaces.
-  **Direct Connection**
Connects directly to the motherboard for lower latency and better performance.

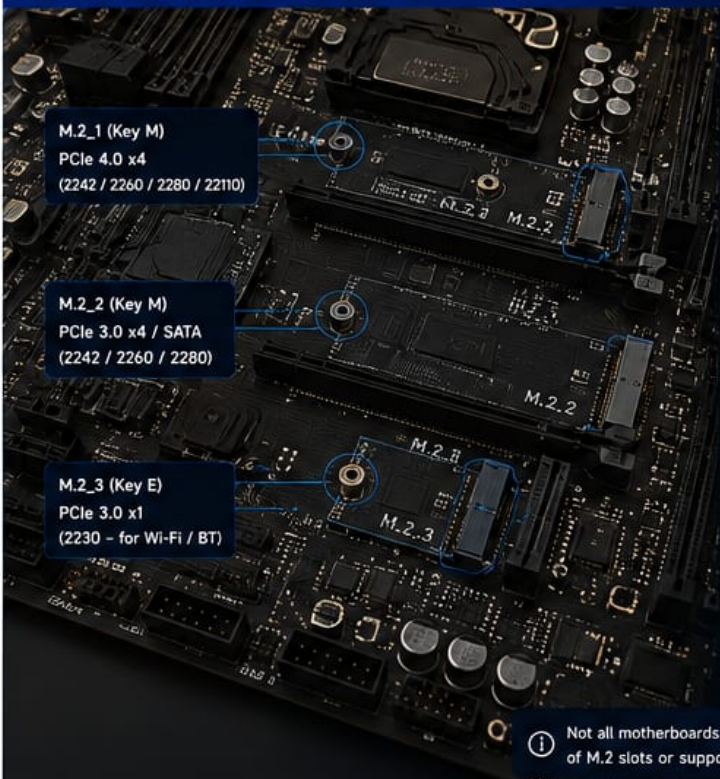
HOW TO INSTALL AN M.2 SSD

- 1 Locate the M.2 Slot**
Find the M.2 slot on your motherboard.
- 2 Remove Screw**
Remove the screw from the mounting hole (standoff).
- 3 Insert SSD**
Insert the M.2 SSD into the slot at a 30° angle.
- 4 Press Down**
Gently press the SSD down flat so the hole aligns with the standoff.
- 5 Secure with Screw**
Secure the SSD with the screw you removed.

 Handle M.2 SSDs by the edges and avoid touching the gold contacts.

DID YOU KNOW?
 M.2 SSDs can be up to 7x faster than SATA SSDs in real-world performance (NVMe PCIe 4.0 vs SATA).

M.2 SLOTS ON A MOTHERBOARD (EXAMPLES)



M.2_1 (Key M)
PCIe 4.0 x4
(2242 / 2260 / 2280 / 22110)

M.2_2 (Key M)
PCIe 3.0 x4 / SATA
(2242 / 2260 / 2280)

M.2_3 (Key E)
PCIe 3.0 x1
(2230 - for Wi-Fi / BT)

PARTS OF AN M.2 SLOT

- 1 M.2 Connector**
The slot where the M.2 SSD is inserted.
- 2 Mounting Hole / Standoff**
Used to secure the SSD at different lengths.
- 3 Screw**
Keeps the M.2 SSD firmly in place.
- 4 Heatsink (Optional)**
Helps maintain optimal temperatures for better performance.

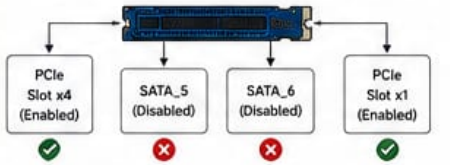
 Not all motherboards have the same number of M.2 slots or support all M.2 types.

M.2 LANE SHARING

Some M.2 slots share bandwidth with other ports. Using certain M.2 slots may disable SATA ports or PCIe slots.

Example:

- Using M.2_1 (PCIe 4.0 x4) disables SATA_5 and SATA_6.

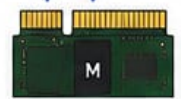


 Always check your motherboard manual for lane sharing details.

M.2 KEYING (NOTCHES)

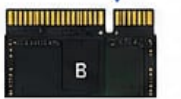
The notch on the connector determines compatibility.

Key M
(Socket 3)



For NVMe (PCIe) & SATA SSDs

Key B
(Socket 2)



For SATA SSDs

Key E
(Socket 1)



For Wi-Fi / Bluetooth and some PCIe devices

M.2 SIZES (LENGTH)

2230 (30mm)



2242 (42mm)



2260 (60mm)



2280 (80mm)



22110 (110mm)



 The motherboard will list which sizes are supported for each M.2 slot.

M.2 INTERFACES

PCIe (NVMe)



Uses PCI Express lanes for maximum performance.
Common: PCIe 3.0 x4 / PCIe 4.0 x4
PCIe 5.0 x4

SATA



Uses SATA bus for data transfer.
Typical speed: Up to 600 MB/s

TIPS & BEST PRACTICES

-  Use an M.2 SSD in the highest bandwidth slot for best performance.
-  Ensure the M.2 SSD is compatible (key, interface, and size).
-  Use a heatsink if your motherboard does not include one.
-  Keep your BIOS/UEFI updated for maximum compatibility and performance.
-  Do not overtighten the screw.



INTERFACE SPEED COMPARISON (THEORETICAL MAXIMUM)

Interface	PCIe 3.0 x4 / SATA III	PCIe 4.0 x4	PCIe 5.0 x4
Bandwidth	~3.94 GB/s	~7.88 GB/s	~15.75 GB/s
Typical SSD Speed	Up to 560 MB/s	Up to 7,000 MB/s	Up to 14,000+ MB/s

- ### COMMON M.2 SLOT ISSUES
-  SSD not detected: Check seating, compatibility, and BIOS settings.
 -  Slow speeds: Ensure the SSD is in a PCIe slot, not a SATA M.2 slot.
 -  Overheating: Add or improve heatsink / airflow.
 -  Lane sharing: Other ports may be disabled when using certain slots.

 **PRO TIP**
For best performance, install your OS and most used applications on an M.2 NVMe SSD.



SATA PORTS

CONNECTING STORAGE DEVICES

SATA (Serial ATA) ports on the motherboard allow you to connect 2.5"/3.5" hard drives, SSDs, and optical drives for data storage and transfer.

KEY FUNCTIONS



Data Connection
Provides a data interface between the motherboard and SATA devices.



Power Efficient
SATA is more power efficient compared to older IDE (PATA).

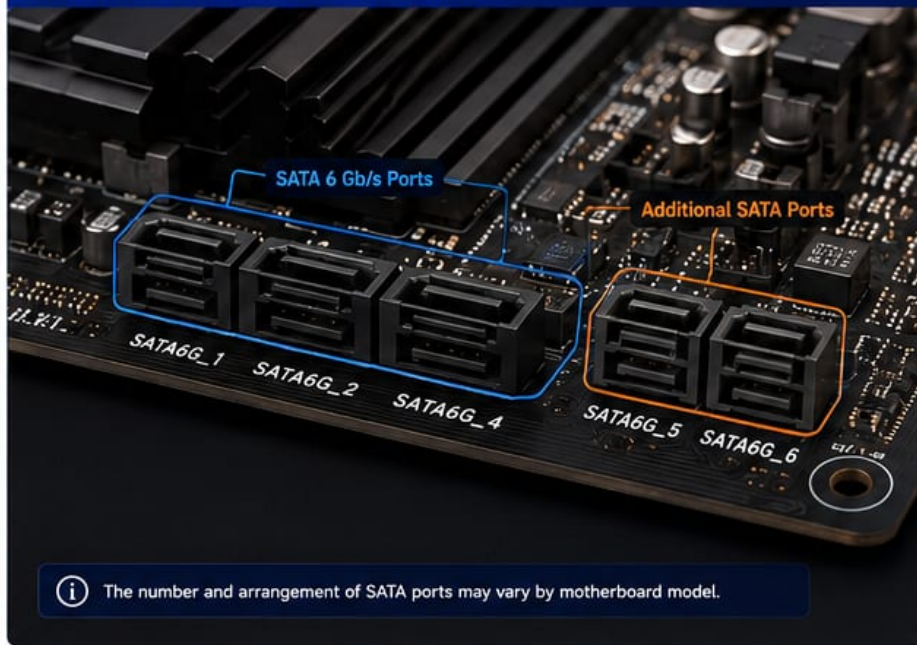


High Speed
Supports up to 6 Gb/s (SATA III) for fast data transfer.



Hot Swappable (AHCI)
Allows drives to be connected/disconnected while the system is running (with AHCI enabled).

SATA PORTS ON A MOTHERBOARD



SATA PORT TYPES (BY SPEED)

SATA Version	Max Speed	Color (Common)	Backward Compatibility
SATA I	1.5 Gb/s	Gray	Yes
SATA II	3 Gb/s	Black	Yes
SATA III	6 Gb/s	Blue	Yes

i SATA III ports are backward compatible with SATA II and SATA I devices.

SATA CONNECTORS (ON MOTHERBOARD)

7-Pin Data Connector (SATA Port)



L-shaped key ensures correct connection.

15-Pin Power Connector (from PSU)



Provides power to the SATA device.

SATA DEVICES YOU CAN CONNECT



3.5" HDD



2.5" SSD / HDD

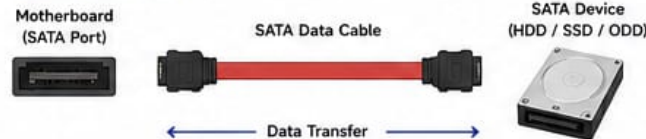


Optical Drive (CD/DVD/Blu-ray)



SATA SSD

HOW SATA WORKS



i SATA uses two connections:
1. Data Cable (7-pin) • 2. Power Cable (15-pin)

SATA PORT LABELING



Always refer to your motherboard manual for exact port functions.

HOW TO CONNECT A SATA DEVICE

1

Connect SATA Data Cable
Attach one end of the data cable to the motherboard SATA port.



2

Connect to the Device
Attach the other end of the data cable to the SATA port on the device.



3

Connect Power Cable
Connect the 15-pin power connector from the PSU to the device.



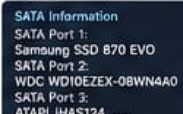
4

Secure Cables
Ensure both connectors are firmly seated.



5

Power On & Detect
Power on the system. The device should be detected in BIOS/UEFI.



SATA Information
SATA Port 1: Samsung SSD 870 EVO
SATA Port 2: WD10EZEX-08WN4A0
SATA Port 3: ATAPI iH45124

! Always use the correct cables and ensure they are fully seated to avoid connection issues.

SATA CABLES

SATA Data Cable (7-pin)



- Thin
- L-shaped
- Transfers data

SATA Power Cable (15-pin)



- Wider
- Flat connector
- Provides power

TIPS & BEST PRACTICES

- ✓ Use SATA III (6 Gb/s) ports and cables for best performance.
- ✓ Enable AHCI mode in BIOS for maximum speed and features.
- ✓ Keep cables tidy and away from fans and heat sources.
- ✓ Check your motherboard manual for port limitations (e.g., some ports may be disabled when using M.2 SATA).
- ✓ For SSDs, place them on the primary SATA ports (often SATA1/2).



DID YOU KNOW?

SATA III (6 Gb/s) is 4x faster than SATA II (3 Gb/s) and 6x faster than SATA I (1.5 Gb/s).

SATA GENERATION SPEED COMPARISON

Generation	SATA I	SATA II	SATA III
Max Speed	1.5 Gb/s	3 Gb/s	6 Gb/s
Real-World Speed (approx.)	~150 MB/s	~300 MB/s	~600 MB/s

COMMON SATA PORT ISSUES

- ⚠ Device not detected: Check cables and power connection.
- ⚠ Slow performance: Ensure AHCI mode is enabled.
- ⚠ Port not working: Try another SATA port or cable.
- ⚠ M.2 SATA in use: Some SATA ports may be disabled.



PRO TIP

For the best performance, use an SSD with SATA III on a SATA III (6 Gb/s) port and enable AHCI in BIOS/UEFI.





DATA UNITS EXPLAINED FROM 1 KILOBYTE TO ZETTABYTES

Measuring the Digital Universe



Smaller

Larger

**KILOBYTE
(KB)**



= 10^3 bytes

1,000
bytes

A few lines
of text

**MEGABYTE
(MB)**



= 10^6 bytes

1,000
kilobytes

A high-resolution
photo

**GIGABYTE
(GB)**



= 10^9 bytes

1,000
megabytes

A movie
(HD)

**TERABYTE
(TB)**



= 10^{12} bytes

1,000
gigabytes

Store on a
large hard drive

**PETABYTE
(PB)**



= 10^{15} bytes

1,000
terabytes

Data of a large
data center

**EXABYTE
(EB)**



= 10^{18} bytes

1,000
petabytes

Cloud providers
scale

**ZETTABYTE
(ZB)**



= 10^{21} bytes

1,000
exabytes

The digital
universe

BYTES 10^0 10^1 10^2 10^3 10^4 10^5 10^6 10^7 10^8 10^9 10^{10} 10^{11} 10^{12} 10^{13} 10^{14} 10^{15} 10^{16} 10^{18} 10^{19} 10^{20} 10^{21}

QUICK REFERENCE TABLE

UNIT	ABBREVIATION		EQUALS (BYTES)
Kilobyte	KB	10^3	1,000
Megabyte	MB	10^6	1,000,000
Gigabyte	GB	10^9	1,000,000,000
Terabyte	TB	10^{12}	1,000,000,000,000
Petabyte	PB	10^{15}	1,000,000,000,000,000
Exabyte	EB	10^{18}	1,000,000,000,000,000,000
Zettabyte	ZB	10^{21}	1,000,000,000,000,000,000,000

We use the decimal system (base 10) in data storage.

REAL-WORLD EXAMPLES

1 KB		A short sentence ($\approx$ 1 page of text)
1 MB		A large photo ($\approx$ 1–2 MB)
1 GB		$\approx$ 250 songs (MP3)
1 TB		$\approx$ 500 hours of 4K video
1 PB		$\approx$ 1 million hours of HD video
1 EB		$\approx$ All internet data in a month (est.)
1 ZB		$\approx$ All data ever created by humans (est.)

HOW BIG IS 1 ZETTABYTE?

**1 ZETTABYTE (ZB) =
1,000 EXABYTES**

= 1,000,000 PETABYTES

= 1,000,000,000 TERABYTES

= 1,000,000,000,000 GIGABYTES

= 1,000,000,000,000,000 MEGABYTES

= 1,000,000,000,000,000,000 KILOBYTES

= 1,000,000,000,000,000,000,000 BYTES

DID YOU KNOW?

The global datasphere is estimated to reach
221 ZETTABYTES in 2026 and keep growing!

STORAGE DEVICES COMPARISON

	USB Flash Drive	16 GB
	Smartphone (128 GB)	128 GB
	Laptop SSD	1 TB
	Enterprise SSD	30.72 TB
	Data Center Storage	PB – EB
	Global Datasphere	ZB+



From tiny text files to the entire digital universe—
data powers our world.



Note: Some systems use binary (base 2):
1 KiB = 1,024 bytes. We use base 10 here.

Sources: IEC 80000-13:2008, Statista, IDC DataAge 2024,
Seagate Datasphere Report 2024



THE HISTORY OF STORAGE DEVICES

From the Beginning Till the Very Latest – A Journey of Innovation and Capacity

From bulky magnetic drums to ultra-fast solid-state drives, storage technology has evolved dramatically to meet the world's growing data needs.

KEY



Speed



Capacity



Relative Cost

1950s The Beginning	1960s Early Magnetic Storage	1970s Improvement & Miniaturization	1980s Personal Computing Era	1990s High Capacity Growth	2000s Speed & Reliability	2010s Solid-State Revolution	2020s NVMe & Beyond	Future The Next Frontier
Magnetic Drum	Magnetic Tape	Floppy Disk	Hard Disk Drive (HDD)	CD-ROM / Zip Drive	DVD / Blu-ray / HDD	SATA SSD	NVMe SSD	Next-Gen Storage
								
The first computer storage. Used magnetic drums to store data.	Sequential access tape storage used for backups and large data archives.	Removable and reusable storage for early personal computers.	First HDDs for PCs. Growing capacity and faster access.	Optical discs for distribution; Zip drives for larger removable storage.	Higher capacity optical discs; HDDs reach terabytes.	Solid-state drives become mainstream with huge speed improvement.	NVMe SSDs deliver extreme speed with higher capacities at lower latency.	Pushing the limits with petabyte capacities and blazing speeds.
 Very Slow	 Slow	 Slow	 Medium	 Medium	 Medium	 Fast	 Very Fast	 Ultra Fast
 5 KB	 MBs – GBs	 100 KB – 1.2 MB	 10 MB – 200 MB	 650 MB – 750 MB	 4.7 GB – 1 TB+	 120 GB – 4 TB	 250 GB – 30 TB	 1 PB+
 Extremely High	 High	 Affordable	 Moderate	 Affordable	 Affordable	 Falling	 Moderate	 Very High
Real World Example IBM 305 RAMAC (1956) 5 MB, refrigerator-sized	Real World Example IBM 729 Magnetic Tape (1956) 1/2 inch tape	Real World Example IBM 3740 Floppy Disk (1976) 8-inch, 1.2 MB	Real World Example Seagate ST-225 (1980) 20 MB, 5.25-inch	Real World Example Iomega Zip 100 (1994) 100 MB	Real World Example WD Caviar 250GB HDD (2005)	Real World Example Samsung 840 EVO (2013) 1 TB, SATA SSD	Real World Example Samsung 990 PRO (2023) 2 TB, NVMe PCIe 4.0	Real World Example Research Prototypes 1 PB+ NVMe SSDs*
Typical Price (ZAR) R200,000+ (≈ R2,000,000 in today's money)	Typical Price (ZAR) R2,000 – R5,500 per tape (≈ R20,000+ in today's money)	Typical Price (ZAR) R300 – R600 per disk (≈ R3,000+ in today's money)	Typical Price (ZAR) R1,000 – R2,500 (≈ R10,000+ in today's money)	Typical Price (ZAR) R250 – R400 (≈ R3,000+ in today's money)	Typical Price (ZAR) R600 – R1,200 (≈ R7,000+ in today's money)	Typical Price (ZAR) R1,600 – R3,000 (≈ R15,000+ in today's money)	Typical Price (ZAR) R2,000 – R4,000 (≈ R20,000+ in today's money)	Typical Price (ZAR) Not yet commercially available

HOW MUCH IS 1 PB?

1 Petabyte (PB) = 1,000 Terabytes (TB)

1 PB = 1,000 TB

= 1,000,000 GB

= 1,000,000,000 MB

= 1,000,000,000,000 KB

= 1,000,000,000,000,000 bytes

TO PUT 1 PB INTO PERSPECTIVE

 ≈ 250 million photos (4 MB each)

 ≈ 200,000 hours of HD video

 ≈ 500 million PDF documents

 ≈ 1,000 × 1 TB SSDs

MODERN STORAGE DEVICES – REAL PARTS & REAL PRICES (SOUTH AFRICAN RAND)

SATA SSD (2.5-inch)	M.2 SATA SSD	M.2 NVMe SSD (PCIe 3.0)	M.2 NVMe SSD (PCIe 4.0)	M.2 NVMe SSD (PCIe 5.0)	U.2 / E1.S SSD (Enterprise)	Enterprise SAS SSD
						
Interface: SATA III (6 Gb/s) Read Speed: Up to 560 MB/s Capacity: 250 GB – 2 TB Best For: Everyday PCs, laptops, upgrades	Interface: SATA III (6 Gb/s) Read Speed: Up to 560 MB/s Capacity: 250 GB – 2 TB Best For: Laptops, upgrades	Interface: PCIe 3.0 x4 Read Speed: Up to 3,500 MB/s Capacity: 250 GB – 2 TB Best For: Faster boot, gaming, creators	Interface: PCIe 4.0 x4 Read Speed: Up to 7,000 MB/s Capacity: 500 GB – 4 TB Best For: High performance computing, content creation	Interface: PCIe 5.0 x4 Read Speed: Up to 14,800 MB/s Capacity: 1 TB – 4 TB Best For: Extreme speed workloads, future-ready.	Interface: PCIe 4.0 x4 (U.2) Read Speed: Up to 6,200 MB/s Capacity: 800 GB – 15.36 TB Best For: Servers, data centers, 24/7 workloads	Interface: SAS 12Gb/s Read Speed: Up to 2,200 MB/s Capacity: 480 GB – 15.36 TB Best For: Enterprise apps, databases, mission critical
Crucial MX500 1TB R1,699	WD Blue SA510 1TB R1,699	Samsung 970 EVO Plus 1TB R2,299	WD Black SN850 1TB R2,799	Samsung 990 PRO 2TB R5,499	Micron 7450 PRO 3.84TB R16,999	Samsung PM893 3.84TB R22,999

KEY TAKEAWAYS



Storage capacity has grown from KBs to Petabytes (1,000,000+ GB).



Speed has increased from KB/s to over 14,000 MB/s (and beyond).



SSDs have replaced HDDs in most consumer and enterprise systems.



NVMe over PCIe 5.0 is the new standard for top-tier performance.

THE LARGEST COMMERCIAL SSD AVAILABLE (AS OF 2024)

Solidigm D5-P5336
122.88 Terabytes (TB)
PCIe 4.0 NVMe SSD

 **SOLIDIGM.**

-  **CAPACITY:** 122.88 TB
-  **INTERFACE:** PCIe 4.0 x4, NVMe 1.4
-  **SEQUENTIAL READ:** Up to 6,800 MB/s
-  **SEQUENTIAL WRITE:** Up to 6,800 MB/s
-  **RANDOM READ (4K):** Up to 1.3M IOPS
-  **RANDOM WRITE (4K):** Up to 300K IOPS
-  **ENDURANCE (DWPD):** 1 DWPD (5 years).
-  **BEST FOR:** AI, Big Data, Cloud, Hyperscale, Mission-critical workloads

TYPICAL PRICE (ZAR)

R120,000 – R160,000+
(Depending on vendor & configuration)



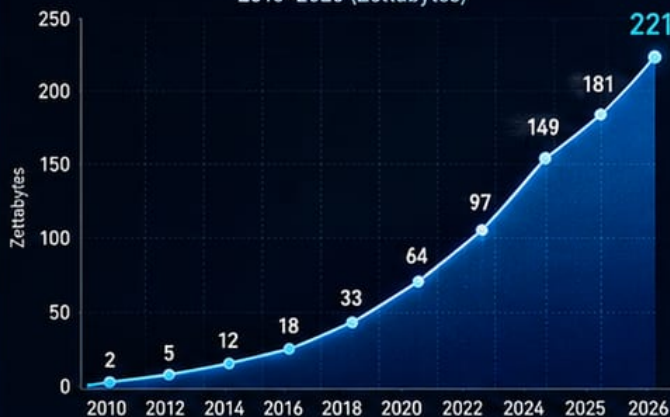
A true 1 PB SSD is not yet a standard retail product. It would cost well over R1 million using today's technology.

* Prototype / enterprise research products shown for reference only.

How Much Information Do We Have in the World?

Global Digital Datasphere (2026)

GROWTH OF THE GLOBAL DATASPHERE
2010-2026 (Zettabytes)



The global datasphere is growing at an unprecedented pace.

221 ZETTABYTES (ZB)

Estimated global datasphere in 2026
(221,000,000,000,000,000,000 bytes)

WHAT IS 221 ZB?

EB	221,000 EB
PB	221,000,000 PB
TB	221 BILLION TB
GB	221 TRILLION GB
MB	221 QUADRILLION MB

WHERE DOES THE DATA COME FROM?



WHAT CREATES THE MOST DATA



FUN COMPARISONS

221 ZB could store approximately:

- 55 TRILLION HD MOVIES
- 44 QUADRILLION PHOTOS (5 MB)
- HUNDREDS OF BILLION YEARS OF HD VIDEO
- OVER 221 BILLION 1 TB SSDs



EVERY DAY IN 2026

Approximately **403 MILLION TB** of new data are created every day.



LOOKING AHEAD

Projected 2028: **320+ ZB**
Projected 2030: **500+ ZB**

POWER SUPPLY UNIT (PSU)

The Heart That Powers Your PC

The Power Supply Unit (PSU) converts electricity from your wall outlet into the clean, regulated DC power that your computer's components need to run.

WHAT DOES A PSU DO?

It converts high-voltage AC (Alternating Current) from your wall outlet into low-voltage DC (Direct Current) and supplies it to all the components inside your computer safely and efficiently.

HOW A PSU WORKS

- AC power enters the PSU from the wall outlet.
- The EMI filter removes electrical noise.
- The rectifier converts AC to high-voltage DC.
- The inverter switches the DC at high frequency.
- The transformer steps the voltage down.
- The rectifier & filters produce stable DC outputs.
- The PSU regulates the voltages and sends power to your PC components.

AC INPUT (Wall Outlet)

AC to DC Conversion

Voltage Transformation

DC Output & Regulation

Power to Components

TYPES OF PSUS

NON-MODULAR

All cables are fixed to the PSU.

- ✓ Lower cost
- ✗ More cables to manage

SEMI-MODULAR

Some cables are detachable.

- ✓ Better cable management
- ✓ More affordable

FULLY MODULAR

All cables are detachable.

- ✓ Best cable management
- ✗ Higher cost

KEY FUNCTIONS

- Converts AC to DC power
- Regulates voltage for stable performance
- Protects components from power issues (OCP, OVP, SCP, etc.)
- Keeps the system cool with built-in fans

EFFICIENCY RATINGS (80 PLUS)

PSUs are rated for energy efficiency. Higher efficiency means less wasted energy, less heat, and lower electricity costs.

Standard
80%

Bronze
82%

Silver
85%

Gold
87%

Platinum
90%

Titanium
92%

Typical Efficiency at 50% Load

120mm FAN
Cooling fan draws air in and pushes hot air out to keep the PSU cool.

AC INPUT
Connects to your wall outlet using the included power cable.

POWER SWITCH
Turns the PSU on or off.

VOLTAGE SWITCH (If Available)
Selects input voltage (115V / 230V) based on your region.

DURABLE BUILD
Built with high-quality components for reliable and long-term use.

CABLES & CONNECTORS
Deliver power to all components through specific connectors.

COMMON PSU CONNECTORS		
CONNECTOR	USED FOR	IMAGE
24-Pin ATX	Motherboard	
4+4 Pin EPS (8-Pin CPU)	CPU Power	
6+2 Pin PCIe (8-Pin)	Graphics Cards	
SATA Power	SSDs, HDDs, Optical Drives	
Molex (4-Pin)	Older Drives, Fans, Accessories	

COMMON DC OUTPUTS		
RAIL	VOLTAGE	USED FOR
+3.3V	+3.3 Volts	Motherboard, RAM, Chipset
+5V	+5 Volts	USB, HDDs, SSDs, Fans
+12V	+12 Volts	CPU, GPU, Motors, Fans
-12V	-12 Volts	Legacy/Serial ports (rarely used)
+5VSB	+5 Volts Standby	Power for standby features (Wake-on-LAN, etc.)

Note:
The +12V rail is the most important for modern PCs as it powers the CPU and GPU.

HOW TO CHOOSE THE RIGHT PSU

- ✓ Check your system's power requirements (use a PSU calculator).
- ✓ Choose a reputable brand for reliability and safety.
- ✓ Pick the right wattage (don't go too low or unnecessarily high).
- ✓ Look for 80 PLUS certification for better efficiency.
- ✓ Choose the right type (non-modular, semi, or fully modular).

SAFETY PROTECTIONS (BUILT INTO PSUS)

OCP
Over Current Protection

OVP
Over Voltage Protection

UVP
Under Voltage Protection

SCP
Short Circuit Protection

OPP
Over Power Protection

OTP
Over Temp Protection

PRO TIP

A high-quality PSU protects your expensive components and ensures system stability. Never compromise on the power supply!

CAUTION

Using a low-quality or underpowered PSU can cause crashes, data loss, or even damage to your components.

STILL WIDELY USED

Desktops & Workstations

Servers & Data Centers

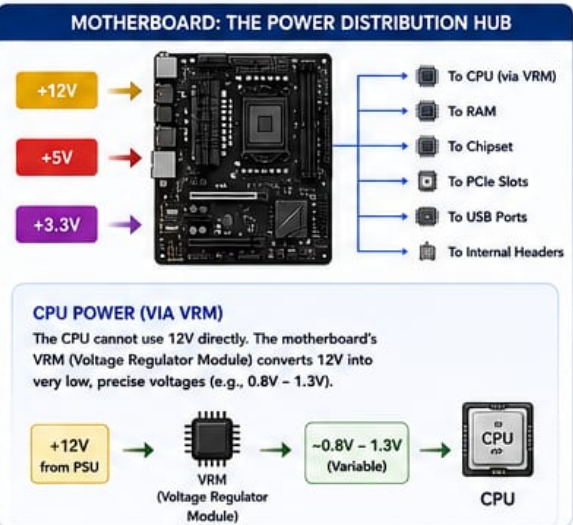
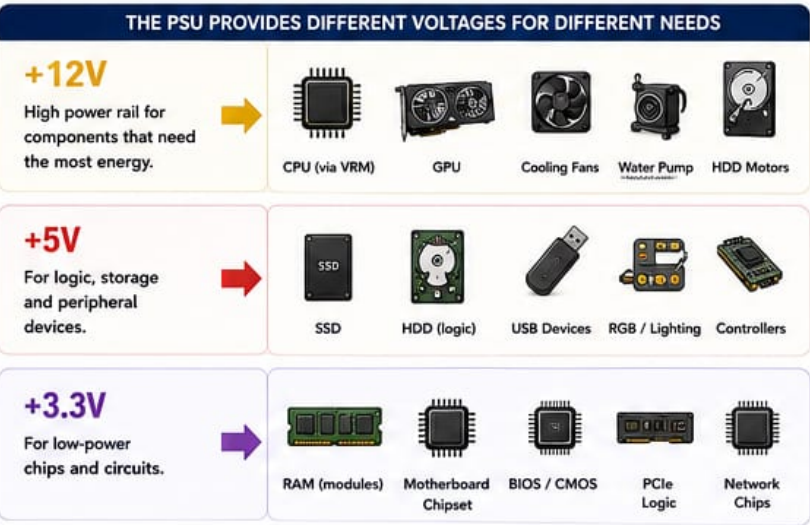
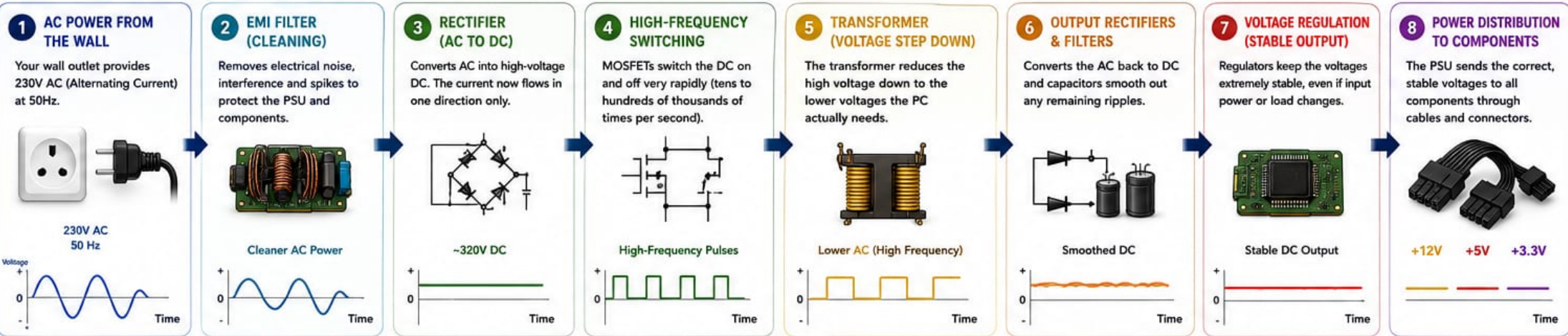
Gaming PCs

Industrial Systems

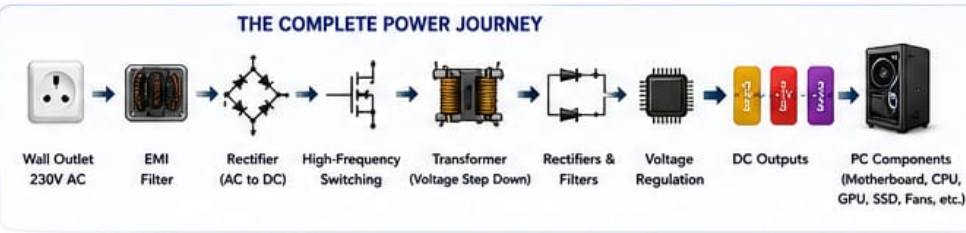
PSUs may not be the most exciting part of a PC, but they are one of the most important!

HOW A PSU CONVERTS AND DELIVERS POWER IN A PC

From the Wall Outlet to Every Component



HOW DIFFERENT COMPONENTS USE POWER				
	INPUT FROM PSU	ON MOTHERBOARD	OUTPUT TO COMPONENT	USED FOR
 CPU	+12V	VRM	0.8V – 1.3V (Precision DC)	CPU Cores, Cache, Integrated Controllers
 GPU	+12V (through PCIe cable)	Internal Voltage Regulators	~0.8V – 1.2V	GPU Core, Memory, VRAM, Controllers
 SSD	+5V	On SSD Controller	3.3V / 1.8V (Internal)	Controller IC, NAND Flash
 HDD	+12V (for motor) +5V (for logic)	On Drive PCB Controller	12V (Motor) 5V / 3.3V (Logic)	Spins Platter (12V) Electronics (5V)
 Fans	+12V	Via Fan Header (PWM Control)	12V (Variable Speed)	Runs Fan Motor



 **KEY TAKEAWAY:**
The PSU converts high-voltage AC power from the wall into multiple stable DC voltages (+12V, +5V, +3.3V). It delivers the right power to the right components at the right time, keeping your PC safe, stable, and reliable.

 **NOTE:** Efficiency, quality components, and proper wattage ensure better performance, lower heat, and longer lifespan.

COMPUTER CASE (CHASSIS)

THE FOUNDATION OF YOUR PC

A computer case houses and protects all the internal components of your PC, provides structure, enables airflow for cooling, and determines compatibility, expandability, and aesthetics.

WHAT DOES A CASE DO?



PROTECTS COMPONENTS

Shields delicate parts from dust, impacts, and damage.



SUPPORTS HARDWARE

Provides mounts and space for motherboard, GPU, drives, PSU, cooling, and more.



MANAGES AIRFLOW

Designed with vents and fan mounts to keep components cool.



ENHANCES BUILD EXPERIENCE

Good cable management and layout make building and upgrading easier.



SHOWCASES YOUR STYLE

From minimal to RGB, your case reflects your personality.

CASE ANATOMY (TYPICAL ATX MID-TOWER)



COMMON CASE TYPES



FULL TOWER

Largest size. Maximum space, extreme cooling support, ideal for high-end builds and workstations.



MID TOWER

Most popular. Great balance of size, features, expandability, and price.



MINI TOWER

More compact. Fewer expansion options, good for basic to mid-range builds.



MICRO ATX

Compact and space-saving. Supports mATX motherboards.



ITX / MINI ITX

Smallest size. Supports Mini ITX motherboards. Great for small form factor (SFF) builds.

CASE SIZE GUIDE (BY MOTHERBOARD SUPPORT)

CASE SIZE	MOTHERBOARD SUPPORT	BEST FOR
Full Tower	E-ATX, ATX, mATX, Mini ITX	High-end builds, workstations, custom water cooling
Mid Tower	ATX, mATX, Mini ITX	Gaming, productivity, all-purpose builds
Mini Tower	mATX, Mini ITX	Budget builds, office PCs
Micro ATX	mATX, Mini ITX	Compact builds
ITX / Mini ITX	Mini ITX only	Small form factor builds

MATERIALS & CONSTRUCTION



STEEL (SPCC)

Strong, durable, and affordable. Most common.



ALUMINUM

Lightweight, premium look and feel. Better for portability.



TEMPERED GLASS

Shows off your build. Usually on the side panel.



PLASTIC

Used in some panels and front designs. Lighter, less durable.



AIRFLOW & COOLING

OPTIMAL AIRFLOW SETUP (EXAMPLE)



FAN & RADIATOR SUPPORT (EXAMPLE: MID TOWER)



FRONT
• 3x 120mm / 2x 140mm fans
• Radiator: 120 / 240 / 280 / 360mm



TOP
• 2x 120mm / 2x 140mm fans
• Radiator: 120 / 240 / 280mm

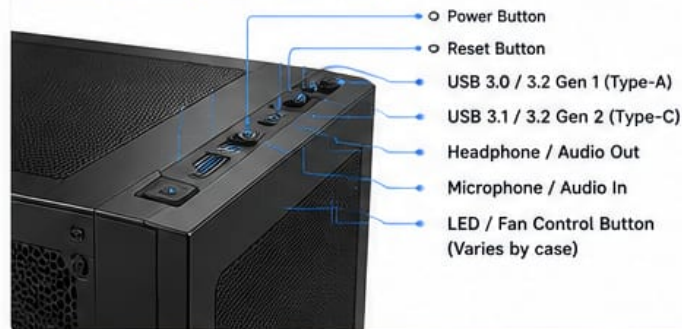


REAR
• 1x 120mm / 1x 140mm fan
• Radiator: 120mm



BOTTOM
• 2x 120mm fans (PSU shroud)
(not all cases support this)

FRONT I/O PANEL (TYPICAL EXAMPLE)



KEY FEATURES TO CONSIDER



COMPATIBILITY

Ensure the case supports your motherboard size, GPU length, CPU cooler height, and PSU size.



AIRFLOW

Look for mesh panels, ventilation, and support for multiple fans or radiators.



EXPANDABILITY

More drive bays, expansion slots, and fan mounts = more upgrade options.



CABLE MANAGEMENT

Good routing space and tie-down points make builds cleaner and airflow better.



BUILD QUALITY

Solid construction means durability, less vibration, and better protection.



AESTHETICS

Choose a design, color, and features (RGB, glass, minimal, etc.) you love.



PRO TIP

A good case with excellent airflow can lower temperatures, reduce noise, and extend the life of your components.



REAR I/O PANEL

OLD VS. LATEST

The Rear I/O (Input/Output) Panel is the section on the back of a computer where you connect external devices such as monitors, keyboards, mouse, USB drives, network cables, audio speakers, and more.

KEY FUNCTIONS

- Connects External Devices**
Allows connection of peripherals such as USB devices, monitors, network, audio, and more.
- Provides Communication**
Enables data transfer between your computer and external devices.
- Expands Capabilities**
Different ports and versions support faster speeds, higher resolutions, and more features.

OLD REAR I/O PANEL (2005–2012)



						
PS/2	Parallel Port (DB-25)	Serial Port (COM)	VGA (D-Sub)	USB 2.0	RJ45 Ethernet (10/100)	3.5mm Audio Jacks
Keyboard / Mouse	Printers and older devices	Legacy devices and modems	Analog video output	Up to 480 Mb/s	Wired network connection	Line In / Line Out / Microphone

OLD PANEL HIGHLIGHTS

- Focus on legacy ports (PS/2, Parallel, Serial, VGA)
- USB 2.0 is the fastest available
- Ethernet up to 100 Mb/s
- Analog video and audio
- Limited speed and functionality by today's standards

LATEST REAR I/O PANEL (2023–2024)



								
DisplayPort (DP)	HDMI 2.1	USB4 / Type-C (Thunderbolt™ 4)	USB 3.2 Gen 2x2 (Type-C)	USB 3.2 Gen 2 (Type-A)	USB 3.2 Gen 1 (Type-A)	2.5G / 10G Ethernet	Wi-Fi 6E / 7 (External Antenna)	Audio + Optical S/PDIF Out
Digital video and audio	High-definition video and audio	Up to 40 Gb/s video, data, and power	Up to 20 Gb/s	Up to 10 Gb/s	Up to 5 Gb/s	Faster wired network	Wireless connectivity	7.1 Surround sound and digital audio output

LATEST PANEL HIGHLIGHTS

- High-speed USB (up to 20 Gb/s and 40 Gb/s)
- Modern video outputs (DP, HDMI 2.1)
- 2.5G / 10G Ethernet for ultra-fast networking
- Wi-Fi 6E/7 for advanced wireless
- High-quality audio with optical output
- More bandwidth, speed, and features for modern devices

QUICK COMPARISON

Feature	Old Panel (2005–2012)	Latest Panel (2023–2024)
USB Version	USB 2.0 (up to 480 Mb/s)	USB 3.2 Gen 2x2 / USB4 (up to 20/40 Gb/s)
Video Outputs	VGA (Analog)	DisplayPort 1.4/2.1, HDMI 2.1
Network	10/100 Ethernet	2.5G / 10G Ethernet, Wi-Fi 6E/7
Audio	3.5mm Analog	High-Definition Audio, Optical S/PDIF
Legacy Ports	PS/2, Parallel, Serial	Removed (Rarely used)
Performance	Slower	Much Faster
Overall	Basic connectivity	High speed, more features, future-ready

EVOLUTION OVER TIME



WHY IT MATTERS

- Speed**
Newer ports offer much faster data transfer.
- Compatibility**
Supports modern devices and higher resolutions.
- Future-Ready**
Latest I/O panels ensure your system stays up to date.
- Better Experience**
Improved audio, network, and connectivity for gaming, streaming, and productivity.

NEW PORTS & CONNECTIONS

MODERN PORTS FOR TODAY'S DEVICES

Technology is always evolving. New ports and connections deliver faster speeds, more power, higher video resolutions, and greater versatility than ever before.

KEY BENEFITS

- Faster Data Transfer**
Move large files in seconds.
- Higher Video & Audio Quality**
Support for 4K, 8K and beyond.
- More Power Delivery**
Charge laptops and devices with a single cable.
- Universal Connectivity**
One port for many devices and uses.
- Future-Ready**
Built for modern devices and tomorrow's demands.

LATEST PORTS & CONNECTIONS OVERVIEW

USB4® (40Gb/s)	THUNDERBOLT™ 4 (40Gb/s)	USB 3.2 GEN 2x2 (20Gb/s)	USB-C® (USB 3.x / USB4)	HDMI® 2.1	DISPLAYPORT™ 2.1	2.5G / 5G / 10G ETHERNET	WI-FI 6E / WI-FI 7 (BLUETOOTH® 5.3+)
• Up to 40 Gb/s • Data, video, audio • Power Delivery (up to 100W) • Compatible with Thunderbolt 3/4	• Up to 40 Gb/s • Dual 4K or single 8K video support • Power Delivery (up to 100W) • Daisy-chain up to 6 devices	• Up to 20 Gb/s • Backward compatible with USB 3.2/2.0 • Ideal for fast external storage	• Reversible connector • Data, video, audio • Power Delivery • Many protocols in one small port	• Up to 48 Gbps • 4K @ 120Hz, 8K @ 60Hz • Dynamic HDR, eARC • VRR, ALLM, QFT for gaming	• Up to 80 Gbps • Up to 16K @ 60Hz • DSC, HDR, VRR, MST • High refresh rate support	• 2.5 / 5 / 10 Gbps speeds • Backward compatible with 1G/100M • Better for streaming, gaming & transfers	• 6 GHz band (6E) or Multi-Link Operation (7) • Lower latency • Higher throughput • Bluetooth 5.3+ for better range & audio

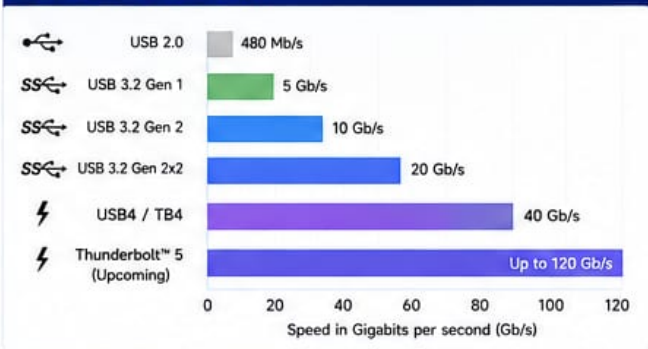
WHAT CAN THEY CONNECT?

USB4 / Thunderbolt 4	USB 3.2 Gen 2x2	USB-C (PD)	HDMI 2.1	DisplayPort 2.1	2.5G / 5G / 10G Ethernet	Wi-Fi 6E / Wi-Fi 7
Docks, monitors, external GPUs, storage, eGPUs, network adapters, and more.	Fast external SSDs/HDDs, cameras, smartphones, 10Gb adapters.	Laptops, tablets, phones, power banks, and accessories with power delivery.	TVs, monitors, projectors, soundbars, AV receivers, consoles, streaming devices.	High-resolution monitors, VR headsets, professional displays, daisy-chained setups.	Routers, switches, NAS, network attached storage, internet connections.	Wireless routers, smart home devices, AR/VR, Bluetooth audio & more.

COMMON CONNECTOR SHAPES

USB-A (3.x) 	USB-C 	Micro-USB 	Mini-DisplayPort 	HDMI 	DisplayPort
RJ45 Ethernet 	3.5mm Audio Jack 	Optical Audio (TOSLINK) 	USB-B (3.x) 	USB-C (PD) 	Thunderbolt™ 4

PORT SPEED COMPARISON



QUICK TIPS

- ✓ Always use certified cables for best performance.
- ✓ USB-C ports can support different features. Check your device specifications.
- ✓ For high refresh rates and resolutions, use HDMI 2.1 or DisplayPort 1.4/2.1.
- ✓ Use wired Ethernet for the most stable and fastest internet connection.
- ✓ Keep your drivers and firmware up to date for maximum compatibility.



ONE PORT CAN DO IT ALL!

Modern ports like USB-C, USB4, and Thunderbolt 4 combine data, video, audio, and power in a single connection—making your setup cleaner, faster, and more powerful.

SS = SuperSpeed USB PD = Power Delivery VRR = Variable Refresh Rate
eARC = Enhanced Audio Return Channel MST = Multi-Stream Transport

PORTS & CONNECTORS

The gateways that connect your devices and expand what your computer can do.



Transfer Data



Display Video



Deliver Power



Connect to Network



Transmit Audio



Read/Write Storage

1 USB-A



WHAT IT DOES



Transfers data and delivers power to your devices.

WHY IT'S USED



Connects older and common devices like keyboards, mice, flash drives and external hard drives.



2 USB-C



WHAT IT DOES



Transfers data, video, audio and power over a reversible plug.

WHY IT'S USED



One port for almost everything—charging, displays, storage, phones and more.



3 THUNDERBOLT 4/5



WHAT IT DOES



Ultra-fast data transfer, video output and power delivery—all in one.

WHY IT'S USED



For high-performance devices, multiple 4K/8K displays, fast storage and docking stations.



4 HDMI



WHAT IT DOES



Transmits high-definition video and audio to external displays.

WHY IT'S USED



Connects your computer to TVs, monitors, projectors and home theater systems.



5 DISPLAYPORT



WHAT IT DOES



Delivers high-resolution video and audio with high refresh rates.

WHY IT'S USED



Ideal for gaming monitors, professional screens and multi-monitor setups.



6 ETHERNET (RJ-45)



WHAT IT DOES



Connects your device to a wired network or the internet.

WHY IT'S USED



Provides faster, more stable and secure internet than Wi-Fi.



7 AUDIO JACK (3.5mm)



WHAT IT DOES



Transmits analog audio for input (mic) or output (headphones/speakers).

WHY IT'S USED



Connects headsets, microphones and speakers easily.



8 SD CARD READER



WHAT IT DOES



Reads and writes data from SD memory cards.

WHY IT'S USED



Quickly transfers photos, videos and files from cameras and devices.



WHY PORTS & CONNECTORS MATTER

They let your computer communicate with the outside world.

The right port makes the right connection—fast, clear and reliable.



Not all ports do the same thing.

Check the icon and label.



Look for speed symbols.

Thunderbolt ⚡ means high performance.



Quality cables make a difference.

Use good cables for best performance.

ONE PORT. MANY POSSIBILITIES.

Connect, create, communicate.



EXPANSION HARDWARE

Add more features, performance, and connectivity to your computer.

WHAT IS EXPANSION HARDWARE?

Expansion hardware are additional cards or modules installed in a computer to add or enhance capabilities beyond the built-in features of the motherboard.

KEY



PCIe Slot



M.2 Slot



USB Header / Internal

1. WI-FI CARD

Adds wireless internet connectivity to your PC.



Interface:

PCIe x1

- Wi-Fi 6 / Wi-Fi 6E / Wi-Fi 7
- High-speed wireless
- Better range and stability
- Supports 2.4GHz, 5GHz and 6GHz bands

Common Uses



Wireless Internet



Online Gaming



Streaming



Large File Downloads

2. BLUETOOTH MODULE

Adds Bluetooth connectivity to your PC.



Interface:

M.2 (Key E)
or USB

- Bluetooth 5.0 / 5.2 / 5.3+
- Connects mouse, keyboard, headphones, speakers, etc.
- Low power consumption
- Compact and easy to install

Common Uses



Wireless Peripherals



Headphones / Audio



File Transfer



IoT / Smart Devices

3. CAPTURE CARD

Captures video and audio from another device.



Interface:

PCIe x4
(or x1)

- Record or stream gameplay
- Supports HDMI / SDI / Component / Composite
- 4K / 1080p high quality
- Low latency passthrough

Common Uses



Game Streaming



Video Recording



Live Broadcasting



Content Creation

4. SOUND CARD

Enhances or adds high-quality audio to your system.



Interface:

PCIe x1

- Better audio quality
- 7.1 Surround sound
- Lower latency
- Dedicated audio processing

Common Uses



Gaming



Music Production



Home Theater



Voice Recording

5. RAID CONTROLLER

Manages multiple drives for better performance or redundancy.



Interface:

PCIe x4 / x8

- RAID 0, 1, 5, 10, etc.
- Hardware RAID (offloads CPU)
- More reliable data storage
- Supports SAS / SATA drives

Common Uses



Servers



Data Protection



Workstations



High-Speed Storage

6. THUNDERBOLT EXPANSION CARD

Adds Thunderbolt ports for ultra-fast data, video, and power.



Interface:

PCIe x4

- Thunderbolt 3 / 4 support
- Up to 40 Gbps bandwidth
- Supports 4K/8K displays, docks, NVMe drives
- Daisy-chain multiple devices

Common Uses



High-Speed Storage



4K / 8K Monitors



Thunderbolt Docks



Professional Workflows

7. USB EXPANSION CARD

Adds more USB ports to your PC.



Interface:

PCIe x1

- USB 3.0 / 3.1 / 3.2 / 3.2 Gen 2x2
- Add multiple USB ports
- Supports front panel USB
- Internal header support

Common Uses



More USB Ports



Peripherals

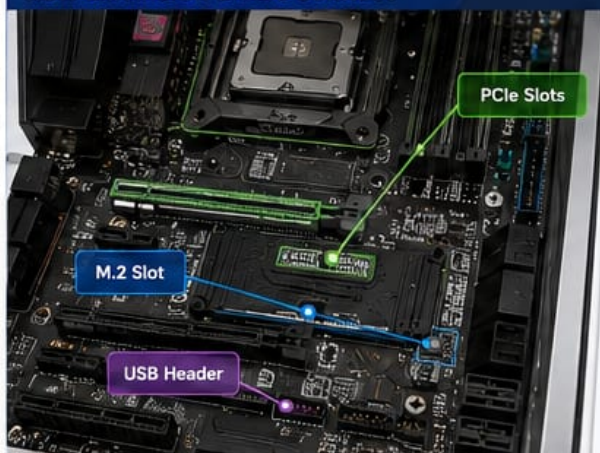


Front Panel Connections



Data Transfer

HOW EXPANSION CARDS CONNECT



EXPANSION SLOTS & CONNECTORS



PCIe (Peripheral Component Interconnect Express)

Used for high-performance cards like graphics cards, RAID controllers, sound cards, and more.



M.2 Slot (Key E / Key M)

Used for smaller modules like Wi-Fi, Bluetooth, and NVMe SSDs.



USB Header (Internal)

Connects USB ports to the front panel or adds internal USB support.



BENEFITS OF EXPANSION HARDWARE

- ✓ Extend system capabilities
- ✓ Improve performance
- ✓ Add modern connectivity
- ✓ Customise your PC for your needs
- ✓ Future-proof your system



TIPS

- Check compatibility (slot type, size, bandwidth).
- Install the latest drivers for best performance.
- Use quality cards from reputable brands.
- Follow ESD safety practices during installation.

INPUT DEVICES

The tools we use to input data and control our digital world



KEYBOARD



WHAT IT DOES



Lets you type letters, numbers and commands.

WHY IT'S USED



Essential for typing, coding, gaming and operating your computer.



MOUSE



WHAT IT DOES



Moves the cursor, selects items and opens programs.

WHY IT'S USED



Provides speed, precision and ease of navigation.



TOUCHPAD



WHAT IT DOES



Detects finger movement and gestures on a pad.

WHY IT'S USED



Built into laptops for convenient cursor control and gestures.



TOUCHSCREEN



WHAT IT DOES



Detects touch input directly on the screen.

WHY IT'S USED



Allows direct interaction—ideal for tablets, phones and kiosks.



GRAPHICS TABLET



WHAT IT DOES



Lets you draw, write or design with a stylus on a sensitive surface.

WHY IT'S USED



Perfect for digital artists, designers and photo editors.



WEBCAM



WHAT IT DOES



Captures video and streams it to your computer.

WHY IT'S USED



Used for video calls, online classes, meetings and live streaming.



MICROPHONE



WHAT IT DOES



Captures your voice and converts it to digital audio.

WHY IT'S USED



Used for calls, recording, voice commands and content creation.



FINGERPRINT READER



WHAT IT DOES



Scans your fingerprint to verify your identity.

WHY IT'S USED



Provides fast, secure access to devices and protects your data.



All these devices act as a bridge between you and your computer. They make it possible to communicate, create, control and get things done.



Communicate



Create



Control



Navigate



Secure

INPUT DEVICES

The tools we use to input data and control our digital world

Input devices send data and commands to your computer for processing.

Type Point Touch Capture Control

1 KEYBOARD



Types keys, numbers and commands.

BEST FOR

- Typing
- Gaming
- Shortcuts
- Type

2 MOUSE



Moves the pointer, selects items and opens programs.

BEST FOR

- Navigation
- Selection
- Gaming
- Point

3 TOUCHPAD



Built-in pointing device for cursor control and gestures.

BEST FOR

- Laptops
- Gestures
- Travel
- Point

4 TOUCHSCREEN



Detects touch input directly on the screen.

BEST FOR

- Tablets
- Kiosks
- Interactive apps
- Touch

5 GRAPHICS TABLET



Draw, write or design with a stylus on a sensitive surface.

BEST FOR

- Digital Art
- Design
- Photo Editing
- Point

6 WEBCAM



Captures video and streams it to your computer.

BEST FOR

- Video Calls
- Streaming
- Conferencing
- Capture

7 MICROPHONE



Captures your voice and converts it to digital audio.

BEST FOR

- Recording
- Voice Calls
- Streaming
- Touch

8 FINGERPRINT READER



Scans your fingerprint to verify your identity.

BEST FOR

- Security
- Login
- Access Control
- Control

9 STYLUS / DIGITAL PEN



Writes or draws on touchscreens or tablet devices.

BEST FOR

- Note Taking
- Drawing
- Signing Documents
- Control

10 TRACKBALL



Moves the pointer using a stationary ball.

BEST FOR

- Precision Work
- Ergonomic Use
- Limited Space
- Point

11 JOYSTICK



Controls movement and actions in games or simulations.

BEST FOR

- Flight Sims
- Racing Games
- Gaming
- Control

12 GAME CONTROLLER



Input device for controlling games and applications.

BEST FOR

- Console Gaming
- PC Gaming
- Emulators
- Control

13 LIGHT GUN



Detects screen position using light sensors.

BEST FOR

- Arcade Games
- Shooting Games
- Retro Gaming
- Control

14 SCANNER



Scans documents and images into your computer.

BEST FOR

- Documents
- Photos
- Archiving
- Capture

15 BARCODE / QR CODE READER



Reads barcodes or QR codes and inputs data.

BEST FOR

- Retail
- Inventory
- Check-ins
- Capture

16 SMART CARD READER



Reads data from smart cards and identity cards.

BEST FOR

- Security
- Banking
- ID Verification
- Control

17 MIDI KEYBOARD / MUSIC CONTROLLER



Inputs musical notes and controls to software.

BEST FOR

- Music Production
- Composition
- Live Performance
- Control

18 EYE TRACKER



Tracks eye movement and gaze direction.

BEST FOR

- Research
- Accessibility
- Gaming
- Control

19 MOTION SENSOR



Detects body movement and gestures.

BEST FOR

- Gaming
- Gesture Control
- Interaction
- Control

20 FORCE FEEDBACK WHEEL



Provides force feedback for realistic driving experience.

BEST FOR

- Racing Sims
- Driving Games
- Training
- Control

21 VOICE RECOGNITION DEVICE



Recognizes voice commands and inputs data.

BEST FOR

- Smart Home
- Automation
- Hands-free Control
- Capture

22 BIOMETRIC IRIS SCANNER



Scans your iris pattern for high-level security.

BEST FOR

- High Security
- Access Control
- Authentication
- Control

23 DATA GLOVE



Captures hand movements and finger positions.

BEST FOR

- VR / AR
- Research
- Animation
- Control

24 FOOT PEDAL / FOOT SWITCH



Activates commands using foot pressure.

BEST FOR

- Hands-free Use
- Accessibility
- Gaming / Control
- Control



WHY INPUT DEVICES MATTER

They are the bridge between you and your computer. Good input devices improve accuracy, speed, comfort and productivity.

HOW INPUT DEVICES WORK



YOU ACT
You press, move, touch or speak.



DEVICE CAPTURES
The input device captures the action.



DATA SENT
Data is sent to your computer.



COMPUTER PROCESSES
The computer understands and executes the command.



RESULT / OUTPUT
You see or hear the result on the output device.

CHOOSE WHAT FITS YOU

- For work: Comfort & efficiency
- For gaming: Precision & response
- For creativity: Control & sensitivity
- For accessibility: Ease & inclusion

★ From simple clicks to advanced controls, input devices power everything you do.



Choose the right input device for the right task!

OUTPUT DEVICES

The devices that present information from your computer to your senses



Output devices take digital data from your computer and convert it into something you can see, hear or experience.



See



Hear



Feel/Experience



1 MONITOR



Displays visual output such as text, images, videos and applications.

BEST FOR

- Work & Productivity
- Design & Editing
- Gaming



2 SPEAKERS



Plays audio from your computer, including music, videos, alerts and system sounds.

BEST FOR

- Music & Entertainment
- Movies & Videos
- Presentations



3 HEADPHONES



Delivers audio directly to your ears for a personal and immersive listening experience.

BEST FOR

- Focus & Privacy
- Gaming
- Live Streaming



4 PROJECTOR



Projects visual content onto a large surface such as a wall or screen.

BEST FOR

- Presentations
- Classrooms
- Home Cinema



5 VR HEADSET



Displays 3D visuals and creates an immersive virtual environment that surrounds you.

BEST FOR

- VR Gaming
- Training & Simulation
- Virtual Tours



6 PRINTER



Produces hard copies of digital documents and images on paper.

BEST FOR

- Documents
- Photos
- Reports



7 PLOTTER



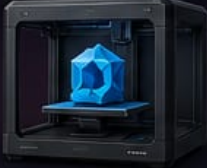
Creates large, precise prints for technical drawings, blueprints and posters.

BEST FOR

- Architecture
- Engineering
- CAD Drawings



8 3D PRINTER



Transforms digital 3D models into physical objects layer by layer.

BEST FOR

- Prototyping
- Product Design
- Education



9 LED DISPLAY / DIGITAL SIGNAGE



Displays bright, dynamic content suitable for public viewing in large spaces.

BEST FOR

- Advertising
- Events & Venues
- Information Displays



10 BRAILLE DISPLAY



Outputs text in Braille for users who are visually impaired.

BEST FOR

- Accessibility
- Education
- Inclusive Workspaces



11 HAPTIC FEEDBACK DEVICE



Provides touch feedback, vibrations or force to simulate the sense of touch.

BEST FOR

- Gaming
- Training & Simulation
- Virtual Reality



12 SMART GLASSES



Displays information in your field of view without blocking the real world.

BEST FOR

- Navigation
- Industrial Work
- Augmented Reality



13 HOLOGRAPHIC DISPLAY (EMERGING)



Projects 3D holograms in mid-air for a futuristic visual experience.

BEST FOR

- Advanced Presentations
- Museums & Exhibits
- Product Showcases



WHY OUTPUT DEVICES MATTER

They turn data into meaningful experiences. The right output device improves understanding, engagement and productivity.



VISION

See images, videos and information.



HEARING

Hear sound, music and voice.



TOUCH

Feel feedback, vibrations and interactions.



IMMERSION

Experience virtual and real-world environments.

UNUSUAL BUT POWERFUL



TACTILE DISPLAY

Uses pins to create touch sensations on the skin to display images or text.



SCENT OUTPUT DEVICE

Releases scents to enhance digital experiences (e.g. in gaming or VR).



★ From simple screens to immersive technologies, output devices bring digital content to life.



Choose the right output device for the right experience!

NETWORKING HARDWARE 2026

The Latest & Best Devices – What They Do and Why They Matter

🌐 The backbone of a fast, secure and always-connected world.

1

ETHERNET CONTROLLER



Built into the motherboard or server.

WHAT IT DOES

Handles wired network communication between your device and the network.

WHY IT'S USED

Provides fast, reliable and efficient data transfer for everything connected to a network.



EXAMPLE (2026)
Intel Ethernet 800 Series

2

NETWORK INTERFACE CARD (NIC)



Add-on card that connects a device to a network.

WHAT IT DOES

Adds high-speed wired connectivity to desktops, servers or workstations.

WHY IT'S USED

Boosts performance, adds advanced features and more reliability than built-in ports.



EXAMPLE (2026)
NVIDIA ConnectX-7

3

WI-FI 7 ADAPTER



Connects your device to wireless networks.

WHAT IT DOES

Delivers ultra-fast Wi-Fi, lower latency and more stable connections.

WHY IT'S USED

Perfect for gaming, streaming, AR/VR, remote work and smart homes.



EXAMPLE (2026)
Intel BE200
Wi-Fi 7 Adapter

4

NETWORK SWITCH



Connects multiple devices within a network.

WHAT IT DOES

Directs data between devices on a local network (LAN) quickly and efficiently.

WHY IT'S USED

Keeps your network organized, fast and running smoothly as your devices grow.



EXAMPLE (2026)
Cisco Catalyst 9600X
Series

5

ROUTER



Connects your network to the internet and other networks.

WHAT IT DOES

Routes data between different networks and protects your network with built-in security.

WHY IT'S USED

Shares internet securely, manages traffic and connects offices or branches together.



EXAMPLE (2026)
Cisco Catalyst 8300
Series

6

MODEM



Connects your internet service to your network.

WHAT IT DOES

Converts the internet signal from your ISP (cable, fiber, 5G, DSL) so your network can use it.

WHY IT'S USED

Provides the link between your home or office and your internet provider.



EXAMPLE (2026)
Arris SURFboard
S34 DOCSIS 4.0

7

ACCESS POINT



Creates Wi-Fi coverage for wireless devices.

WHAT IT DOES

Broadcasts Wi-Fi signal and connects many wireless devices seamlessly.

WHY IT'S USED

Expands Wi-Fi coverage in homes, offices and large spaces with strong, reliable signal.



EXAMPLE (2026)
HPE Aruba AP-775
Wi-Fi 7 (802.11be)

WHY THIS HARDWARE MATTERS TO YOU



SPEED

Get faster downloads, smarter streaming and smoother gaming.



RELIABILITY

Stay connected with stable, consistent performance.



SECURITY

Protect your data with built-in security and threat defense.



SCALABILITY

Easily support more devices, users and future growth.



PERFORMANCE

Built to handle today's demands and tomorrow's innovations.



EFFICIENCY

Lower latency, less power usage, better experience.

BUILT FOR THE FUTURE

Wi-Fi 7, 10G/25G/100G speeds, AI-driven networks, and advanced security make these devices future-ready.



Together, these devices create a powerful network that keeps you connected, productive and secure.



For homes, businesses, schools and data centers.

INTERNAL CONNECTIONS

The hidden links inside your computer that make everything work together.



Transfer Data



Deliver Power



Enable Cooling



Provide Control



Ensure Everything Works in Sync

1 PCI EXPRESS (PCIe)



WHAT IT DOES

Provides a high-speed connection for expansion cards like GPUs, network cards, and SSD adapters.

WHY IT'S USED

Offers fast data transfer and bandwidth for high-performance devices.



Data

2 SATA CABLE



WHAT IT DOES

Connects storage devices (HDDs, SSDs, optical drives) to the motherboard.

WHY IT'S USED

Transfers data between storage devices and your computer.



Data

3 ATX POWER CONNECTOR



WHAT IT DOES

Supplies main power from the power supply to the motherboard.

WHY IT'S USED

Powers the motherboard and enables all components to operate.



Power

4 EPS CPU POWER CONNECTOR



WHAT IT DOES

Provides dedicated power to the CPU through the motherboard.

WHY IT'S USED

Ensures stable power delivery for the processor, especially under heavy load.



Power

5 GPU POWER CONNECTOR (12V-2x6 / PCIe)



WHAT IT DOES

Delivers high power to the graphics card for maximum performance.

WHY IT'S USED

Provides the power needed for demanding GPUs and high-end graphics.



Power

6 FRONT PANEL CONNECTORS



WHAT IT DOES

Connects case buttons and LEDs (power, reset, HDD LED, power LED).

WHY IT'S USED

Allows you to control and monitor your PC from the case.



Control

7 USB INTERNAL HEADER



WHAT IT DOES

Connects front panel USB ports to the motherboard.

WHY IT'S USED

Enables front USB ports to transfer data to and from your system.



Data

8 FAN HEADERS



WHAT IT DOES

Connects case and CPU fans to the motherboard.

WHY IT'S USED

Allows the motherboard to control fan speed for optimal cooling and quiet operation.



Cooling



WHY INTERNAL CONNECTIONS MATTER

They ensure data moves fast, power flows safely, and your system stays cool and stable.

Good connections = Better performance, reliability and a longer-lasting PC.

THINGS TO REMEMBER



Proper Connection
Always connect cables securely.



Cable Management
Keep cables tidy for better airflow and easy troubleshooting.



Check Compatibility
Use the right connector for the right port.



Handle with Care
Don't force connectors. It can damage pins or components.

BUILT TO WORK TOGETHER



PSU



Motherboard



GPU



Fans



Storage

Power, data and control — all connected inside.



From cables to connectors — small details that power big performance.



Understand your connections. Build better. Troubleshoot smarter.

SOFTWARE MODULES: THE PROGRAMS THAT POWER YOUR PC

Software tells the hardware what to do and helps us work, learn, create and play.



Makes hardware useful
Software turns hardware into powerful tools.



Improves productivity
Helps you get more done faster and easier.



Enhances security
Protects your data, privacy and system.



Enables communication
Connects you with people and the world.



Drives innovation
New software creates new possibilities every day.

1 OPERATING SYSTEM (OS)



Manages hardware, system resources and provides common services for applications.

EXAMPLES

Windows 11, macOS, Ubuntu, Android, iOS

KEY FEATURES



2 APPLICATION SOFTWARE



Programs designed to help users perform specific tasks and solve problems.

EXAMPLES

Microsoft Office, Chrome, Photoshop, VLC

KEY FEATURES



3 UTILITY SOFTWARE



Maintains, optimizes and protects the system for smooth performance.

EXAMPLES

Antivirus, Backup Tools, Disk Cleanup, Compression Tools

KEY FEATURES



4 PROGRAMMING SOFTWARE



Tools used by developers to create, test, debug and maintain software.

EXAMPLES

Visual Studio Code, IntelliJ IDEA, Python, GCC, Git

KEY FEATURES



5 DRIVER SOFTWARE

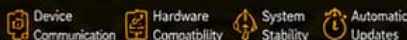


Acts as a translator between the operating system and hardware devices.

EXAMPLES

NVIDIA GPU Driver, Realtek Audio Driver, Printer Drivers

KEY FEATURES



6 FIRMWARE



Low-level software stored in ROM/Flash that controls hardware before the OS loads.

EXAMPLES

BIOS, UEFI, Embedded Firmware

KEY FEATURES



7 MIDDLEWARE



Connects different software components and enables them to work together.

EXAMPLES

Database Managers, Web Servers, Message Brokers

KEY FEATURES



8 DATABASE SOFTWARE



Stores, organizes, retrieves and manages large amounts of structured data.

EXAMPLES

MySQL, Microsoft SQL Server, Oracle DB, MongoDB

KEY FEATURES



9 CLOUD SOFTWARE / SaaS



Software hosted on the internet and accessed from anywhere.

EXAMPLES

Google Workspace, Microsoft 365, Dropbox, Salesforce

KEY FEATURES



10 GAME SOFTWARE



Interactive entertainment software designed for fun and engagement.

EXAMPLES

Minecraft, Fortnite, EA Sports FC, Steam

KEY FEATURES



HOW SOFTWARE WORKS TOGETHER



You give a command (click, type or tap) → Application Software receives your request → Operating System processes and manages the request → Drivers & Firmware communicate with hardware → Data is stored or retrieved (if needed) → Result is shown back to you

TYPES OF SOFTWARE



System Software
Runs the computer and manages hardware.
Examples: OS, Drivers, Utilities, Firmware



Application Software
Helps users complete tasks and solve problems.
Examples: Office, Browsers, Media Players, Games



Developer Software
Helps developers build, test and maintain programs.
Examples: IDEs, Compilers, Debuggers, Tools

WHY SOFTWARE MATTERS

- ✓ Turns hardware into useful tools
- ✓ Makes tasks easier, faster and safer
- ✓ Enables creativity, learning and communication
- ✓ Keeps systems running smoothly
- ✓ Drives technology forward



Great software + reliable hardware = Powerful experiences



Software is everywhere — in your PC, your phone, and your future.



Keep learning. Keep building. The possibilities are endless!

OPERATING SYSTEMS (SYSTEM SOFTWARE)

The foundation that manages hardware and provides services for applications and users.



SYSTEM SOFTWARE
Runs the computer and manages resources.



MANAGES HARDWARE
CPU, memory, storage, devices and more.



PROVIDES SERVICES
File management, security, networking, UI and more.



RUNS APPLICATIONS
Provides a platform for apps and programs.



MULTI-USER SUPPORT
Allows multiple users to work securely.



KEEPS SYSTEM STABLE
Ensures reliability, updates and system protection.

WHAT IS AN OPERATING SYSTEM?

An Operating System (OS) is system software that acts as an intermediary between users, applications, and computer hardware.

It manages resources, provides services, and allows applications to run efficiently and securely.



USERS



APPLICATIONS



OPERATING SYSTEM



HARDWARE

MAIN FUNCTIONS

- ✓ Manages hardware resources
- ✓ Provides user interface
- ✓ Runs and manages applications
- ✓ Handles files and storage
- ✓ Provides security and protection
- ✓ Enables networking and communication

Microsoft Windows 11



- Modern, secure and user-friendly OS for PCs and laptops.
- Designed for productivity, gaming and creativity.

Features: Snap Layouts, Widgets, DirectStorage, Copilot in Windows.

Used in: Home, Office, Education, Gaming, Business.

Windows Server 2025



- Enterprise server operating system for managing networks, users and applications.
- Built-in security, virtualization and cloud integration.

Features: Hotpatching, Windows Containers, Azure Arc, SMB over QUIC.

Used in: Data Centers, Cloud, Enterprise Environments.

Linux



- Open-source OS kernel and family of operating systems.
- Highly secure, stable and customizable.
- Features: Multi-user, Multi-tasking, Strong permissions.
- Used in: Servers, Supercomputers, Cloud, Development.

Ubuntu



- Popular Linux distribution (based on Debian).
- User-friendly, secure and community supported.
- Features: Ubuntu Software Center, Snap packages, Long Term Support.
- Used in: Desktop, Server, Cloud, IoT, Development.

Red Hat Enterprise Linux (RHEL)



- Enterprise-grade Linux distribution from Red Hat.
- High performance, security and long-term support.
- Features: SELinux, KVM, Containers, Automation.
- Used in: Enterprises, Cloud, Finance, Government.

macOS



- Operating system for Apple Mac computers.
- Seamless integration with Apple ecosystem.
- Features:** Safari, Handoff, Time Machine, Gatekeeper.
- Used in:** Creative Work, Education, Business, Development.

Android



- Mobile operating system by Google for smartphones and tablets.
- Open-source based on Linux kernel.
- Features:** Google Play Store, Widgets, Multi-tasking.
- Used in:** Mobile Devices, Tablets, Smart TVs, Wearables.

iOS



- Mobile operating system by Apple for iPhone.
- Secure, smooth and optimized for Apple hardware.
- Features:** App Store, Face ID, iMessage, AirDrop.
- Used in:** iPhones, iPod Touch.

ChromeOS



- Lightweight OS by Google for Chromebooks.
- Fast, secure and cloud-first.
- Features:** Automatic updates, Google Apps, Sandbox security.
- Used in:** Education, Cloud Computing, Web-based Tasks.

10 Virtual Operating Systems



- Run multiple OS environments on a single physical machine.
- Ideal for testing, development and server consolidation.
- Examples:** VMware, VirtualBox, Hyper-V, Parallels.
- Used in:** Developers, IT Labs, Testing, Cloud, Businesses.

POPULAR OS COMPARISON

OS	Best For	Security	Ease of Use	Customization	Cost
Windows 11	General Use	★★★★	★★★★	★★★★	Paid
Windows Server 2025	Servers / Enterprise	★★★★	★★★★	★★★★	Paid
Linux (General)	Servers / Dev / Cloud	★★★★	★★★★	★★★★	Free
macOS	Creative / Business	★★★★	★★★★	★★★★	Paid
Android	Mobile / Tablets	★★★★	★★★★	★★★★	Free
iOS	Mobile (iPhone)	★★★★	★★★★	★★★★	Paid
ChromeOS	Education / Web	★★★★	★★★★	★★★★	Free

HOW AN OS WORKS (HIGH LEVEL)



OS CORE COMPONENTS



WHERE OS ARE USED

- Personal Computers (Laptops, Desktops)
- Servers & Data Centers
- Mobile Devices (Smartphones, Tablets)
- Embedded Systems (Smart TVs, IoT, Cars)
- Cloud Computing (Azure, AWS, Google Cloud)



DID YOU KNOW?

The first widely used OS for personal computers was MS-DOS (1981). Today, billions of devices run on different operating systems tailored for their needs!



KNOWLEDGE CHECK

Which operating system is open-source, highly secure, and widely used in servers and cloud environments?

- A) Windows 11 B) macOS C) Linux D) iOS

OPERATING SYSTEM COMPONENTS

The building blocks that work together to run your computer and applications.

1 KERNEL



What is it?
The core of the operating system. It manages hardware resources and provides essential services for programs.

How it works



Example
Windows NT Kernel, Linux Kernel

2 USER INTERFACE (GUI)



What is it?
The graphical interface that allows users to interact with the computer using windows, icons, menus and pointers.

How it works



Example
Windows 11 Desktop, GNOME (Linux), macOS Finder

3 COMMAND LINE INTERFACE (CLI)



What is it?
A text-based interface where users type commands to perform tasks.



Example
Command Prompt (Windows), Bash (Linux)

4 FILE SYSTEM



What is it?
Organizes and stores data on storage devices in files and directories.

How it works



Example
NTFS (Windows), ext4 (Linux), APFS (macOS)

5 DEVICE DRIVERS



What is it?
Software that allows the operating system to communicate with hardware devices.

How it works



Example
NVIDIA Graphics Driver, Intel Chipset Driver

6 PROCESS MANAGER

Name	CPU	Memory
App 1	2.1%	120 MB
App 2	1.3%	80 MB
App 3	0.6%	60 MB
Background Process	0.2%	30 MB
System	1.1%	50 MB

What is it?
Manages running programs (processes), allocates CPU time and terminates processes when needed.

How it works



Example
Windows Task Manager, Linux top/htop

7 MEMORY MANAGER



What is it?
Controls how memory (RAM) is allocated and deallocated to running processes.

How it works



Example
Windows Memory Management, Linux Kernel MM

8 TASK SCHEDULER



What is it?
Schedules tasks and processes to run at specific times or events.

How it works



Example
Windows Task Scheduler, cron (Linux)

9 REGISTRY (WINDOWS)



What is it?
A hierarchical database that stores system settings and options in Windows.

How it works



Example
Windows Registry Editor (regedit)

10 SERVICES



What is it?
Background programs that run without user interaction to perform system tasks.

How it works



Example
Windows Services (services.msc), systemd (Linux)

11 WINDOWS REGISTRY VS LINUX CONFIGURATION



WINDOWS REGISTRY

Centralized hierarchical database storing settings for system, applications, hardware and users.



Stored in:
C:\Windows\System32\config
Edited with:
Registry Editor (regedit)
Example Use:
Software settings, hardware config, user preferences

VS



LINUX CONFIGURATION

Decentralized text-based configuration files stored in directories.



Stored in:
/etc/ and other directories
Edited with:
Text Editors (nano, vim)
Example Use:
System settings, services, network, users

HOW THEY WORK TOGETHER



Applications

Operating System Components



Hardware

- ✓ The kernel communicates with hardware through drivers.
- ✓ System services run in the background to keep everything working.
- ✓ Files are stored and organized by the file system.
- ✓ Processes, memory, and tasks are managed for smooth performance.
- ✓ Settings and configurations are stored securely.

WHY IT MATTERS



Makes the system stable and reliable



Allows applications to run efficiently



Helps manage resources like CPU, memory, and storage



Provides security and user management



Enables customization and automation

KEY TAKEAWAY



An operating system is a collection of components that work together invisibly behind the scenes to make your computer usable, powerful and reliable.



KNOWLEDGE CHECK

Which component allows the OS to communicate with a printer?

- A) Kernel B) Device Driver C) File System D) Registry

KERNEL vs REGISTRY (Windows)

Two different parts of the operating system that work together to keep your computer running.

KERNEL – The Brain and Traffic Controller

The kernel is the core (heart) of the operating system.



WHAT IS THE KERNEL?

It is the first major part of the OS that starts after boot and stays in memory while the computer is running. Without the kernel, the OS cannot work.

WHAT DOES THE KERNEL DO?

The kernel is the manager of the computer.



CPU
Management



Memory
Management



Device
Management



File
Management



Security &
Protection

IMAGINE A SCHOOL

The computer is a school.

- Students = Programs
- Teachers = Applications
- Classrooms = RAM
- Library = Storage
- Principal = Kernel

The principal decides who goes where, who may enter, who must wait, and keeps everything running smoothly.



MAIN JOBS OF THE KERNEL

- Decides which program uses the CPU.
- Allocates RAM to programs and protects them from each other.
- Communicates with hardware through device drivers.
- Loads files from storage and manages file access.
- Protects the system, files and resources with security rules.

HOW THE KERNEL WORKS (EXAMPLE)



KEY TAKEAWAY (KERNEL)

The kernel is the manager (in memory), and makes Windows powerful and runs smoothly.

HOW THEY WORK TOGETHER

Example: You connect a new printer

- 1 Windows detects the new printer.
↓
- 2 The Kernel communicates with the printer through the device driver.
↓
- 3 Windows stores the printer configuration in the Registry (name, settings, defaults).
↓
- 4 Next time you print: Kernel controls the hardware communication and the Registry provides the saved settings.
↓

KERNEL vs REGISTRY

KERNEL	VS	REGISTRY
Part of the OS core		Database of settings
Runs continuously in memory		Stored on disk and loaded as needed
Controls CPU, RAM, devices and hardware		Stores configuration information
Talks directly to hardware		Stores information about hardware and software
Required for Windows to run		Windows can boot even if some registry entries are damaged
Executes instructions		Stores information

REGISTRY – The Notebook or Settings Database

The Windows Registry is a large database of configuration settings.



WHAT IS THE REGISTRY?

It stores information about Windows, programs, hardware, and user preferences. Think of it as Windows' master settings book.

WHAT DOES THE REGISTRY STORE?



Windows Settings
Desktop background, time zone, language, display resolution and more.



Installed Programs
Where programs are installed, versions, licenses, and file associations.



Hardware Settings
Printers, monitors, keyboard layout, sound devices and more.



User Accounts
Desktop icons, taskbar settings, themes, accessibility options. Each user can have different settings.



System Configuration
Startup programs, services, hardware profiles, security policies and more.

WHERE IS THE REGISTRY?

The Registry is stored on disk in special files called registry hives.



When Windows starts, these hives are loaded into memory so Windows and applications can quickly read and update settings.

REGISTRY STRUCTURE

The Registry looks like folders.



Each folder contains Keys, Subkeys, and Values (similar to folders and files).

KEY TAKEAWAY (REGISTRY)

The Registry stores information! It is a database of settings that tells Windows and programs how things should work.



QUICK MEMORY TIP

Kernel = Does the work
Registry = Stores the information

Kernel is the brain (manager).

Registry is the notebook (settings).

Together they make Windows run perfectly!



Kernel
(Does the work)

+



Registry
(Stores information)

=



Windows Works Smoothly!



DID YOU KNOW?

The kernel is different for every operating system (Windows, Linux, macOS), but the registry is unique to Windows.

KERNEL vs REGISTRY (Windows)

Two different parts of the operating system that work together to keep your computer running.

KERNEL – The Brain and Traffic Controller

The kernel is the core (heart) of the operating system.

It is the first major part of the OS that starts after boot and stays in memory while the computer is running.

Without the kernel, the OS cannot work.



WHAT DOES THE KERNEL DO?

The kernel is the manager of the computer.



Imagine a School

The computer is a school.

- Students = Programs
- Teachers = Applications
- Classrooms = RAM
- Library = Storage
- Principal = Kernel



The principal decides who goes where, who may enter, who must wait, and keeps everything running smoothly.

MAIN JOBS OF THE KERNEL

- CPU Management**
Decides which program uses the CPU and for how long.
- Memory Management**
Allocates RAM to programs and protects them from each other.
- Device Management**
Communicates with hardware through device drivers.
- File Management**
Finds files on storage, loads them to RAM, and saves changes.
- Security & Protection**
Controls access to resources, files, and system permissions.

HOW THE KERNEL WORKS (EXAMPLE)



KEY TAKEAWAY (KERNEL)
The kernel does the work!

The kernel is the manager (in memory).
It runs in memory, controls hardware, and manages how they make Windows powerful and reliable.

HOW THEY WORK TOGETHER

Example: You connect a new printer

- Windows detects the printer.
- Kernel communicates with the printer through the device driver.
- Windows stores the printer configuration in the Registry (name, settings, defaults).
- Next time you print: Kernel controls the hardware communication, and the Registry provides the saved settings.

KERNEL vs REGISTRY

KERNEL	VS	REGISTRY
Part of the operating system core		Database of settings
Runs continuously in memory		Stored on disk and loaded as needed
Controls CPU, RAM and hardware		Stores configuration information
Talks directly to hardware		Stores settings for hardware and software
Required for Windows to run		Windows can still boot even if some registry entries are damaged, though problems may occur
Executes instructions		Stores information

REGISTRY – The Notebook or Settings Database

The Windows Registry is a large database of configuration settings.



It stores information about Windows, programs, hardware, and user preferences. Think of it as Windows' master settings book.

WHAT DOES THE REGISTRY STORE?

- Windows Settings**
Desktop background, time zone, language, display resolution, and more.
- Installed Programs**
Where programs are installed, versions, licenses, and file associations.
- Hardware Settings**
Printers, monitors, keyboard layout, sound devices, and more.
- User Accounts**
Desktop icons, taskbar settings, themes, accessibility options. Each user can have different settings.
- System Configuration**
Startup programs, services, hardware profiles, security policies, and more.

WHERE IS THE REGISTRY?

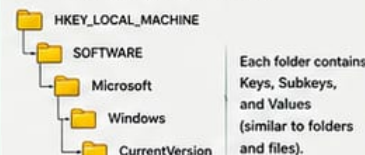
The Registry is stored on disk in special files called registry hives.



When Windows starts, these hives are loaded into memory so Windows and applications can quickly read and update settings.

REGISTRY STRUCTURE

The Registry looks like folders.



KEY TAKEAWAY (REGISTRY)

The Registry stores information!
It is a database of settings that tells Windows and programs how things should work.



QUICK MEMORY TIP

Kernel = Does the work
Registry = Stores the information

SUMMARY

Kernel is the manager (in memory).
Registry is the notebook (on disk).
Together they make Windows powerful and reliable.



Kernel
(Brain / Manager)

+



Registry
(Notebook / Database)

=



Windows Works
Smoothly!



DID YOU KNOW?

The kernel is different for every operating system (Windows, Linux, macOS), but the registry is unique to Windows.

WHEN WOULD YOU WANT TO USE THE **REGISTRY**?

The Windows Registry is a powerful database that stores settings for Windows, hardware, software, and users.



Use the Registry only when there isn't a safer or easier option in Windows Settings or Control Panel.
Incorrect Registry changes can cause serious problems. Always back up the Registry or create a System Restore point before making changes.



**Backup First,
Stay Safe!**



1 CHANGE HIDDEN WINDOWS SETTINGS

Some Windows features aren't available in Settings but can be enabled or disabled through the Registry.

Examples:

Enable/disable Windows features, customize File Explorer, change login behavior, taskbar, Start menu and more.



2 FIX WINDOWS PROBLEMS

If Windows or a program isn't working correctly, the Registry may contain incorrect settings.

Examples:

Repair broken file associations, reset corrupted program configurations, fix login or Windows Update issues.



3 MANAGE STARTUP PROGRAMS

Windows stores information about programs that start automatically.

Example:

Remove an unwanted application that starts every time you log in.



4 CONFIGURE HARDWARE

Windows saves settings for many hardware devices in the Registry.

Examples:

Printers, monitors, sound devices, keyboard layouts, mouse settings and more.



5 APPLY COMPANY POLICIES

IT departments use the Registry to enforce rules and standardize computers.

Examples:

Disable Control Panel, prevent USB storage, restrict software installation, block features.



**WINDOWS
REGISTRY**

REAL-WORLD EXAMPLE

An IT administrator configures 100 new laptops by deploying Registry settings to:

- ✓ Set company wallpaper
- ✓ Enable Windows Defender
- ✓ Disable USB storage
- ✓ Add security settings
- ✓ Configure Microsoft Edge
- ✓ Hide unnecessary features

Saves many hours of manual work!

6 SOFTWARE DEVELOPMENT

Programmers store application settings in the Registry.

Examples:

Installation path, license information, user preferences, recent files, application configuration.



7 TROUBLESHOOTING

Support technicians check the Registry to diagnose issues.

Examples:

Missing drivers, corrupted software settings, incorrect network configuration, performance issues.



8 RESTORE SETTINGS

The Registry can be backed up before making changes.

Examples:

Restore a Registry backup, or use a System Restore point to recover settings.



9 AUTOMATE WINDOWS CONFIGURATION

Organizations deploy the same Registry settings to many computers.

Example:

Configure hundreds of PCs with the same wallpaper, security settings, and network options.



10 SECURITY HARDENING

Administrators use the Registry to improve security.

Examples:

Disable insecure features, enable advanced security options, configure Windows Defender, restrict storage.



WHEN TO USE THE REGISTRY

- ✓ Change a hidden Windows setting
- ✓ Troubleshoot Windows problems
- ✓ Fix a corrupted Windows setting
- ✓ Store application settings (developers)
- ✓ Configure company computers
- ✓ Apply advanced security settings



WHEN NOT TO USE THE REGISTRY

- ✗ Install everyday software
- ✗ Change Wi-Fi password
- ✗ Change desktop wallpaper
- ✗ Use normal Settings or Control Panel
- ✗ Change screen brightness
- ✗ Do not modify unless you understand



KEY TAKEAWAY

The Registry is the "behind-the-scenes" configuration database of Windows. Use it wisely, back it up, and it can be a powerful tool for control, automation, and troubleshooting.



QUICK TIP

- Before making any changes:
- Back up the Registry
 - Create a System Restore point
 - Know what the setting does
 - Change only what you must



IMPORTANT: The Registry is powerful but sensitive. A wrong change can cause instability or prevent Windows from starting. Always be careful!

Safe. Smart. Secure.

Learn • Understand • Configure • Protect



FIRMWARE & LOW-LEVEL SOFTWARE 2026

The invisible software that powers and controls hardware.

Firmware is specialized software embedded in hardware that provides core control, initializes devices, and enables communication between hardware and operating systems.

WHY IT MATTERS



Enables & Controls Hardware



Security & Trust Foundation



Performance & Stability



Updates & Compatibility



Critical for Business Continuity

1 BIOS

Basic Input/Output System



Legacy firmware that initializes hardware and starts the boot process.

Key Functions

- Power-On Self Test (POST)
- Hardware initialization
- Boot device selection
- Basic system configuration

2 UEFI

Unified Extensible Firmware Interface



Modern firmware interface that replaces BIOS with advanced features and security.

Key Functions

- Faster boot and shutdown
- Supports large drives (>2TB)
- Secure Boot and TPM integration
- Graphical interface & mouse support

3 Device Firmware (General)



Firmware embedded in individual devices to control their specific functions.

Key Functions

- Device initialization & control
- Manage device features
- Ensure hardware compatibility
- Provide bug fixes & improvements

4 SSD Firmware

Solid State Drive Firmware



Controls SSD operations, flash memory management and data reliability.

Key Functions

- Wear leveling & garbage collection
- Error correction & data integrity
- Performance optimization
- Self-monitoring (SMART)

5 GPU Firmware (VBIOS)

Video BIOS



Firmware on the graphics card that controls hardware operation and settings.

Key Functions

- Initialize GPU hardware
- Manage clocks, voltages & fans
- Display output configuration
- Ensure stability & compatibility

6 Motherboard Firmware (Board Firmware)



Firmware that manages the motherboard's components and built-in devices.

Key Functions

- Chipset & component control
- Power management
- Hardware monitoring
- Feature enablement & updates

7 TPM Firmware

Trusted Platform Module Firmware



Firmware that runs on the TPM chip to provide hardware-based security.

Key Functions

- Secure key generation & storage
- Platform integrity measurement
- Encryption & authentication
- Secure boot support

8 Embedded Controllers (EC Firmware)



Firmware in embedded controllers that manage low-level system functions.

Key Functions

- Power button & sleep control
- Keyboard, fan & thermal control
- Battery & charging management
- Sensor monitoring

9 Secure Boot (UEFI Feature)



Security feature that ensures only trusted firmware and OS components run.

Key Functions

- Verifies digital signatures
- Blocks malicious bootloaders
- Protects boot process integrity
- Works with TPM for trust

10 Firmware Updates



Updates to firmware that add features, fix bugs and close security issues.

Key Functions

- Bug fixes & stability improvements
- Security vulnerability patches
- Performance enhancements
- Hardware compatibility updates

ACRONYM EXPLANATIONS

BIOS – Basic Input/Output System
UEFI – Unified Extensible Firmware Interface
SSD – Solid State Drive
GPU – Graphics Processing Unit
VBIOS – Video BIOS (Firmware on GPU)

TPM – Trusted Platform Module
POST – Power-On Self Test
SMART – Self-Monitoring, Analysis and Reporting Technology

COMMONLY USED BY ORGANIZATIONS LIKE INVESTEC FOR



System Security



Hardware Performance



Data Integrity



System Reliability



Regulatory Compliance

BEST PRACTICES

- ✓ Keep firmware up to date from official sources.
- ✓ Enable Secure Boot and TPM for stronger security.
- ✓ Test updates in a controlled environment first.
- ✓ Monitor release notes for fixes and known issues.
- ✓ Ensure power stability during firmware updates.



IMPORTANT: Incorrect firmware updates can cause system instability or hardware failure. Always follow vendor instructions and backup configurations.



PRO TIP: Regular firmware maintenance improves security, stability, performance and hardware lifespan.



2026 Stay updated. Stay secure. Stay productive.



APPLICATION SOFTWARE

Application software are programs designed to help users perform specific tasks. They run on top of system software (like Windows) and make computers more useful and productive.

WHY APPLICATION SOFTWARE MATTERS



Helps you be productive



Improves communication



Enhances creativity



Provides access to information



Entertainment and fun

PRODUCTIVITY & OFFICE

- Microsoft Word**
Create and edit documents
- Microsoft Excel**
Spreadsheets and data analysis
- Microsoft PowerPoint**
Create presentations
- Microsoft Outlook**
Email and calendar management
- Microsoft OneNote**
Note-taking and organization
- Google Docs**
Online documents
- Google Sheets**
Online spreadsheets
- LibreOffice**
Open-source office suite

COMMUNICATION & COLLABORATION

- Microsoft Teams**
Chat, meetings, calls, collaboration
- Zoom**
Video conferencing
- Slack**
Team messaging
- Discord**
Community chat and voice
- Skype**
Voice and video calls
- WhatsApp Desktop**
Messaging and file sharing
- Telegram Desktop**
Cloud-based messaging
- Mailbird**
Email management

CREATIVE & DESIGN

- Adobe Photoshop**
Image editing and design
- Adobe Illustrator**
Vector graphics and illustration
- Adobe InDesign**
Page layout and publishing
- CorelDRAW**
Graphic design
- Canva**
Online design and templates
- GIMP**
Free image editing
- Figma**
UI/UX design and prototyping
- Blender**
3D modeling and animation

MULTIMEDIA & ENTERTAINMENT

- Adobe Premiere Pro**
Video editing
- Adobe After Effects**
Motion graphics and VFX
- DaVinci Resolve**
Video editing and color grading
- CyberLink PowerDirector**
Easy video editing
- Audacity**
Audio recording and editing
- Spotify**
Music streaming
- iTunes / Apple Music**
Music library and player
- Netflix / YouTube**
Video streaming

WEB BROWSERS

- Google Chrome**
Fast and popular web browser
- Microsoft Edge**
Web browser by Microsoft
- Mozilla Firefox**
Secure and customizable
- Opera**
Feature-rich browser
- Brave**
Privacy-focused browser
- Vivaldi**
Customizable and powerful
- Tor Browser**
Anonymous browsing
- Safari**
Web browser by Apple

DATA MANAGEMENT & ANALYSIS

- Microsoft Access**
Database management
- MySQL Workbench**
Database design and admin
- Tableau**
Data visualization
- Power BI Desktop**
Business intelligence
- SPSS Statistics**
Statistical analysis
- RStudio**
Data analysis with R

DEVELOPMENT & PROGRAMMING

- Visual Studio Code**
Code editing
- Visual Studio 2022**
IDE for development
- PyCharm**
Python development
- IntelliJ IDEA**
Java and Kotlin IDE
- Eclipse IDE**
Multi-language IDE
- XAMPP**
Local web development

EDUCATION & REFERENCE

- Microsoft OneNote**
Digital note-taking
- Grammarly**
Writing assistant
- Dictionary.com**
Online dictionary
- Wolfram Alpha**
Computational knowledge
- GeoGebra**
Math and science tools
- Zotero**
Research and citation

PROUOCATIVITY TOOLS

- Notion**
All-in-one workspace
- Evernote**
Notes and organization
- Trello**
Project management
- Todoist**
Task management
- AnyDesk**
Remote desktop access
- 7-Zip**
File compression

UTILITIES & SYSTEM TOOLS

- CCleaner**
System cleaning
- WinRAR**
File archiver
- Malwarebytes**
Malware protection
- Revo Uninstaller**
Uninstall programs
- Everything**
Fast file search
- ShareX**
Screenshots and sharing

OTHER POPULAR APPLICATIONS

- Adobe Acrobat Reader**
View and annotate PDFs
- VLC Media Player**
Play almost any media file
- uTorrent Web**
Download torrents
- Steam**
Gaming platform
- OBS Studio**
Screen recording & streaming
- BlueStacks**
Android apps on PC



DID YOU KNOW?

Application software can be installed locally on your computer or accessed online through the internet (web applications).

EXAMPLES OF WEB APPLICATIONS



Gmail
Email



Google Drive
Cloud storage



Microsoft 365
Online Office



Trello
Project tracking



Spotify



Canva
Online design



Figma
UI/UX design



KEY TAKEAWAY

Application software helps users do specific tasks more efficiently. There are thousands of applications available for work, study, creativity, communication, entertainment, and more!



REMEMBER

- Use official sources
- Download safely
- Keep software updated
- Use the right tool for the right task



System Software (Operating System)



Application Software



You (User)



Get Things Done!

Choose the right app. Be productive. Be creative. Have fun!



ENTERPRISE & BUSINESS SOFTWARE 2026

Powering Digital Transformation. Driving Business Success.

Enterprise & business software solutions help organizations streamline operations, enhance collaboration, make data-driven decisions, and deliver exceptional customer value.

WHY ENTERPRISE SOFTWARE MATTERS



Improve Efficiency



Enhance Collaboration



Make Better Decisions



Ensure Compliance & Security



Drive Growth & Innovation

1

SAP ERP



Comprehensive ERP solution to manage core business processes in real time.

Key Features

- ✓ Finance & accounting
- ✓ Supply chain & procurement
- ✓ Human capital management
- ✓ Production & operations
- ✓ Real-time analytics & reporting

2

Microsoft Dynamics 365



Smart business applications that unify data, processes, and teams.

Key Features

- ✓ ERP & CRM in one platform
- ✓ Sales, marketing & customer service
- ✓ Finance & operations
- ✓ AI-powered insights
- ✓ Seamless Microsoft 365 integration

3

Salesforce



#1 CRM platform for sales, service, marketing and more.

Key Features

- ✓ Lead & opportunity management
- ✓ Customer service & support
- ✓ Marketing automation
- ✓ Analytics & dashboards
- ✓ AppExchange ecosystem

4

Oracle ERP



Robust ERP suite for managing complex business operations globally.

Key Features

- ✓ Financial management
- ✓ Project portfolio management
- ✓ Procurement & risk management
- ✓ Supply chain management
- ✓ Global compliance & reporting

5

ServiceNow



Digital workflow platform that helps businesses automate and modernize operations.

Key Features

- ✓ IT Service Management (ITSM)
- ✓ HR Service Delivery
- ✓ Workflow automation
- ✓ Asset & incident management
- ✓ AI & predictive intelligence

6

Jira



Agile project management tool for planning, tracking and delivering software.

Key Features

- ✓ Agile boards & Scrum/Kanban
- ✓ Issue & bug tracking
- ✓ Roadmaps & reports
- ✓ Custom workflows
- ✓ Integration with DevOps tools

7

Confluence



Team collaboration and knowledge management software.

Key Features

- ✓ Create & manage documentation
- ✓ Team collaboration in real time
- ✓ Knowledge bases
- ✓ Templates & organized spaces
- ✓ Deep integration with Jira

8

Trello



Visual collaboration tool that helps teams organize projects and tasks.

Key Features

- ✓ Boards, lists & cards
- ✓ Drag-and-drop simplicity
- ✓ Checklists & due dates
- ✓ Power-ups & automation
- ✓ Team collaboration

9

monday.com



Work OS that helps teams plan, track and deliver work more efficiently.

Key Features

- ✓ Customizable workflows
- ✓ Project & task management
- ✓ Dashboards & reporting
- ✓ Automation & integrations
- ✓ Scalable for any team

10

Power BI



Business analytics service for visualizing data and sharing insights.

Key Features

- ✓ Interactive dashboards
- ✓ Data modeling & transformation
- ✓ Real-time data connectivity
- ✓ AI insights & natural language Q&A
- ✓ Secure sharing & collaboration

COMMON BUSINESS FUNCTIONS SUPPORTED



Finance & Accounting



Human Resources



Sales & CRM



Supply Chain & Operations



IT Service Management



Project & Work Management



Analytics & Reporting

TOP BENEFITS



Unified data & processes



Improved productivity



Better customer experience



Faster decision making



Scalability & flexibility



Cost savings & efficiency

2026 TRENDS

- AI-powered automation across enterprise platforms
- Low-code/no-code for faster app development
- Deeper integrations and unified ecosystems
- Real-time analytics and predictive insights
- Enhanced security, privacy & compliance



BEST PRACTICES FOR SUCCESS



Align software with business goals



Invest in user training & adoption



Keep data clean, secure & accessible



Integrate tools for seamless workflows



Monitor performance & optimize



Stay updated with new features & trends



2026



WEB & INTERNET SOFTWARE 2026

Essential Tools for Browsing, Building, Hosting & Connecting

From browsing the web to building websites, managing servers, and testing APIs, these software solutions power the modern internet.

WHY THESE TOOLS MATTER

Access the Web Faster

Secure & Private Browsing

Build & Host Websites

Streamline Dev & Operations

Boost Productivity & Efficiency

Power Digital Experiences

1

Google Chrome

Fast, secure, and feature-rich web browser developed by Google.

Key Features

- High performance & speed
- Google account sync & profiles
- Built-in security (Safe Browsing)
- Extensive extensions library
- Cross-platform (Windows, macOS, Linux, Android, iOS)

2

Microsoft Edge

Secure and modern browser by Microsoft built on Chromium.

Key Features

- Collections, vertical tabs & productivity tools
- Microsoft account & sync
- Tracking prevention & SmartScreen
- Integration with Microsoft 365
- Cross-platform compatibility

3

Mozilla Firefox

Open-source browser focused on privacy and user control.

Key Features

- Enhanced tracking protection
- Open-source & community driven
- Customizable interface
- Add-ons and developer tools
- Cross-platform support

4

Brave

Privacy-first browser that blocks ads and trackers by default.

Key Features

- Built-in ad & tracker blocker
- Brave Shields for privacy
- Crypto wallet & Web3 support
- Fast performance
- Rewards for private ads (opt-in)

5

Apache Web Server

Open-source web server software, widely used around the world.

Key Features

- Highly reliable and secure
- Supports many platforms (Windows, Linux, Unix)
- Flexible configuration
- Large community support
- Supports HTTP/2 and more

6

Nginx

High-performance web server, reverse proxy, load balancer & more.

Key Features

- High performance & low resource usage
- Reverse proxy & load balancing
- HTTP/2, HTTPS & TCP/UDP proxy
- Scalable and event-driven
- Ideal for high-traffic sites

7

IIS (Internet Information Services)

Microsoft's web server for hosting websites and web applications.

Key Features

- Seamless Windows integration
- Supports ASP.NET, PHP, & more
- Advanced security & authentication
- Application pools & isolation
- Easy management with IIS Manager

8

WordPress

Popular open-source CMS for building websites and blogs.

Key Features

- Easy to use & customize
- Thousands of themes & plugins
- SEO friendly
- Large global community
- Suitable for blogs, business & eCommerce

9

FileZilla FTP

Popular FTP client for transferring files between client & server.

Key Features

- Supports FTP, FTPS & SFTP
- Easy site management
- Drag & drop interface
- Cross-platform (Windows, macOS, Linux)
- Resume & transfer queue support

10

Postman

API platform for building, testing, documenting and sharing APIs.

Key Features

- Send & test API requests easily
- Collections & environment variables
- Automated testing & monitors
- API documentation
- Integrations & team collaboration

COMMON USE CASES

- Browse the web securely
- Develop & test websites
- Host & manage websites
- Transfer files to/from servers
- Test & integrate APIs

WEB STACK WORKFLOW

All these tools work together to deliver powerful web experiences.

2026 TRENDS

- AI-powered browsing & search enhancements
- Stronger privacy & ad-free experiences
- Edge computing & serverless architectures
- Faster, more secure web protocols (HTTP/3)
- Low-code/no-code web development growth

BEST PRACTICES

- Keep software up to date
- Use HTTPS for secure connections
- Regular backups of websites & data
- Monitor performance & security
- Follow least privilege access



MULTIMEDIA & CREATIVE SOFTWARE 2026

Powering Creativity. Inspiring Ideas. Delivering Impact.

From photo editing and vector design to video production, animation, audio recording and live streaming—these tools help creators bring their ideas to life.

WHY THESE TOOLS MATTER



Unleash Creativity



Boost Productivity



Enhance Collaboration



Save Time & Increase Efficiency



Create Professional Quality Content



Grow Your Brand & Audience

1 Adobe Photoshop



Industry-standard tool for photo editing and raster graphics.

KEY FEATURES

- ✓ Advanced photo editing & retouching
- ✓ Layers, masks & smart objects
- ✓ AI-powered tools (Generative Fill, etc.)
- ✓ Wide range of filters & effects
- ✓ Supports RAW & 3D editing

2 Adobe Illustrator



Professional vector graphics software for illustrations, logos and typography.

KEY FEATURES

- ✓ Vector illustration & design
- ✓ Scalable artwork without quality loss
- ✓ Advanced typography tools
- ✓ Artboards & responsive assets
- ✓ Integration with Adobe Creative Cloud

3 Adobe Premiere Pro



Professional video editing software for film, TV and online content.

KEY FEATURES

- ✓ Multi-track video editing
- ✓ High-performance playback
- ✓ Color correction & audio tools
- ✓ AI-powered features (Auto Reframe)
- ✓ Seamless Adobe ecosystem integration

4 Adobe After Effects



Motion graphics & visual effects for stunning animations and compositions.

KEY FEATURES

- ✓ Motion graphics & visual effects
- ✓ Keying, tracking & compositing
- ✓ 3D animation & camera tools
- ✓ Hundreds of effects & presets
- ✓ Integration with Premiere Pro & more

5 Audacity



Free, open-source audio editor and recorder.

KEY FEATURES

- ✓ Record & edit audio easily
- ✓ Multi-track editing
- ✓ Noise reduction & audio restoration
- ✓ Supports many audio formats
- ✓ Plugins & effect support

6 OBS Studio



Free and open-source software for live streaming and screen recording.

KEY FEATURES

- ✓ Live streaming to multiple platforms
- ✓ Scene & source management
- ✓ Real-time audio mixing
- ✓ High performance & lightweight
- ✓ Plugins & extensive customization

7 Blender



Powerful open-source 3D creation suite for modeling, animation, rendering and more.

KEY FEATURES

- ✓ 3D modeling, sculpting & animation
- ✓ Rendering with Cycles & Eevee
- ✓ Simulation (physics, fluids, smoke)
- ✓ Video editing & VFX
- ✓ Completely free & open-source

8 DaVinci Resolve



Professional video editing, color grading, visual effects and audio post-production.

KEY FEATURES

- ✓ Advanced color correction & grading
- ✓ Edit, VFX, motion graphics & audio in one tool
- ✓ High-end professional quality
- ✓ Collaboration & cloud workflows
- ✓ Free version available (powerful)

9 Canva



User-friendly design platform for graphics, presentations, social media and more.

KEY FEATURES

- ✓ Drag-and-drop design tools
- ✓ Thousands of templates & assets
- ✓ Brand kits & team collaboration
- ✓ Video & animation maker
- ✓ Web, print & social content support

10 Figma



Collaborative UI/UX design tool for teams and digital products.

KEY FEATURES

- ✓ Real-time collaboration
- ✓ UI/UX design & prototyping
- ✓ Design systems & components
- ✓ Developer handoff & inspect mode
- ✓ Cloud-based & platform-independent

COMMON USE CASES



Photo Editing & Retouching



Graphic Design & Illustration



Video Editing & Production



Motion Graphics & Visual Effects



Audio Recording & Editing



Live Streaming & Webcasts

KEY BENEFITS



Enhances Creativity



Saves Time & Increases Efficiency



Professional Quality Output



Better Collaboration & Workflow



Cost-Effective Solutions

2026 TRENDS

- 1 AI-powered creative assistance
- 2 Cloud collaboration & real-time workflows
- 3 Higher quality, 8K & immersive content
- 4 Cross-platform & integrated ecosystems
- 5 Creator economy & digital storytelling growth



CREATIVE WITH PURPOSE

The right tools. The right ideas. The right impact.



KEEP LEARNING. KEEP CREATING. KEEP INSPIRING.

Master the tools. Share your vision. Shape the future.



2026

Create today. Inspire tomorrow.

DATABASE SOFTWARE

Store. Manage. Protect. Analyze.

Database software helps organizations store, manage, retrieve and analyze data efficiently and securely.

WHAT DATABASE SOFTWARE DOES



STORE
Organize data efficiently



MANAGE
Create, update and maintain



RETRIEVE
Search and access data



ANALYZE
Generate insights and reports



PROTECT
Ensure security and reliability

POPULAR DATABASE SOFTWARE



Enterprise-grade relational database from Microsoft.

- High performance & scalability
- Advanced security
- Business intelligence
- Integration with Microsoft ecosystem

Best for: Enterprise applications, data warehousing, BI solutions



Popular open-source relational database.

- Fast, reliable and easy to use
- Cost-effective
- Large community support
- Supports SQL standards

Best for: Web applications, SMBs, startups



Advanced open-source relational database.

- ACID compliant
- Extensible and powerful
- Supports complex queries
- Strong security

Best for: Complex applications, analytics, GIS, developers



Enterprise-grade database for mission-critical workloads.

- High availability
- Advanced security
- Powerful performance
- Comprehensive tools

Best for: Large enterprises, mission-critical systems



Leading NoSQL document database.

- Flexible document model
- High scalability
- Schema-less design
- Fast development

Best for: Big data, agile apps, real-time applications



Lightweight, serverless relational database.

- No server required
- Zero configuration
- Small, fast, reliable
- ACID compliant

Best for: Mobile apps, embedded systems, IoT



In-memory key-value data structure store.

- Extremely fast
- Caching and sessions
- Pub/Sub messaging
- High availability

Best for: Caching, real-time analytics, session management

DBMS (DATABASE MANAGEMENT SYSTEMS)

A DBMS is software that acts as an interface between users/applications and the database. It helps you define, create, manipulate, control and secure data.



Define
Create database structures



Manipulate
Insert, update, delete data



Control
Manage access and permissions



Monitor
Track performance and usage



Backup & Recover
Protect and restore data



Multi-user Support
Allow multiple users to work safely

SQL vs NoSQL

	SQL (Relational)	NoSQL (Non-Relational)
Data Model	Tables (rows & columns)	Documents, key-value, graphs, wide-columns
Schema	Fixed schema	Flexible / Schema-less
Query Language	SQL	Varies (JSON, Queries, etc.)
Relationships	Joins, foreign keys	Embedded or referenced
Scalability	Vertical scaling	Horizontal scaling
Best For	Structured data, complex transactions	Big data, high scale, real-time apps

DBMS FEATURES



Data Storage



Data Security



Data Integrity



Backup & Recovery



User Management



Query Processing



Performance Optimization

SQL vs NoSQL – WHEN TO USE WHAT?

Use SQL When:

- ✓ You need ACID compliance
- ✓ Your data is structured
- ✓ You need complex transactions
- ✓ You have strict consistency needs



SQL

Use NoSQL When:

- ✓ You need high scalability
- ✓ Your data is unstructured or semi-structured
- ✓ You need fast development
- ✓ You handle large volumes of data



NoSQL

COMMON USE CASES



Web Applications



E-commerce Platforms



Business Intelligence



Mobile Apps



IoT & Devices



Banking & Finance



Healthcare



Gaming

DID YOU KNOW?



The global database software market is growing rapidly, driven by cloud computing, AI, analytics, and real-time data needs.

TRENDS TO WATCH



Cloud Databases



AI-Powered Databases



Real-time Analytics



Multi-model Databases



Advanced Security

BEST PRACTICES

- ✓ Choose the right database for your needs
- ✓ Normalize data (SQL) / Model properly (NoSQL)
- ✓ Index your data for faster queries
- ✓ Regularly backup your databases
- ✓ Monitor performance and optimize queries
- ✓ Keep database software up to date



Better Data.



Better Decisions.



Stronger Security.



Smarter Future.

Choose the right database. Build the right future.



PROGRAMMING & DEVELOPMENT SOFTWARE

Tools, languages and platforms developers use to build, test, debug and deploy software.



DEVELOPMENT ENVIRONMENTS (IDEs & EDITORS)



Visual Studio

Microsoft's powerful IDE



- ✓ Full-featured IDE for .NET and Windows development
- ✓ Advanced debugging
- ✓ IntelliSense
- ✓ Integrated tools & designers
- ✓ Supports multiple languages

Best for: .NET, C#, Windows apps, enterprise solutions



Visual Studio Code

Lightweight, fast and extremely extensible



- ✓ Built-in Git & terminal
- ✓ Huge extension marketplace
- ✓ IntelliSense & smart code completion
- ✓ Supports many languages
- ✓ Cross-platform

Best for: Web development, scripts, multi-language projects



JetBrains IDEs

Smart, intelligent IDEs for professionals



- ✓ Intelligent code assistance
- ✓ Advanced refactoring
- ✓ Built-in debugging & testing
- ✓ Supports many languages
- ✓ IDEs: IntelliJ IDEA, PyCharm, WebStorm, Rider, GoLand, CLion

Best for: Professional development across many languages

VERSION CONTROL & COLLABORATION



Git

Distributed version control system

- ✓ Track changes in code
- ✓ Branching & merging
- ✓ History & rollback
- ✓ Works offline
- ✓ Open source & fast

Common Git Commands

```
git init           # Initialize repository
git clone <url>    # Clone repository
git add .          # Stage changes
git commit -m "msg" # Commit changes
git push          # Push to remote
git pull          # Pull latest changes
```



GitHub Desktop

Easy GUI for Git and GitHub

- ✓ Simple and intuitive
- ✓ Commit, push, pull with clicks
- ✓ Branch management
- ✓ View history & changes
- ✓ Integrates with GitHub

Best for: Developers who prefer a visual interface for Git

OTHER ESSENTIAL TOOLS



Postman
API development & testing



Insomnia
REST & GraphQL client



Swagger / OpenAPI
API documentation



MongoDB Compass
GUI for MongoDB



MySQL Workbench
Database design & admin



Figma
UI/UX design & prototyping



Firebase
Backend services platform



Jira
Project & issue tracking

PROGRAMMING LANGUAGES



Python

High-level, versatile

- ✓ Easy syntax
- ✓ Huge standard library
- ✓ AI, ML, Data Science
- ✓ Web, Automation, Scripting



Java

Robust, platform independent

- ✓ Write once, run anywhere*
- ✓ Large ecosystem
- ✓ Enterprise, Android, Cloud



C#

Modern, type-safe and powerful

- ✓ .NET development
- ✓ Windows, Web, Games
- ✓ Strong tooling (Visual Studio)



C/C++

High performance and efficient

- ✓ System programming
- ✓ Game engines
- ✓ Embedded systems



JavaScript

The language of the web

- ✓ Runs in browsers
- ✓ Front-end & back-end
- ✓ Huge ecosystem (npm)



TypeScript

Typed superset of JavaScript

- ✓ Safer code
- ✓ Better tooling
- ✓ Scalable apps

RUNTIME & PLATFORM TOOLS



Node.js

JavaScript runtime environment

- Run JS on server
- npm ecosystem
- Scalable network apps

CONTAINERIZATION & ORCHESTRATION



Docker Desktop

Build, share and run applications in containers

- ✓ Consistent environments
- ✓ Great for Dev, Test & Prod
- ✓ Supports Docker Compose
- ✓ Cross-platform (Windows, Mac, Linux)



Kubernetes

Orchestrate and manage containerized applications

- Automated deployment
- Scaling & self-healing
- Load balancing
- Designed for production

DEVELOPER WORKFLOW



1. Plan

Define requirements and design



2. Code

Write code in IDE or editor



3. Commit

Save changes with Git



4. Test

Run tests and fix issues



5. Build

Build application or package



6. Deploy

Deploy to server or cloud



7. Monitor

Monitor, log and improve

WHY THESE TOOLS MATTER

- ⌚ Increase productivity
- 🛡️ Improve code quality
- 👥 Enable collaboration
- 🚀 Speed up development
- 📦 Ensure reliable deliveries
- ⚡ Power modern applications

QUICK TIPS

- Use version control (Git) for every project.
- Write clean code and test often.
- Automate builds and deployments.
- Use containers for consistent environments.
- Keep learning and explore new tools!



Great tools + Good practices = Amazing software!



Build. Test. Commit. Push. Deploy. Repeat.



Learn continuously. Build the future.



UTILITY SOFTWARE 2026

Essential tools that keep modern organizations secure, efficient and productive.

Used across IT, security, finance and operations by leading companies like Investec to maintain systems, protect data, automate tasks and improve performance.

WHY UTILITY SOFTWARE MATTERS



Improves
Performance



Protects Data
& Systems



Saves Time &
Automates Tasks



Simplifies IT
Management



Ensures Compliance
& Reliability



Reduces Costs &
Optimizes Resources

1 Microsoft PowerToys



Utilities to customize
Windows for power users.

- ✓ Advanced keyboard manager
- ✓ Window layouts & shortcuts
- ✓ File Explorer add-ons
- ✓ Power rename, image resizer

2 Microsoft Sysinternals Suite



Advanced system
utilities for IT pros.

- ✓ Process Explorer
- ✓ Autoruns
- ✓ Sysmon
- ✓ TCPView & more

3 Notepad++



Advanced text editor
for developers and
IT professionals.

- ✓ Syntax highlighting
- ✓ Code folding
- ✓ Multi-language support
- ✓ Lightweight & fast

4 Everything



Fast file search tool
for Windows.

- ✓ Instant search results
- ✓ Lightweight
- ✓ Search by name or content
- ✓ Filters & advanced search

5 TeamViewer



Remote access and
remote support.

- ✓ Remote control devices
- ✓ File transfer
- ✓ Unattended access
- ✓ Cross-platform

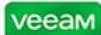
6 AnyDesk



Fast, secure remote
desktop software.

- ✓ Low latency
- ✓ Easy to use
- ✓ File transfer
- ✓ Cross-platform

7 Veeam Backup & Replication



Data backup and disaster
recovery solution.

- ✓ Virtual, physical & cloud backup
- ✓ Instant recovery
- ✓ Data deduplication
- ✓ High availability

8 SolarWinds Toolset



Network & system
management tools.

- ✓ Network monitoring
- ✓ Performance analysis
- ✓ Configuration management
- ✓ IT infrastructure visibility

9 ManageEngine ADManager Plus



Active Directory management
and reporting.

- ✓ User management
- ✓ Password reset
- ✓ AD reporting
- ✓ Change tracking

10 Malwarebytes



Advanced malware
detection and removal.

- ✓ Real-time protection
- ✓ Ransomware protection
- ✓ Adware & PUP removal
- ✓ Threat scanning

11 ESET PROTECT (Endpoint Security)



Endpoint protection
for businesses.

- ✓ Antivirus & antispyware
- ✓ Device control
- ✓ Web & email protection
- ✓ Centralized management

12 Acronis Cyber Protect Cloud



Backup and cybersecurity
in one integrated solution.

- ✓ Image backup
- ✓ Anti-malware protection
- ✓ Cloud storage
- ✓ Disaster recovery

13 WinSCP



Secure file transfer
(SFTP, SCP, FTP).

- ✓ Secure file transfers
- ✓ File management
- ✓ Script automation
- ✓ Integration with PuTTY

14 PuTTY



SSH and Telnet
client for remote
connections.

- ✓ Secure SSH connections
- ✓ Lightweight
- ✓ Session management
- ✓ Scripting support

15 FileZilla



FTP client for fast
and reliable file
transfers.

- ✓ Easy site management
- ✓ Resume transfers
- ✓ Cross-platform
- ✓ Drag-and-drop interface

16 BleachBit



Cleans unnecessary
files and frees up
space.

- ✓ System cleaner
- ✓ Privacy protection
- ✓ Browser cache cleaning
- ✓ Open source

17 Revo Uninstaller Pro



Advanced uninstall
and system cleanup.

- ✓ Complete program removal
- ✓ Leftover file scanning
- ✓ Browser extension removal
- ✓ Forced uninstall

18 Advanced IP Scanner



Fast network scanner
and IP discovery.

- ✓ Scan IP ranges
- ✓ Detect devices
- ✓ Remote access
- ✓ Export results

19 SpaceSniffer



Disk space analyzer
for Windows.

- ✓ Visualize disk usage
- ✓ Find large files
- ✓ Easy to use
- ✓ Portable

20 HashTab



Calculate and verify
file hashes.

- ✓ Supports MD5, SHA-1, SHA-256
- ✓ File integrity verification
- ✓ Right-click integration
- ✓ Lightweight

21 Print Management (Windows)



Manage printers and
print servers.

- ✓ Deploy printers
- ✓ Monitor print queues
- ✓ Set permissions
- ✓ Track usage

22 Windows Task Scheduler



Automate tasks and
run programs.

- ✓ Schedule tasks
- ✓ Triggers & actions
- ✓ System maintenance
- ✓ Improves productivity

23 Glary Utilities (Pro)



All-in-one system
optimization suite.

- ✓ Registry cleaner
- ✓ Disk cleaner
- ✓ Startup manager
- ✓ Performance boost

24 7-Zip File Manager



File manager with
built-in compression.

- ✓ 7z, ZIP, RAR, TAR support
- ✓ High compression ratio
- ✓ File encryption (7z)
- ✓ Open source

WHAT THEY HELP ACHIEVE



Stronger
Security



Data Protection
& Backup



System
Reliability



Operational
Efficiency



IT Management
& Automation



Cost Savings



Compliance
& Governance

USED BY ORGANIZATIONS LIKE INVESTEC FOR



Secure
Operations



Risk
Management



IT Infrastructure
Monitoring



Business
Continuity



Regulatory
Compliance



Productivity
& Efficiency

BEST PRACTICES

- ✓ Keep software up to date.
- ✓ Use strong passwords and encryption.
- ✓ Schedule regular backups and test restores.
- ✓ Monitor systems and performance regularly.
- ✓ Remove unused software and old files.





CLOUD COMPUTING SOFTWARE

Work. Collaborate. Store. Secure. Manage. Anywhere.

Cloud software delivers powerful services that help individuals, teams and organizations be more productive, secure and efficient—anytime, anywhere.



Anywhere
Access from any device, anywhere



Secure
Enterprise-grade security and compliance



Scalable
Grow and scale resources as you need



Reliable
High availability and disaster recovery



Cost Efficient
Pay for what you use and save more

CLOUD PLATFORMS (INFRASTRUCTURE & SERVICES)



Microsoft Azure

Build, deploy and manage applications and services through Microsoft's global cloud.



Amazon Web Services

Broad and deep cloud platform with the most comprehensive services and global reach.



Google Cloud Platform

Secure, data analytics and machine learning solutions that run on the same infrastructure as Google.



PRODUCTIVITY & COLLABORATION SUITES



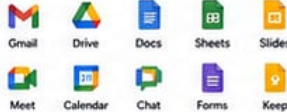
Microsoft 365

Productivity apps and cloud services to help you and your organization achieve more.



Google Workspace

Smart, secure and innovative tools to collaborate and get work done.



CLOUD STORAGE & FILE SHARING



OneDrive

Microsoft's cloud storage for files and backups.

- 5 GB free storage
- Access anywhere
- Files on-demand
- Integrated with Windows & 365



SharePoint

Team sites, content management and intranet solutions.

- Document libraries
- Workflows
- Permissions
- Version control



Dropbox

Secure file storage and sharing made simple.

- Easy file sharing
- File recovery
- Team collaboration
- Cross-device sync

MANAGEMENT, IDENTITY & SECURITY



Microsoft Intune

Cloud-based endpoint management to secure and manage devices, apps and data.



Microsoft Entra ID (Azure AD)

Identity and access management to protect users and resources.



KEY BENEFITS

- ✓ Access powerful tools from anywhere, on any device
- ✓ Enhance collaboration and communication
- ✓ Protect data with enterprise-grade security
- ✓ Scale resources up or down in minutes
- ✓ Reduce costs and increase operational efficiency
- ✓ Stay updated with automatic updates and new features

WHY CLOUD SOFTWARE?



Flexibility
Adapt quickly to change and scale with ease



Collaboration
Work together in real-time from anywhere



Security
Protect data with advanced security



Reliability
99.9% uptime and disaster recovery



Innovation
Access the latest technology and innovations



Cost Savings
Lower IT costs with pay-as-you-go pricing

Cloud software empowers people and organizations to do more—anywhere, securely and efficiently. | Secure. Scalable. Reliable. | The cloud is more than technology—it's a better way to work.

MAJOR CLOUD SERVICE PROVIDERS

Powering the World's Digital Transformation



Amazon Web Services

Compute, Storage, Database, Networking, Analytics, AI/ML, IoT, DevOps

Largest global presence, mature services, scalability

Startups to enterprises, highly scalable workloads

34+ regions
108+ Availability Zones



Microsoft Azure

Compute, Storage, Database, AI/ML, Analytics, IoT, DevOps, Hybrid Cloud

Hybrid cloud leader, Microsoft ecosystem, enterprise-ready

Enterprises, .NET apps, Windows workloads

60+ regions
140+ Availability Zones



Google Cloud

Compute, Storage, Database, AI/ML, Data Analytics, Kubernetes, DevOps

Data & AI leader, innovation, open source, Kubernetes (GKE)

Data-driven apps, AI/ML, developers, startups

40+ regions
121+ Zones



IBM Cloud

Compute, Storage, Database, AI/ML, Networking, Hybrid Cloud

Hybrid cloud leader, security, compliance, enterprise solutions

Enterprises, regulated industries

20+ regions
60+ Data Centers



Oracle Cloud

Compute, Storage, Database, Autonomous DB, AI/ML, Networking, Apps

Strong in databases, enterprise apps, performance

Enterprises, Oracle customers

36+ regions
96+ Availability Domains



Alibaba Cloud

Compute, Storage, Database, AI/ML, Big Data, IoT, Networking, Security

Strong in Asia, competitive pricing, e-commerce solutions

Businesses in Asia, e-commerce, startups

28+ regions
86+ Availability Zones



Tencent Cloud

Compute, Storage, Database, AI/ML, Gaming, CDN, Networking, Security

Strong in gaming, social, media, and mobile ecosystem

Gaming, media, internet companies, startups

27+ regions
70+ Availability Zones



DigitalOcean

Compute, Storage, Database, Kubernetes, App Platform, Networking

Simplicity, developer friendly, cost-effective

Developers, startups, SMBs

15+ regions
40+ Data Centers



VMware Cloud

Compute, Storage, Networking, Disaster Recovery, Desktop as a Service

Hybrid cloud, VMware ecosystem, migration solutions

VMware customers, hybrid cloud needs

25+ regions
Global Data Centers



Global Reach
Hundreds of regions worldwide



High Availability
Built for reliability and uptime



Security First
Advanced security and compliance



Scalable
Scale up or down in minutes



Cost Optimized
Pay for what you use



CLOUD COMPUTING SOFTWARE

Power. Productivity. Collaboration. Security. Anywhere.

Cloud software delivers services, applications and infrastructure over the internet—on demand, scalable and secure.



CLOUD PLATFORMS (INFRASTRUCTURE & SERVICES)



Microsoft Azure

Microsoft's cloud platform for building, deploying and managing applications and services.



Best for: Hybrid cloud, enterprise solutions, Microsoft integration, AI, analytics and DevOps.



Amazon Web Services (AWS)

The world's most comprehensive and widely adopted cloud platform.



Best for: Scalability, global reach, startups, enterprise, and innovative cloud solutions.



Google Cloud Platform (GCP)

A suite of cloud computing services that runs on the same infrastructure as Google.



Best for: Data analytics, machine learning, Kubernetes, and modern application development.

PRODUCTIVITY & COLLABORATION SUITES



Microsoft 365

Cloud-powered productivity suite with familiar apps and intelligent cloud services.



Best for: Businesses, education, communication, and document collaboration.



Google Workspace

Cloud-native productivity and collaboration tools designed for teams.



Best for: Real-time collaboration, simplicity, and cloud-first organizations.

CLOUD STORAGE & FILE SHARING



OneDrive

Microsoft's cloud storage for files, photos and backups.

- 5 GB free storage
- File sync & access anywhere
- Integrated with Windows & 365



SharePoint

Team sites, content management and intranet solutions.

- Document libraries
- Workflows
- Permissions & version control



Dropbox

Secure file storage and easy sharing across devices.

- Simple file sharing
- File recovery
- Team collaboration tools

MANAGEMENT, IDENTITY & SECURITY



Microsoft Intune

Cloud-based endpoint management for organizations.



Best for: Managing devices, apps and security across your organization.



Microsoft Entra ID (Azure AD)

Identity and access management for secure access to resources and applications.



Best for: Secure access, identity protection and compliance.

BENEFITS OF CLOUD COMPUTING SOFTWARE



Scalability

Scale resources up or down instantly.



Cost Efficiency

Pay for what you use. Reduce infrastructure costs.



Anywhere Access

Access your data and apps from any device, anywhere.



Security

Built-in security, compliance and advanced threat protection.



Collaboration

Work together in real time and boost productivity.



The Cloud. Limitless Possibilities.



Secure your data.



Scale with confidence.



Collaborate without limits.



Innovate faster.



VIRTUALIZATION & CONTAINERS 2026

Powering Modern IT Infrastructure, Development & Cloud

Virtualization and containerization technologies enable you to run more workloads, use resources efficiently, and deliver applications faster and more securely.

WHY IT MATTERS



Maximize
Resource
Utilization



Enable Cloud
and Hybrid
Environments



Speed Up
Development
& Delivery



Improve Security
& Isolation



Reduce Costs
& Complexity



Hyper-V

1 Hyper-V

Microsoft's native hypervisor for creating and managing virtual machines on Windows.

Key Features

- ✓ Create and manage virtual machines
- ✓ Dynamic memory and resource allocation
- ✓ Virtual switches and networking
- ✓ High availability and failover clustering
- ✓ Seamless integration with Windows Server and Azure



2 VMware Workstation

Powerful desktop virtualization software for running multiple operating systems on a single Windows or Linux PC.

Key Features

- ✓ Run multiple OS simultaneously
- ✓ Snapshots and cloning
- ✓ Advanced virtual networking
- ✓ 3D graphics acceleration
- ✓ Support for a wide range of OS



3 VMware ESXi

Enterprise-grade bare-metal hypervisor for building and managing virtualized data center environments.

Key Features

- ✓ High performance and scalability
- ✓ Centralized management with vCenter
- ✓ Live migration (vMotion)
- ✓ High availability and fault tolerance
- ✓ Strong security and role-based access



4 Oracle VM VirtualBox

Open-source desktop virtualization software for running virtual machines across platforms.

Key Features

- ✓ Free and open source
- ✓ Cross-platform (Windows, Linux, macOS, Solaris)
- ✓ Snapshots and cloning
- ✓ Shared folders and seamless mode
- ✓ Extensive guest OS support



5 Docker

Platform for developing, shipping, and running applications in lightweight containers.

Key Features

- ✓ Containerize applications and dependencies
- ✓ Consistent environments
- ✓ Fast deployment and portability
- ✓ Large ecosystem and registry (Docker Hub)
- ✓ Easy integration in CI/CD pipelines



6 Kubernetes

Open-source container orchestration platform for automating deployment, scaling, and management of containerized apps.

Key Features

- ✓ Automated deployment and scaling
- ✓ Self-healing and high availability
- ✓ Load balancing and service discovery
- ✓ Secret and configuration management
- ✓ Extensible with a rich ecosystem



7 Windows Sandbox

Lightweight, isolated desktop environment for safely running applications in a temporary virtual space.

Key Features

- ✓ Clean and disposable environment
- ✓ Built into Windows 10/11 Pro, Enterprise, Education
- ✓ Ideal for testing and running untrusted apps
- ✓ No impact on host system
- ✓ Quick start and low resource usage



8 Windows Subsystem for Linux (WSL)

Run a Linux environment natively on Windows for development and administration.

Key Features

- ✓ Run Linux distributions on Windows
- ✓ High performance and system integration
- ✓ Access Windows files and Linux tools
- ✓ Great for developers and DevOps
- ✓ Supports WSL 2 (real Linux kernel)



9 Azure Virtual Machines

Scalable, on-demand virtual machines in the Microsoft Azure cloud platform.

Key Features

- ✓ Provision Windows or Linux VMs in minutes
- ✓ Global availability and scalability
- ✓ Integration with Azure services
- ✓ Pay-as-you-go pricing
- ✓ Built-in security, monitoring, and backups

VIRTUALIZATION vs CONTAINERS



VIRTUAL MACHINES

- ✓ Full OS virtualization
- ✓ Heavier and more isolated
- ✓ Ideal for running different OS
- ✓ Higher resource usage
- ✓ Examples: Hyper-V, VMware, VirtualBox



CONTAINERS

VS

- ✓ OS-level virtualization
- ✓ Lightweight and portable
- ✓ Share host OS kernel
- ✓ Faster startup and efficient
- ✓ Examples: Docker, Kubernetes

COMMON USE CASES



Development & Testing



Server Consolidation



Cloud & Hybrid Environments



CI/CD Pipelines



Legacy Application Support



Disaster Recovery & Backup



Edge & IoT Workloads



Learning & Training Labs

2026 TRENDS



Cloud-Native Everything
More workloads moving to containers and serverless platforms.



Stronger Security & Isolation
Enhanced runtime security, zero trust, and confidential computing.



AI-Optimized Infrastructure
GPUs and accelerators integrated in VMs and containers.



Edge & Distributed Computing
Virtualization and containers extending to the edge.

BEST PRACTICES

- ✓ Keep your platforms and tools updated.
- ✓ Use least privilege access and strong authentication.
- ✓ Monitor performance and resource usage continuously.
- ✓ Back up your images, VMs, and configurations regularly.
- ✓ Automate deployments with infrastructure as code.



Virtualization and containers are the foundation of modern IT.

Run smarter. Scale faster. Secure everything.



2026 Build the future with virtualized and containerized solutions.



NETWORKING SOFTWARE 2026

Essential tools to design, configure, manage and troubleshoot modern networks.

From network simulation and device configuration to secure remote access and file transfer, these tools help IT professionals build, manage and secure connectivity.

NETWORKING SOFTWARE HELPS YOU



Design & Simulate



Configure & Manage



Monitor & Analyze



Secure & Protect



Connect & Access



Transfer & Share



1 Windows Network Manager

Built-in Windows tool for viewing and managing network connections, status, and settings.

KEY FEATURES

- ✓ View network status and properties
- ✓ Configure IP, DNS, and adapter settings
- ✓ Manage Wi-Fi and Ethernet connections
- ✓ Troubleshoot connectivity issues
- ✓ Monitor network performance



2 pfSense

Open-source firewall and router platform based on FreeBSD, ideal for network security and management.

KEY FEATURES

- ✓ Stateful firewall and NAT
- ✓ VPN (IPsec, OpenVPN, WireGuard)
- ✓ Intrusion detection & prevention (IDPS)
- ✓ Traffic shaping and bandwidth control
- ✓ High availability and failover



3 Cisco Packet Tracer

Network simulation and visualization tool for learning, teaching, and prototyping network designs.

KEY FEATURES

- ✓ Simulate complex network topologies
- ✓ Configure routers, switches, PCs, etc.
- ✓ Supports real Cisco CLI commands
- ✓ Visualize protocols and traffic flow
- ✓ Ideal for training and certification



4 Cisco IOS

Cisco Internetwork Operating System that powers Cisco routers and switches.

KEY FEATURES

- ✓ Reliable and secure network OS
- ✓ Wide range of routing protocols
- ✓ Scalability and high performance
- ✓ Advanced security features
- ✓ Extensive CLI configuration



5 WinBox (MikroTik)

Configuration tool for managing MikroTik RouterOS devices graphically.

KEY FEATURES

- ✓ Full access to RouterOS settings
- ✓ User-friendly GUI
- ✓ Manage users, firewall, routes, wireless and more
- ✓ Supports remote management



6 PuTTY

Lightweight SSH and telnet client for secure remote command-line access.

KEY FEATURES

- ✓ SSH, Telnet, rlogin, raw socket
- ✓ Save and manage session profiles
- ✓ Lightweight and fast
- ✓ Secure remote access
- ✓ Cross-platform (Windows)



7 OpenVPN

Open-source VPN solution for creating secure point-to-point or site-to-site connections.

KEY FEATURES

- ✓ SSL/TLS encryption
- ✓ Remote access and site-to-site VPN
- ✓ Cross-platform support
- ✓ Flexible authentication options
- ✓ High security and performance



8 WireGuard

Modern, fast, and secure VPN software with a lightweight codebase.

KEY FEATURES

- ✓ High performance and low latency
- ✓ Strong encryption (ChaCha20, Curve25519)
- ✓ Small and easy to audit
- ✓ Simple configuration
- ✓ Cross-platform support



9 FileZilla

Popular FTP, FTPS and SFTP client for transferring files between local and remote hosts.

KEY FEATURES

- ✓ Supports FTP, FTPS, SFTP
- ✓ Drag-and-drop interface
- ✓ Site Manager for easy access
- ✓ Transfer queue and resume support
- ✓ Cross-platform (Windows, macOS, Linux)



10 Remote Desktop

Microsoft's remote access tool for connecting to and controlling remote computers.

KEY FEATURES

- ✓ Remote access to Windows PCs
- ✓ File transfer and clipboard sharing
- ✓ Multi-monitor support
- ✓ Secure with NLA and encryption
- ✓ Easy to use and built-in (Windows)

COMMON NETWORKING USE CASES



Network Design & Simulation

Plan and simulate networks before deployment.



Network Security & VPN

Secure networks and enable safe remote connections.



Device Configuration & Management

Configure routers, switches and firewalls efficiently.



Monitoring & Troubleshooting

Monitor performance and resolve network issues.



File Transfer & Sharing

Transfer files securely across the network.



Remote Access & Administration

Access and manage systems from anywhere securely.



DID YOU KNOW?

Modern networks rely on a combination of tools for design, security, management, and access. Using the right software improves efficiency, security, and uptime.



NETWORKING BEST PRACTICES

- ✓ Keep software updated for security and features
- ✓ Use strong authentication and encryption
- ✓ Regularly back up configurations
- ✓ Monitor network traffic and logs
- ✓ Document network designs and changes

2026 NETWORKING TRENDS



Cloud Networking

More tools integrate with cloud platforms and hybrid networks.



Zero Trust Networking

Secure access is the new standard.



AI-Powered Monitoring

AI helps detect issues faster and predict problems.



Software-Defined Networking (SDN)

More automation and centralized network control.

TIPS FOR GETTING STARTED

- ✓ Start with one tool and master the basics
- ✓ Practice in a virtual lab environment
- ✓ Use realistic scenarios and document your work
- ✓ Join online communities and forums
- ✓ Keep learning and stay updated



Stay Connected. Stay Secure. Stay Productive.



Build. Manage. Secure. Optimize.



The right tools. The right connections. The right future.

2026



CYBERSECURITY SOFTWARE

The 2026 Cybersecurity Toolkit

Powerful tools that protect, detect, respond, and recover in an ever-evolving digital world.



Protect
Prevent threats



Detect
Find threats early



Respond
Act fast & contain



Recover
Restore & strengthen



Comply
Meet standards & regulations

**2026
EDITION**

ENDPOINT PROTECTION



Microsoft Defender for Endpoint
Advanced threat protection for Windows, macOS, Linux, iOS & Android.



CrowdStrike Falcon
AI-powered endpoint protection & EDR.



Bitdefender
Next-gen antivirus with ransomware protection.



ESET
Lightweight, fast, effective endpoint security.



Kaspersky
Real-time protection against cyber threats.

NETWORK & INFRASTRUCTURE SECURITY



Cisco Secure
Comprehensive network & cloud security solutions.



Fortinet FortiGate
Firewall, VPN, & threat protection for networks.



Palo Alto Networks Cortex
Advanced network security & threat intelligence.



Check Point Infinity
Prevent threats across networks, cloud & users.



Sophos XGS Firewall
Next-gen firewall with AI-driven protection.

CLOUD SECURITY



Microsoft Entra ID Protection
Secure identities, access & cloud resources.



Google Cloud Security Command Center
Threat detection & risk management in GCP.



AWS GuardDuty
Intelligent threat detection for AWS workloads.



Cloudflare
Protect websites, APIs & networks (DDoS, WAF, CDN).



SailPoint
Identity security & access governance.

APPLICATION & WEB SECURITY



OWASP ZAP
Web app vulnerability scanner (open source).



Burp Suite Professional
Web security testing & vulnerability scanner.



Veracode
Secure code analysis & application security.



Akamai App & API Protector
Protect APIs & web apps from attacks.



Sucuri SiteCheck
Website malware scanning & firewall.

SECURITY OPERATIONS & MONITORING



Splunk Enterprise Security
Security monitoring, analytics & SIEM.



Microsoft Sentinel
Cloud-native SIEM & SOAR solution.



IBM QRadar
Threat detection, investigation & response.



Elastic Security
Search, analyze & respond to threats.



AlienVault OSSIM
Open-source SIEM & threat detection.

IDENTITY & ACCESS MANAGEMENT



Okta
Identity management & single sign-on (SSO).



Ping Identity
Secure digital identity solutions.



Duo Security
Multi-factor authentication & zero-trust access.



CyberArk
Privileged access management.



LastPass
Secure password management.

THREAT INTELLIGENCE & ANALYSIS



Recorded Future
Real-time threat intelligence.



ThreatConnect
Intelligence-driven threat response.



VirusTotal
Analyze suspicious files, URLs & IPs.



MISP
Threat information sharing platform.

EMAIL & PHISHING PROTECTION



Proofpoint
Advanced email threat protection.



Barracuda Email Security Gateway
Stop phishing, spam & malware.



Abnormal Security
AI-powered email threat detection.



Mimecast
Cloud email security & continuity.

DATA PROTECTION & ENCRYPTION



Veeam Backup & Replication
Encrypt files, drives & partitions.



Acronis Cyber Protect
Backup, ransomware protection & recovery.



Boxcryptor
Easy encryption for cloud storage.



SQL Server TDE (Transparent Data Encryption)
Encrypt sensitive data at database level.

MOBILE SECURITY



Lookout
Mobile threat protection for iOS & Android.



Avast Mobile Security
Protects phones from malware & phishing.



Norton Mobile Security
Advanced mobile threat detection.



Zimperium
Mobile device & app security.

OFFENSIVE SECURITY & TESTING



Kali Linux
Penetration testing & ethical hacking.



Metasploit Framework
Exploit development & penetration testing.



Nmap
Network discovery & security auditing.



Aircrack-ng
Wireless network security auditing.

BACKUP, RECOVERY & BUSINESS CONTINUITY



Veeam Backup & Replication
Reliable backups & quick recovery.



Acronis Cyber Protect
All-in-one protection (backup + security).



Carbonite
Cloud backup for individuals & businesses.



Dell PowerProtect Data Manager
& data protection.

ZERO TRUST & PRIVACY TOOLS



Cloudflare Zero Trust
Secure access to & networks.



Zscaler
Secure internet access & zero trust.



Tailscale
Private, secure VPN for teams.



Proton VPN
Encrypted VPN with privacy focus.

COMPLIANCE & GOVERNANCE



Tenable.io
Vulnerability management & compliance.



Qualys
Security, compliance & asset management.



OneTrust
Data privacy, consent & compliance platform.



DID YOU KNOW?

Cyber threats are evolving every day. These tools help you stay one step ahead, protect what matters, and build a more secure future.



Use strong & unique passwords



Enable multi-factor authentication (MFA)



Keep software & systems up to date



Back up data regularly



Be cautious of phishing emails & links



Limit access & follow least privilege



Monitor, detect & respond quickly

EMERGING TRENDS IN 2026

- ✓ AI-powered threat detection
- ✓ Zero Trust architecture
- ✓ Ransomware-as-a-Service (RaaS) countermeasures
- ✓ Cloud-native security solutions
- ✓ Automation & SOAR platforms



Stay Informed. Stay Protected. Stay Secure.

Cybersecurity is everyone's responsibility.

Protect your data. Protect your future.



AI SOFTWARE

The Ultimate AI Software Landscape 2026

The most powerful and up-to-date AI tools to create, code, automate, analyze, design, and innovate.

WHY AI SOFTWARE MATTERS

 Boost Productivity

 Enhance Creativity

 Smarter Decisions

 Automate Workflows

 Improve Accuracy

 Accessible Anywhere

CHAT & CONVERSATIONAL AI

-  **ChatGPT**
Advanced reasoning, research, analysis, and productivity.
-  **Microsoft Copilot**
Your everyday AI assistant across Microsoft 365.
-  **Google Gemini**
Multimodal AI for search, research and creativity.
-  **Claude 3.7**
Anthropic's powerful AI for complex reasoning.
-  **Meta AI**
Real-time information and unfiltered answers.
-  **xAI Grok 3**

WRITING & CONTENT GENERATION

-  **Jasper**
AI copywriting for blogs, ads, and marketing.
-  **Copy.ai**
Generate copy, content and social posts.
-  **Writesonic**
Blogs, ads, landing pages and more.
-  **Notion AI**
Write, summarize, and organize in Notion.
-  **GrammarlyGO**
AI writing assistant and grammar checker.
-  **Rytr**
Affordable AI writing for everyone.

CODE & DEVELOPMENT

-  **GitHub Copilot**
AI pair programmer that writes code with you.
-  **Tabnine**
AI code completion that understands you.
-  **Cursor**
AI-first code editor built for developers.
-  **Sourcegraph Cody**
AI for code search, explain and fix.
-  **Bolt.new**
Build full-stack apps with AI in the browser.
-  **Amazon CodeWhisperer**
AI coding companion by AWS.
-  **Devika AI**
Autonomous AI software engineer.
-  **Replit Agent**
Build, run, test, and deploy

IMAGE GENERATION & DESIGN

-  **Midjourney v7**
State-of-the-art AI image generation.
-  **DALL-E 3**
Create stunning images from text.
-  **Stable Diffusion 3.5**
Open-source image generation.
-  **Ideogram 3.0**
Generate logos, art and typography.
-  **Adobe Firefly**
AI for images, vectors, video and design.
-  **Canva AI**
Design anything with AI-powered tools.
-  **Leonardo AI**
Game assets, art and visual creation.
-  **Runway Gen-3 Alpha**
AI video, image and creative suite.

VIDEO & AUDIO GENERATION

-  **Runway Gen-3**
Next-gen AI video generation.
-  **Pika Labs 2.1**
Text-to-video and video effects.
-  **Synthesia 2.0**
AI avatars and video presentations.
-  **Descript AI**
Edit video & audio like a document.
-  **ElevenLabs**
Realistic AI voice generation.
-  **Lovo AI**
AI voices for any content.

DATA, ANALYTICS & BUSINESS AI

-  **Power BI Copilot**
AI insights and analytics in Power BI.
-  **Tableau with AI**
Smart data insights and visualization.
-  **DataRobot**
Automated machine learning platform.
-  **ThoughtSpot Sage**
AI-powered analytics and search.
-  **Akkio**
No-code AI for data analysis and predictions.
-  **C3 AI**
Enterprise AI applications at scale.

AUTOMATION & PRODUCTIVITY

-  **Zapier AI**
Automate workflows with AI.
-  **Make (Integromat)**
AI-powered workflow automation.
-  **Motion**
AI calendar and task scheduling.
-  **Notion AI**
Summarize, write and organize better.
-  **ClickUp AI**
AI for tasks, docs and projects.
-  **Mem AI**
AI memory assistant for teams.

RESEARCH & KNOWLEDGE

-  **Perplexity AI**
AI search engine with real-time sources.
-  **Elicit**
AI research assistant for papers.
-  **Consensus**
AI search engine for scientific insights.
-  **scite_**
Scite Assistant
AI citation and research verification.
-  **Connected Papers**
AI visualizes research connections.
-  **Semantic Scholar**
AI-powered academic search engine.

AI AGENTS & ASSISTANTS

-  **AgentGPT**
Build and deploy autonomous AI agents.
-  **AutoGPT**
Autonomous AI agents that get things done.
-  **BabyAGI 2.0**
Task-driven autonomous AI agent.
-  **CrewAI**
Multi-agent collaboration framework.
-  **Phala Network**
Privacy-preserving AI computation.
-  **Relevance AI**
Build AI agents for business tasks.

LOCAL & OPEN SOURCE AI

-  **Ollama**
Run LLMs locally on your machine.
-  **LM Studio**
Discover, download and run local models.
-  **Stable Diffusion WebUI**
Run Stable Diffusion locally.
-  **GPT4All**
Run LLMs offline on your device.
-  **Jan**
Open-source ChatGPT alternative.

AI MODEL PLATFORMS & HUBS

-  **Hugging Face**
Models, datasets and AI community hub.
-  **Replicate**
Run and fine-tune open source models.
-  **Together AI**
Fast inference for open models.
-  **fal**
Serverless AI inference platform.
-  **Modal**
Deploy AI models in the cloud.
-  **Arize Phoenix**
AI observability and evaluation.

POPULAR MODELS YOU CAN ACCESS

-  **GPT-4.5**
-  **Claude 3.7 Sonnet**
-  **Gemini 2.0**
-  **Llama 3.3 70B**
-  **Mistral Large 2**
-  **Phi-4 Multimodal**
-  **Gwen 2.5 Max**
-  **DeepSeek R1**

Access via web apps, APIs or local deployments.

DID YOU KNOW?

AI software is evolving fast! New models, features and tools are released every week. Stay curious, keep learning and keep building the future.

WHAT YOU CAN DO WITH AI SOFTWARE

-  Write & create content
-  Code & build faster
-  Analyze data & get insights
-  Automate workflows
-  Generate images, videos & audio
-  Research faster & deeper
-  Collaborate smarter
-  Personalize experiences

TIPS FOR GETTING STARTED

- ✓ Start with free plans and explore
- ✓ Focus on solving real problems
- ✓ Combine tools for maximum impact
- ✓ Keep learning and stay updated
- ✓ Use AI responsibly and ethically